Project: Mansourah Massarah Gold Project

Customer: MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: CALCULATION

3210 G&C - EMERGENCY STOCKPILE RECLAIM HOPPER FOUNDATION & RAMP DESIGN CALCULATION

AS BUILT

MD-216-3210-EG-CV-CAL-1002

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
D01	03-11-2023	ISSUED FOR CONSTRUCTION	RPK	SBY	PRI	
C01	12-12-2020	ISSUED FOR CONSTRUCTION	RPK	SBY	PRI	
A01	03-11-2020	FIRST ISSUE	RPK	SBY	PRI	

PROPRIETARY & CONFIDENTIAL This document and the information contained herein is Ma'aden proprietary know-how. This document or any part thereof should not be reproduced, stored in a retrieval system or transmitted by any means, electronic or otherwise, or divulged without prior written permission of Ma'aden.

Ma'aden Document Number:

MD-216-3210-EG-CV-CAL-1002 D01

Outotec Document Number:

900707 GAU01 US01 CED01 32113 OU100706082 2

L&T Document Number:

O19045 450 C K 24569 001 2

PROPRIETARY & CONFIDENTIAL : The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

PROPRIETARY & CONFIDENTIAL : This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01
Date	03-11-2020	12-12-2020	03-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	03-11-2020	12-12-2020	03-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

REFERENCE DOCUMENTS	
DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
MD-216-3000-EG-CV-RPT-1008	Geotechnical Report
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 304 :2018	Saudi Concrete Structures Code
ACI 318 :2008	Building Code Requirement of Structural Concrete

TABLE OF CONTENTS

DEFINITIONS AND ABBREVIATIONS			3
1	INTENT		4
2	EXECUTIVE SUMMARY		4
3	SCOPE		4
4	CONCRETE STRUCTURE		4
	4.1	Design	4
	4.2	Materials	4
	4.3	Foundations	4
	4.4	Minimum Concrete Cover	4
	4.5	Allowable Soil Bearing Capacity	4
	4.6	Reference	
5	GEOMETRY & DESIGN PHILOSOPHY		5
6	ANALYSIS & DESIGN OF RAMP PART 1		6-18
	6.1	Load Calculation	6
	6.2	Load Combination	8
	6.3	Design of Ramp Part-1	9
7	ANALYSIS & DESIGN OF RAMP PART 2		19
8	ANALYSIS & DESIGN OF RAMP PART 3		24
9	ANALYSIS & DESIGN OF ERH SUPPORT FOOTING		29
ATTACHMENT 1	MECHANICAL GENERAL ARRANGEMENT DRAWINGS		41
ATTACHMENT 2	STAADPRO INPUT FILE OF TC SUPPORT STRUCTURE		46

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	03-11-2020	12-12-2020	03-11-2023				Date	03-11-2020	12-12-2020	03-11-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	RPK	RPK	RPK				Prepared By	RPK	RPK	RPK			
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	03-11-2020	12-12-2020	03-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	03-11-2020	12-12-2020	03-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

1. INTENT

The document is prepared for civil design Calculation of Ramp for CAT loader and Emergency Reclaim hopper building foundation.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.
MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.
The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

This document is prepared for design of Ramp for CAT Loader and also design of foundation for ERH building.
Out of six columns of ESRH building three are designed as isolated foundation while other are supported on the ramp foundation.
Staad FEM model has been developed for Trough portion of Ramp and analysis has been done in Staad. Result for staad has been used to compute the reinforcement required.
Isolated footing design and design of Part 2 & 3 of Ramp have been performed using MS-Excel.
All the loads applied to the structure for analysis & design followed Civil -Structural Design Criteria and SBC 301:2007
Design of Reinforced Concrete members followed Strength Design as per SBC 304:2018 & ACI 318:2008.

4. CONCRETE FOUNDATIONS

4.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304 and ACI 318M.

4.2 Materials

Concrete Strength of 28 days
· Foundation and Structural Members: 35 MPa
· Grade slab and pavement: 30 MPa
· Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade
· Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
· Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

4.3 Foundations for Structures

Foundation shall be Isolated and combined Shallow Concrete foundation. The design calculations are prepared as per recommendation of standard SBC-304 Clause 13.3.

4.4 Minimum Concrete Cover

Items	Minimum Cover(mm)
Foundation (Side & Bottom)	75
Foundation (Top)	50
Column & Pedestal / Wall	50
Beam	50
Slab	40

4.5 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 500 kN/m2

4.6 Reference

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	03-11-2020	12-12-2020	03-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

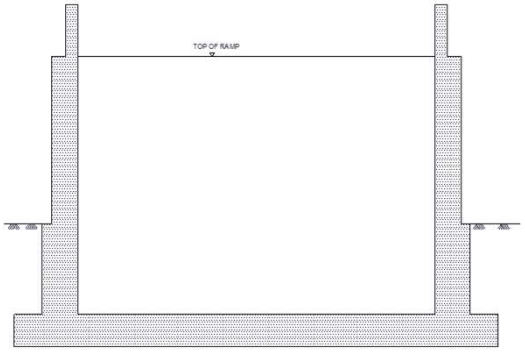
Outotec / L&T Revision No.:	0	1	2
Date	03-11-2020	12-12-2020	03-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

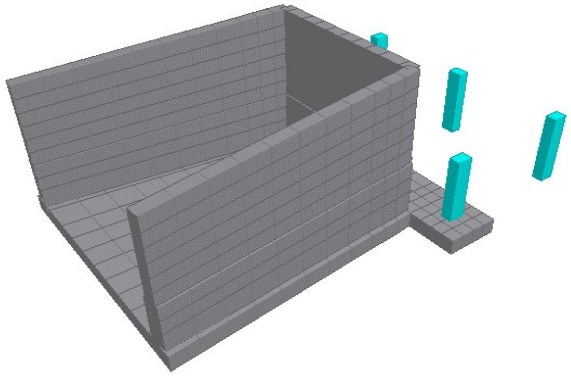
5. GEOMETRY AND DESIGN PHILOSOPHY

The ramp has been divided into three parts.

RAMP PART-1

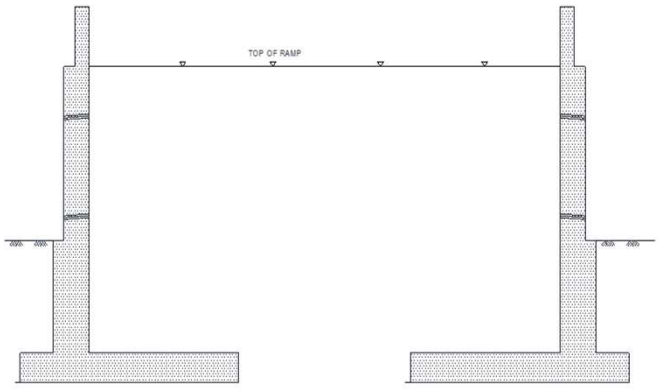
This portion of the ramp ha been designed as a trough wall. This part also supports the columns of Grid 3210/B of emergency reclaim hopper structure. The analysis of this structure has been done in Staad using FEM analysis.





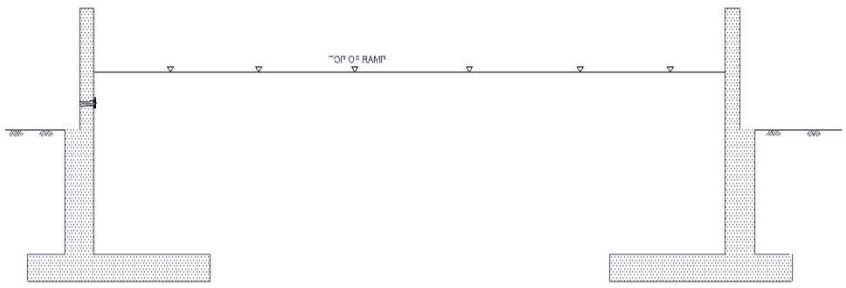
RAMP PART-2

This portion of the the ramp has been design as retaining wall. The design has been done assuming the maximum retaining height of the Ramp portion part-2. The retaining height is taken as 3.85m. The design has been performed in Excel.



RAMP PART-3

This portion of the the ramp has been design as retaining wall. The design has been done assuming the maximum retaining height of the Ramp portion part-2. The retaining height is taken as 2.25m. The design has been performed in Excel.



Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	03-11-2020	12-12-2020	03-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	03-11-2020	12-12-2020	03-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

6. ANALYSIS & DESIGN OF PART1

6.1 Load Calculation

6.1.1 Dead Load

Selfweight of structure is applied through Staad Pro.

Unit weight of concrete = 24.00 KN/m^3
Unit weight of soil = 20.00 KN/m^3

Height of soil in fill side = 7.40 m
Total load on raft = 148.00 KN/m^2

Height of soil in outer side = 2.85 m
Total load on Raft = 57.00 KN/m^2

Height of soil in fill side = 6.30 m
Total load on raft = 126.00 KN/m^2

6.1.2 Earth Pressure

Unit weight of soil = 20.00 KN/m^3
Coefficient of lateral earth pressure = 0.5

Filling side

Height (m)	coeff.	Unit wt KN/m^3	pressure KN/m^2	Avg. Input
0.0	0.5	20.00	0.0	
1.5	0.5	20.00	14.8	7.4
3.0	0.5	20.00	29.6	22.2
4.4	0.5	20.00	44.4	37.0
5.9	0.5	20.00	59.2	51.8
7.4	0.5	20.00	74.0	66.6

Outer side

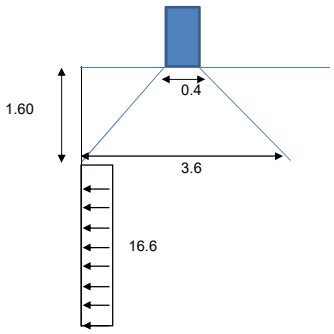
Height (m)	coeff.	Unit wt KN/m^3	pressure KN/m^2	Avg. Input
0.0	0.5	20.00	0.0	
1.5	0.5	20.00	14.8	7.4
3.0	0.5	20.00	29.6	22.2

6.1.3 Surcharge Pressure

Applied as Live Load

Total operating load of CAT loader = 1960 KN
Number of wheels = 4
Length of tyre contact = 0.9
Width tyre contact = 0.4
Distance of tyre and wall = 1.60 m
Effective length of pressure fulcrum = 4.1
Effective width of pressure fulcrum = 3.6
Effective stress per wheel = 33.2 KN/m^2 acts as surcharge from 1.5m below top

Surcharge load from movement of CAT loader = 33.2 KN/m^2
Coefficient of lateral earth pressure = 0.5
Lateral pressure due to surcharge = 16.6 KN/m^2



6.1.4 Water Pressure

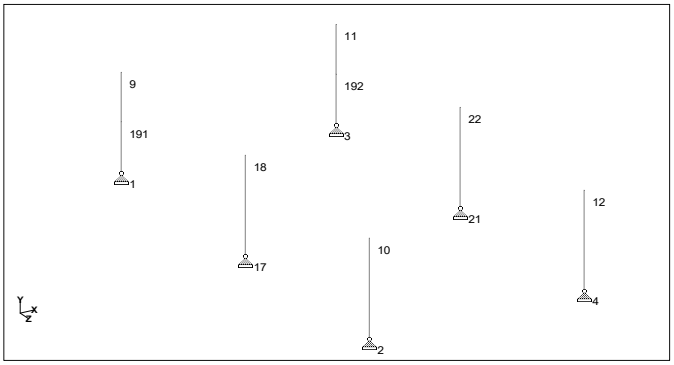
Water pressure behinfd the ramp wall is avoided since weep hole is provided to relieve the pressure.

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	03-11-2020	12-12-2020	03-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	03-11-2020	12-12-2020	03-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

6.1.5 Load Data from Emergency Reclaim Hopper Structure



Basic Load cases

L/C	DESCRIPTION
1	DEAD LOAD
2	LIVE LOAD
3	SDL-1
4	SDL-2
5	WIND LOAD +X
6	WIND LOAD -X
7	WIND LOAD +Z
8	WIND LOAD -Z
9	TEMP LOAD 1
10	TEMP LOAD 2

Node	L/C	FX KN	FY KN	FZ KN	MX KMN	MY KNM	MZ KNM
1	1	1	-42	-10	0	0	0
	2	3	-14	-11	0	0	0
	3	-2	-38	4	0	0	0
	4	-5	-255	-11	0	0	0
	5	36	95	0	0	0	0
	6	-36	-95	0	0	0	0
	7	0	62	29	0	0	0
	8	0	-62	-29	0	0	0
	9	-4	1	-4	0	0	0
	10	4	-1	4	0	0	0
2	1	0	-35	0	0	0	0
	2	1	-7	0	0	0	0
	3	0	-52	0	0	0	0
	4	1	-201	0	0	0	0
	5	36	97	0	0	0	0
	6	-36	-97	0	0	0	0
	7	0	0	1	0	0	0
	8	0	0	-1	0	0	0
	9	-4	0	1	0	0	0
	10	4	0	-1	0	0	0
3	1	-1	-56	-7	0	0	0
	2	-3	-32	-6	0	0	0
	3	1	-34	6	0	0	0
	4	4	-234	-6	0	0	0
	5	38	-95	1	0	0	0
	6	-38	95	-1	0	0	0
	7	0	52	25	0	0	0
	8	0	-52	-25	0	0	0
	9	3	0	-4	0	0	0
	10	-3	0	4	0	0	0
4	1	2	-50	0	0	0	0
	2	2	-43	0	0	0	0
	3	1	-51	0	0	0	0
	4	2	-215	0	0	0	0
	5	36	-97	0	0	0	0
	6	-36	97	0	0	0	0
	7	0	-1	1	0	0	0
	8	0	1	-1	0	0	0
	9	3	1	1	0	0	0
	10	-3	-1	-1	0	0	0
17	1	0	-67	8	0	0	0
	2	0	-41	8	0	0	0
	3	0	-152	32	0	0	0
	4	0	-639	34	0	0	0
	5	3	3	0	0	0	0
	6	-3	-3	0	0	0	0
	7	0	-63	29	0	0	0
	8	0	63	-29	0	0	0
	9	0	-1	4	0	0	0
	10	0	1	-4	0	0	0
21	1	-2	-84	9	0	0	0
	2	-3	-56	10	0	0	0
	3	0	-93	30	0	0	0
	4	-2	-360	33	0	0	0
	5	17	-3	0	0	0	0
	6	-17	3	0	0	0	0
	7	0	-51	24	0	0	0
	8	0	51	-24	0	0	0
	9	2	-1	3	0	0	0
	10	-2	1	-3	0	0	0

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	03-11-2020	12-12-2020	03-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	03-11-2020	12-12-2020	03-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

6.2 Load Combination

Load Combination for Base pressure check

LOAD 301 DL+SDL1
LOAD 302 DL+SDL2
LOAD 303 DL + SDL1 + TL1 + LL + EP
LOAD 304 DL + SDL1 + TL2 + LL + EP
LOAD 305 DL + SDL2 + TL1 + LL + EP
LOAD 306 DL + SDL2 + TL2 + LL + EP
LOAD 307 DL + SDL1 + EP + 0.75LL + 0.75WL(+X)
LOAD 308 DL + SDL1 + EP + 0.75LL + 0.75WL(-X)
LOAD 309 DL + SDL1 + EP + 0.75LL + 0.75WL(+Z)
LOAD 310 DL + SDL1 + EP + 0.75LL + 0.75WL(-Z)
LOAD 311 DL + SDL2 + EP + 0.75LL + 0.75WL(+X)
LOAD 312 DL + SDL2 + EP + 0.75LL + 0.75WL(-X)
LOAD 313 DL + SDL2 + EP + 0.75LL + 0.75WL(+Z)
LOAD 314 DL + SDL2 + EP + 0.75LL + 0.75WL(-Z)
LOAD 315 0.6 (DL + SDL1) + WL(+X) + EP
LOAD 316 0.6 (DL + SDL1) + WL(-X) + EP
LOAD 317 0.6 (DL + SDL1) + WL(+Z) + EP
LOAD 318 0.6 (DL + SDL1) + WL(-Z) + EP
LOAD 319 0.6 (DL + SDL2) + WL(+X) + EP
LOAD 320 0.6 (DL + SDL2) + WL(-X) + EP
LOAD 321 0.6 (DL + SDL2) + WL(+Z) + EP
LOAD 322 0.6 (DL + SDL2) + WL(-Z) + EP

Load Combination for concrete design

LOAD 401 1.4 (DL+SDL1)
LOAD 402 1.4 (DL+SDL2)
LOAD 403 1.4 (DL + SDL1 + TL1) + 1.7 LL + 1.7 EP
LOAD 404 1.4 (DL + SDL1 + TL2) + 1.7 LL + 1.7 EP
LOAD 405 1.4 (DL + SDL2 + TL1) + 1.7 LL + 1.7 EP
LOAD 406 1.4 (DL + SDL2 + TL2) + 1.7 LL + 1.7 EP
LOAD 407 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+X)
LOAD 408 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-X)
LOAD 409 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+Z)
LOAD 410 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-Z)
LOAD 411 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+X)
LOAD 412 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-X)
LOAD 413 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+Z)
LOAD 414 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-Z)
LOAD 415 0.9 (DL + SDL1) + 1.6 WL(+X) + 1.6 EP
LOAD 416 0.9 (DL + SDL1) + 1.6 WL(-X) + 1.6 EP
LOAD 417 0.9 (DL + SDL1) + 1.6 WL(+Z) + 1.6 EP
LOAD 418 0.9 (DL + SDL1) + 1.6 WL(-Z) + 1.6 EP
LOAD 419 0.9 (DL + SDL2) + 1.6 WL(+X) + 1.6 EP
LOAD 420 0.9 (DL + SDL2) + 1.6 WL(-X) + 1.6 EP
LOAD 421 0.9 (DL + SDL2) + 1.6 WL(+Z) + 1.6 EP
LOAD 422 0.9 (DL + SDL2) + 1.6 WL(-Z) + 1.6 EP

Basic Load cases

L/C	DESCRIPTION
DL	DEAD LOAD
LL	LIVE LOAD
SDL1	SDL-1
SDL2	SDL-2
WL(+X)	WIND LOAD +X
WL(-X)	WIND LOAD -X
WL(+Z)	WIND LOAD +Z
WL(-Z)	WIND LOAD -Z
TL1	TEMP LOAD 1
TL2	TEMP LOAD 2
EP	EARTH PRESSURE

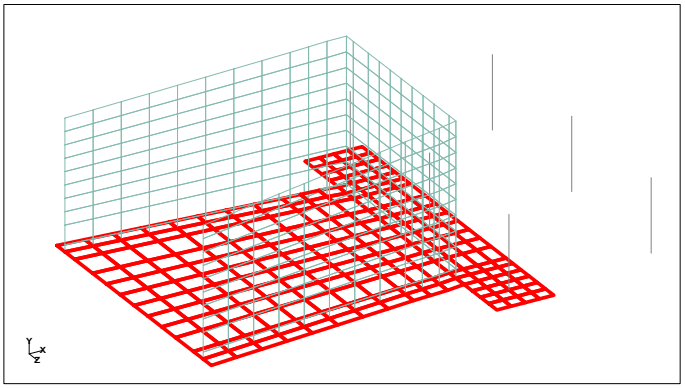
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

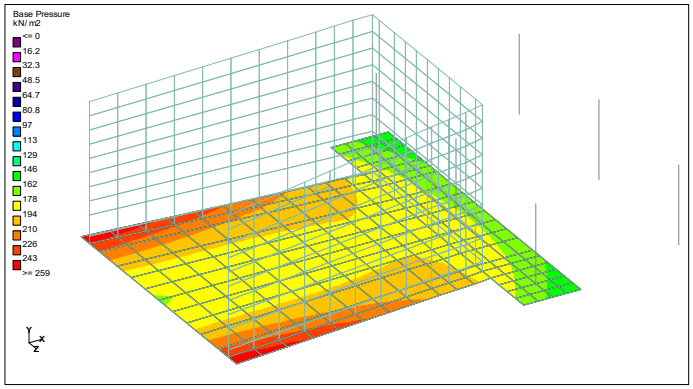
Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
Date	03-11-2020	12-12-2020	03-11-2023	Date	03-11-2020	12-12-2020	03-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK	Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

6.3 Design of Ramp Part-1

6.3.1 Design of Ramp wall Raft

For Calculation of Base pressure and Refer STAAD File **RAMP Part-1**





Finished Ground level	=	912.250	m
Bottom of foudation	=	909.400	m
Unit wt of soil	=	18.0	KN/m^3
Allowable Net Bearing Capacity	=	500	KN/m^2
Gross Bearing Capacity	=	551.3	KN/m^2

1) Check for Gross Base Pressure :

Maximum Base Pressure	=	258.7	KN/m^2	Load Case 303
Minimum Base Pressure	=	0.0	KN/m^2	Load Case 301

2) Design of foundation Raft :

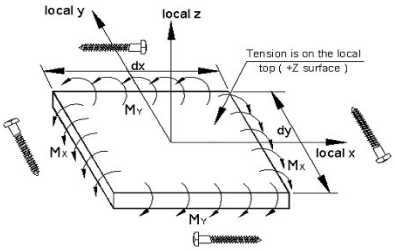
Slab Thickness =	750	mm	Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Width =	1000	mm	Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Clear Cover =	75	mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Compressive Strength of Concrete , fc' =	35	N/mm^2	Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	

STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	671.144	kN-m	Corresponding Mxy	-90.685	kN-m	*(Plate No. - 224 , L/C No. - 405)*
Min. Mx	-437.777	kN-m	Corresponding Mxy	-244.914	kN-m	*(Plate No. - 472 , L/C No. - 404)*
Max. My	89.028	kN-m	Corresponding Mxy	80.590	kN-m	*(Plate No. - 237 , L/C No. - 405)*
Min. My	-536.451	kN-m	Corresponding Mxy	-209.279	kN-m	*(Plate No. - 437 , L/C No. - 403)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	671.144 + 90.685 =	761.829	kN-m
My1 =	My + abs(Mxy) =	89.028 + 80.59 =	169.618	kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	671.144 + 92.373 =	763.517	kN-m
My2 =	My+abs(Mxy^2/Mx) =	89.028 + 9.677 =	98.705	kN-m



DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd =	761.829	kN-m
Myd =	169.618	kN-m

ALONG LOCAL X - DIRECTION :

Mu =	761.829	kN-m					
Φ*Kn =	Mu/bd^2				deff. =	662.5	mm
Mu/bd^2 =	761.829X10^6/(1000X662.5^2) =	1.736	N/mm^2	<	6.9	N/mm^2	
Pt =	0.0048	Reqd. Ast =	0048X1000X662.5/100 =	31.8	cm^2		
Provided	**25** mm @	**150**	C/C	(Provide Ast =	32.72	)cm^2	
				Total Provide Ast =	32.72	cm^2	**O.K.**
				Total Provide Pt =	0.0049		

ALONG LOCAL Y - DIRECTION :

Mu =	169.618	kN-m					
Φ*Kn =	Mu/bd^2				deff. =	637.5	mm
Mu/bd^2 =	169.618X10^6/(1000X637.5^2) =	0.386	N/mm^2	<	6.9	N/mm^2	
Pt =	0.0010	Reqd. Ast =	001X1000X662.5/100 =	6.38	cm^2		
Provided	**25** mm @	**150**	C/C	(Provide Ast =	32.72	)cm^2	
				Total Provide Ast =	32.72	cm^2	**O.K.**
				Total Provide Pt =	0.0051		

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	03-11-2020	12-12-2020	03-11-2023				Date	03-11-2020	12-12-2020	03-11-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	RPK	RPK	RPK				Prepared By	RPK	RPK	RPK		
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 = Mx - abs(Mxy) = -437.777 - 244.914 = -682.691 kN-m
My1 = My - abs(Mxy) = -536.451 - 209.279 = -745.73 kN-m
Mx2 = Mx-abs(Mxy^2/My) = -437.777 - 111.814 = -549.591 kN-m
My2 = My-abs(Mxy^2/Mx) = -536.451 - 100.046 = -436.405 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -682.691 kN-m
Myd = -745.73 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 682.691 kN-m
Φ*Kn = Mu/bd^2 deff. = 662.5 mm
Mu/bd^2 = 82.691X10^6/(1000X662.5^2) = 1.555 N/mm^2 < 6.9 N/mm^2
Pt = 0.0042 Reqd. Ast = 0042X1000X662.5/100 = 27.83 cm^2
Provided **25** mm @ **150** C/C (Provide Ast = 32.72)cm^2

Total Provide Ast = 32.72 cm^2 **O.K.**
Total Provide Pt = 0.0049

ALONG LOCAL Y - DIRECTION :

Mu = 745.73 kN-m
Φ*Kn = Mu/bd^2 deff. = 637.5 mm
Mu/bd^2 = 745.73X10^6/(1000X637.5^2) = 1.835 N/mm^2 < 6.9 N/mm^2
Pt = 0.0050 Reqd. Ast = 0050X1000X637.5/100 = 31.88 cm^2
Provided **25** mm @ **150** C/C (Provide Ast = 32.72)cm^2

Total Provide Ast = 32.72 cm^2 **O.K.**
Total Provide Pt = 0.0051

SHEAR CHECK

(Staad Group : RAFT_SHEAR)
(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = 0.682 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 496 , L/C No. - 415)*
Vu = 451.83 kN

Provided Pt = 0.0049
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0049X451.825X10^3X662.5/682.691/10^6)X1000X662.5/10^3
= 651.302 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0049)X1000X662.5/10^3
= 682.291 kN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X662.5/10^3
= 1136.627 kN
HENCE, Vc = 651.302 kN
Φt*Vc = 0.75X651.302 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 488.477 kN **> 451.825T , Hence O.K.**

ALONG LOCAL Y - DIRECTION :

SQy = 0.538 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 447 , L/C No. - 403)*
Vu = 342.975 kN

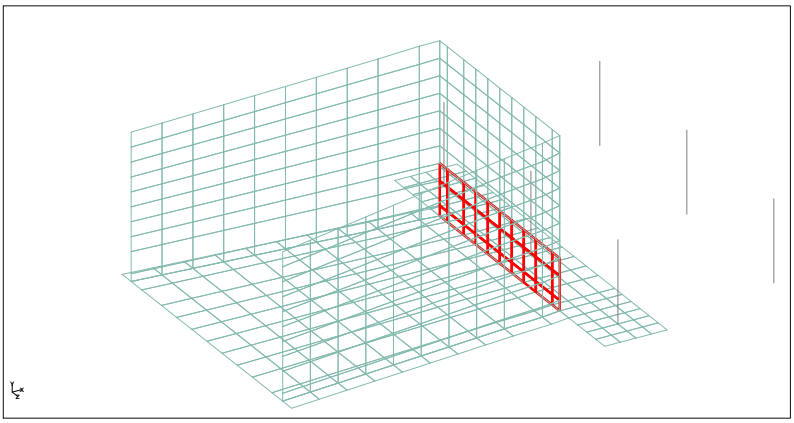
Pt provided = 0.0051
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0051X342.975X10^3X637.5/745.73/10^3)X1000X637.5/10^3
= 619.646 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0051)X1000X637.5/10^3
= 658.711 KN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X637.5/10^3
= 1093.735 kN
HENCE, Vc = 619.646 kN
Φt*Vc = 0.75X619.646 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 464.735 kN **> 342.975T , Hence O.K.**

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	03-11-2020	12-12-2020	03-11-2023				Date	03-11-2020	12-12-2020	03-11-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	RPK	RPK	RPK				Prepared By	RPK	RPK	RPK			
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

6.3.2 Design of Ramp Front wall (700thk)

1) Staad Geometry



2) Design of Wall :

Wall Thickness =	750	mm	Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Width =	1000	mm	Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Clear Cover =	50	mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Compressive Strength of Concrete , fc' =	35	N/mm^2	Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	

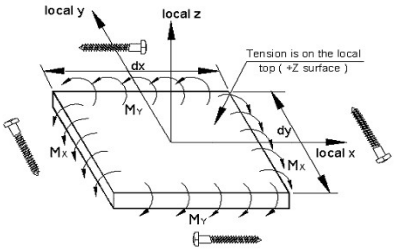
STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : FW_BM_700)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	426.379	kN-m	Corresponding Mxy	6.801	kN-m	*(Plate No. - 404 , L/C No. - 403)*
Min. Mx	-140.785	kN-m	Corresponding Mxy	8.306	kN-m	*(Plate No. - 364 , L/C No. - 402)*
Max. My	499.451	kN-m	Corresponding Mxy	-57.098	kN-m	*(Plate No. - 401 , L/C No. - 406)*
Min. My	-88.620	kN-m	Corresponding Mxy	6.919	kN-m	*(Plate No. - 361 , L/C No. - 406)*

FOR INSIDE FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	426.379 + 6.801 =	433.180	kN-m
My1 =	My + abs(Mxy) =	499.451 + 57.098 =	556.549	kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	426.379 + 0.093 =	426.472	kN-m
My2 =	My+abs(Mxy^2/Mx) =	499.451 + 7.646 =	507.097	kN-m



DESIGN MOMENT CALCULATION FOR INSIDE FACE REINFORCEMENT :

Mxd =	433.18	kN-m
Myd =	556.549	kN-m

ALONG LOCAL X - DIRECTION : VERTICAL

Mu =	433.18	kN-m						
Φ*Kn =	Mu/bd^2					deff. =	662.5	mm
Mu/bd^2 =	433.18X10^6/(1000X662.5^2) =		0.987	N/mm^2	<	6.9	N/mm^2	
Pt =	0.0027		Reqd. Ast =	0027X1000X662.5/100 =		17.89	cm^2	
Provided	**25**	mm @	**150**	C/C	(Provide Ast =	32.72	)cm^2	
					Total Provide Ast =	32.72	cm^2	**O.K.**
					Total Provide Pt =	0.0049		

ALONG LOCAL Y - DIRECTION : HORIZONTAL

Mu =	556.549	kN-m						
Φ*Kn =	Mu/bd^2					deff. =	675	mm
Mu/bd^2 =	556.549X10^6/(1000X675^2) =		1.222	N/mm^2	<	6.9	N/mm^2	
Pt =	0.0033		Reqd. Ast =	0033X1000X662.5/100 =		22.28	cm^2	
Provided	**25**	mm @	**150**	C/C	(Provide Ast =	32.72	)cm^2	
					Total Provide Ast =	32.72	cm^2	**O.K.**
					Total Provide Pt =	0.0048		

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	03-11-2020	12-12-2020	03-11-2023				Date	03-11-2020	12-12-2020	03-11-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	RPK	RPK	RPK				Prepared By	RPK	RPK	RPK			
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

FOR OUTSIDE FACE REINFORCEMENT :

Mx1 = Mx - abs(Mxy) = -140.785 - 8.306 = -149.091 kN-m
My1 = My - abs(Mxy) = -88.62 - 6.919 = -95.539 kN-m
Mx2 = Mx-abs(Mxy^2/My) = -140.785 - 0.778 = -141.563 kN-m
My2 = My-abs(Mxy^2/Mx) = -88.62 - 0.34 = -88.28 kN-m

DESIGN MOMENT CALCULATION FOR OUTSIDE FACE REINFORCEMENT :

Mxd = -149.091 kN-m
Myd = -95.539 kN-m

ALONG LOCAL X - DIRECTION : VERTICAL

Mu = 149.091 kN-m
Φ*Kn = Mu/bd^2 deff. = 670 mm
Mu/bd^2 = 149.091X10^6/(1000X670^2) = 0.332 N/mm^2 < 6.9 N/mm^2
Pt = 0.0009 Reqd. Ast = 0.0009X1000X670/100 = 6.03 cm^2
Provided **20** mm @ **150** C/C (Provide Ast = 20.94)cm^2

Total Provide Ast = 20.94 cm^2 **O.K.**
Total Provide Pt = 0.0031

ALONG LOCAL Y - DIRECTION : HORIZONTAL

Mu = 95.539 kN-m
Φ*Kn = Mu/bd^2 deff. = 690 mm
Mu/bd^2 = 95.539X10^6/(1000X690^2) = 0.201 N/mm^2 < 6.9 N/mm^2
Pt = 0.0005 Reqd. Ast = 0.0005X1000X690/100 = 3.45 cm^2
Provided **20** mm @ **150** C/C (Provide Ast = 20.94)cm^2

Total Provide Ast = 20.94 cm^2 **O.K.**
Total Provide Pt = 0.003

SHEAR CHECK

(Staad Group : FW_SF_700)
(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = 0.113 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 353 , L/C No. - 403)*
Vu = 75.71 kN

Provided Pt = 0.0031
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0031X75.71X10^3X670/149.091/10^6)X1000X670/10^3
= 646.217 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0031)X1000X670/10^3
= 669.513 kN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X670/10^3
= 1149.494 kN
HENCE, Vc = 646.217 kN
Φt*Vc = 0.75X646.217 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 484.663 kN **> 75.71T , Hence O.K.**

ALONG LOCAL Y - DIRECTION :

SQy = 0.315 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 391 , L/C No. - 406)*
Vu = 217.35 kN

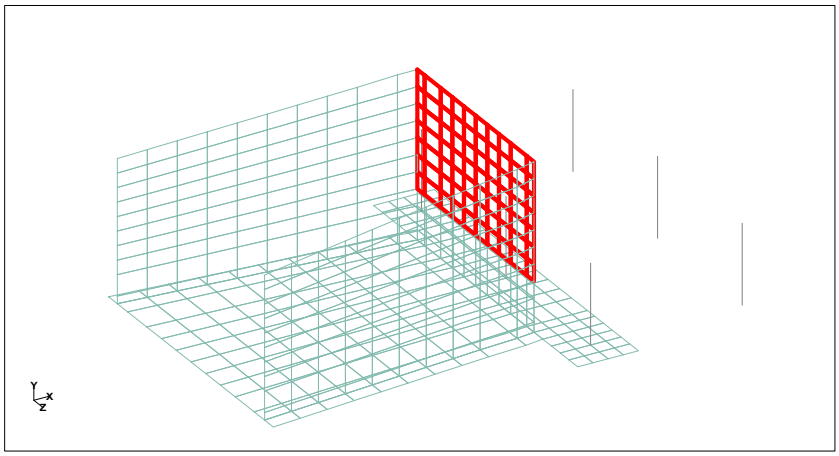
Pt provided = 0.0048
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0048X217.35X10^3X690/95.539/10^3)X1000X690/10^3
= 653.456 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0048)X1000X690/10^3
= 709.439 KN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X690/10^3
= 1183.808 kN
HENCE, Vc = 653.456 kN
Φt*Vc = 0.75X653.456 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 490.092 kN **> 217.35T , Hence O.K.**

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	03-11-2020	12-12-2020	03-11-2023				Date	03-11-2020	12-12-2020	03-11-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	RPK	RPK	RPK				Prepared By	RPK	RPK	RPK			
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

6.3.3 Design of Ramp Front wall (550thk)

1) Staad Geometry



2) Design of Wall :

Wall Thickness =	550	mm	Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Width =	1000	mm	Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Clear Cover =	50	mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Compressive Strength of Concrete , fc' =	35	N/mm^2	Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	

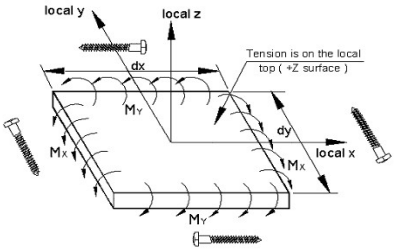
STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : FW_BM_550)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	299.861	kN-m	Corresponding Mxy	31.419	kN-m	*(Plate No. - 365 , L/C No. - 406)*
Min. Mx	-59.931	kN-m	Corresponding Mxy	0.932	kN-m	*(Plate No. - 360 , L/C No. - 419)*
Max. My	473.499	kN-m	Corresponding Mxy	-95.934	kN-m	*(Plate No. - 305 , L/C No. - 403)*
Min. My	-43.259	kN-m	Corresponding Mxy	3.229	kN-m	*(Plate No. - 360 , L/C No. - 406)*

FOR INSIDE FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	299.861 + 31.419 =	331.280	kN-m
My1 =	My + abs(Mxy) =	473.499 + 95.934 =	569.433	kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	299.861 + 2.085 =	301.946	kN-m
My2 =	My+abs(Mxy^2/Mx) =	473.499 + 30.692 =	504.191	kN-m



DESIGN MOMENT CALCULATION FOR INSIDE FACE REINFORCEMENT :

Mxd = 331.28 kN-m

Myd = 569.433 kN-m

ALONG LOCAL X - DIRECTION : VERTICAL

Mu = 331.28 kN-m

Φ*Kn = Mu/bd^2 deff. = 537.5 mm

Mu/bd^2 = 331.28X10^6/(1000X537.5^2) = 1.147 N/mm^2 < 6.9 N/mm^2

Pt = 0.0031 Reqd. Ast = 0031X1000X537.5/100 = 16.66 cm^2

Provided **25** mm @ **150** C/C (Provide Ast = 32.72)cm^2

Total Provide Ast = 32.72 cm^2 **O.K.**

Total Provide Pt = 0.0061

ALONG LOCAL Y - DIRECTION : HORIZONTAL

Mu = 569.433 kN-m

Φ*Kn = Mu/bd^2 deff. = 487.5 mm

Mu/bd^2 = 569.433X10^6/(1000X487.5^2) = 2.396 N/mm^2 < 6.9 N/mm^2

Pt = 0.0067 Reqd. Ast = 0067X1000X537.5/100 = 32.66 cm^2

Provided **25** mm @ **150** C/C (Provide Ast = 32.72)cm^2

Total Provide Ast = 32.72 cm^2 **O.K.**

Total Provide Pt = 0.0067

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	03-11-2020	12-12-2020	03-11-2023				Date	03-11-2020	12-12-2020	03-11-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	RPK	RPK	RPK				Prepared By	RPK	RPK	RPK			
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

FOR OUTSIDE FACE REINFORCEMENT :

Mx1 = Mx - abs(Mxy) = -59.931 - 0.932 = -60.863 kN-m
My1 = My - abs(Mxy) = -43.259 - 3.229 = -46.488 kN-m
Mx2 = Mx-abs(Mxy^2/My) = -59.931 - 0.02 = -59.951 kN-m
My2 = My-abs(Mxy^2/Mx) = -43.259 - 0.174 = -43.085 kN-m

DESIGN MOMENT CALCULATION FOR OUTSIDE FACE REINFORCEMENT :

Mxd = -60.863 kN-m
Myd = -46.488 kN-m

ALONG LOCAL X - DIRECTION : VERTICAL

Mu = 60.863 kN-m
Φ*Kn = Mu/bd^2 deff. = 476 mm
Mu/bd^2 = 60.863X10^6/(1000X476^2) = 0.269 N/mm^2 < 6.9 N/mm^2
Pt = 0.0007 Reqd. Ast = 0.0007X1000X476/100 = 3.33 cm^2
Provided **16** mm @ **150** C/C (Provide Ast = 13.4)cm^2

Total Provide Ast = 13.4 cm^2 **O.K.**
Total Provide Pt = 0.0028

ALONG LOCAL Y - DIRECTION : HORIZONTAL

Mu = 46.488 kN-m
Φ*Kn = Mu/bd^2 deff. = 492 mm
Mu/bd^2 = 46.488X10^6/(1000X492^2) = 0.192 N/mm^2 < 6.9 N/mm^2
Pt = 0.0005 Reqd. Ast = 0.0005X1000X492/100 = 2.46 cm^2
Provided **16** mm @ **150** C/C (Provide Ast = 13.4)cm^2

Total Provide Ast = 13.4 cm^2 **O.K.**
Total Provide Pt = 0.0027

SHEAR CHECK

(Staad Group : FW_SF_550)
(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = -0.191 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 317 , L/C No. - 403)*
Vu = 90.92 kN

Provided Pt = 0.0028
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0028X90.916X10^3X476/60.863/10^6)X1000X476/10^3
= 466.679 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0028)X1000X476/10^3
= 473.226 kN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X476/10^3
= 816.656 kN
HENCE, Vc = 466.679 kN
Φt*Vc = 0.75X466.679 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 350.009 kN **> 90.916T , Hence O.K.**

ALONG LOCAL Y - DIRECTION :

SQy = -0.651 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 315 , L/C No. - 403)*
Vu = 320.292 kN

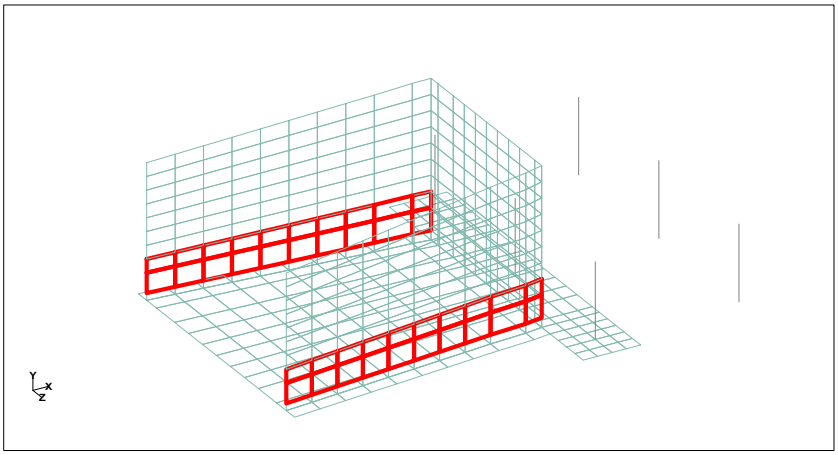
Pt provided = 0.0027
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0027X320.292X10^3X492/46.488/10^3)X1000X492/10^3
= 467.59 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0027)X1000X492/10^3
= 488.297 KN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X492/10^3
= 844.106 kN
HENCE, Vc = 467.59 kN
Φt*Vc = 0.75X467.59 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 350.693 kN **> 320.292T , Hence O.K.**

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
Date	03-11-2020	12-12-2020	03-11-2023	Date	03-11-2020	12-12-2020	03-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK	Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

6.3.4 Design of Ramp Side wall (750thk)

1) Staad Geometry



2) Design of Wall :

Wall Thickness =	750	mm	Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Width =	1000	mm	Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Clear Cover =	50	mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Compressive Strength of Concrete , fc' =	35	N/mm^2	Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	

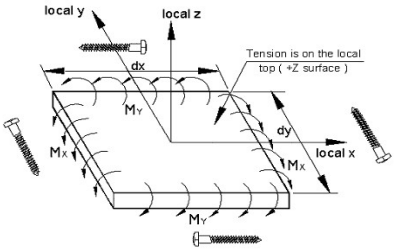
STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : SW_BM_700)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	641.093	kN-m	Corresponding Mxy	146.647	kN-m	*(Plate No. - 24 , L/C No. - 406)*
Min. Mx	-72.215	kN-m	Corresponding Mxy	-40.640	kN-m	*(Plate No. - 144 , L/C No. - 402)*
Max. My	429.576	kN-m	Corresponding Mxy	-174.109	kN-m	*(Plate No. - 111 , L/C No. - 406)*
Min. My	-96.425	kN-m	Corresponding Mxy	-247.682	kN-m	*(Plate No. - 161 , L/C No. - 406)*

FOR INSIDE FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	641.093 + 146.647 =	787.740	kN-m
My1 =	My + abs(Mxy) =	429.576 + 174.109 =	603.685	kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	641.093 + 50.062 =	691.155	kN-m
My2 =	My+abs(Mxy^2/Mx) =	429.576 + 47.285 =	476.861	kN-m



DESIGN MOMENT CALCULATION FOR INSIDE FACE REINFORCEMENT :

Mxd =	787.74	kN-m
Myd =	603.685	kN-m

ALONG LOCAL X - DIRECTION : VERTICAL

Mu =	787.74	kN-m					
Φ*Kn =	Mu/bd^2				deff. =	662.5	mm
Mu/bd^2 =	787.74X10^6/(1000X662.5^2) =	1.795	N/mm^2	<	6.9	N/mm^2	
Pt =	0.0049	Reqd. Ast =	0049X1000X662.5/100 =	32.46	cm^2		
Provided	**25**	mm @	**150**	C/C	(Provide Ast =	32.72	)cm^2
					Total Provide Ast =	32.72	cm^2
					Total Provide Pt =	0.0049	

O.K.

ALONG LOCAL Y - DIRECTION : HORIZONTAL

Mu =	603.685	kN-m					
Φ*Kn =	Mu/bd^2				deff. =	687.5	mm
Mu/bd^2 =	03.685X10^6/(1000X687.5^2) =	1.277	N/mm^2	<	6.9	N/mm^2	
Pt =	0.0035	Reqd. Ast =	0035X1000X662.5/100 =	24.06	cm^2		
Provided	**25**	mm @	**150**	C/C	(Provide Ast =	32.72	)cm^2
					Total Provide Ast =	32.72	cm^2
					Total Provide Pt =	0.0048	

O.K.

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	03-11-2020	12-12-2020	03-11-2023				Date	03-11-2020	12-12-2020	03-11-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	RPK	RPK	RPK				Prepared By	RPK	RPK	RPK		
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

FOR OUTSIDE FACE REINFORCEMENT :

Mx1 = Mx - abs(Mxy) = -72.215 - 40.64 = -112.855 kN-m
My1 = My - abs(Mxy) = -96.425 - 247.682 = -344.107 kN-m
Mx2 = Mx-abs(Mxy^2/My) = -72.215 - 17.128 = -89.343 kN-m
My2 = My-abs(Mxy^2/Mx) = -96.425 - 849.496 = 753.071 kN-m

DESIGN MOMENT CALCULATION FOR OUTSIDE FACE REINFORCEMENT :

Mxd = -112.855 kN-m
Myd = -344.107 kN-m

ALONG LOCAL X - DIRECTION : VERTICAL

Mu = 112.855 kN-m
Φ*Kn = Mu/bd^2 deff. = 670 mm
Mu/bd^2 = 112.855X10^6/(1000X670^2) = 0.251 N/mm^2 < 6.9 N/mm^2
Pt = 0.0007 Reqd. Ast = 0.0007X1000X670/100 = 4.69 cm^2
Provided **20** mm @ **150** C/C (Provide Ast = 20.94)cm^2

Total Provide Ast = 20.94 cm^2 **O.K.**
Total Provide Pt = 0.0031

ALONG LOCAL Y - DIRECTION : HORIZONTAL

Mu = 344.107 kN-m
Φ*Kn = Mu/bd^2 deff. = 690 mm
Mu/bd^2 = 344.107X10^6/(1000X690^2) = 0.723 N/mm^2 < 6.9 N/mm^2
Pt = 0.0019 Reqd. Ast = 0.0019X1000X690/100 = 13.11 cm^2
Provided **20** mm @ **150** C/C (Provide Ast = 20.94)cm^2

Total Provide Ast = 20.94 cm^2 **O.K.**
Total Provide Pt = 0.003

SHEAR CHECK

(Staad Group : SW_SF_700)
(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = 0.556 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 13 , L/C No. - 405)*
Vu = 372.52 kN

Provided Pt = 0.0031
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0031X372.52X10^3X670/112.855/10^6)X1000X670/10^3
= 712.293 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0031)X1000X670/10^3
= 669.513 kN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X670/10^3
= 1149.494 kN
HENCE, Vc = 669.513 kN
Φt*Vc = 0.75X669.513 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 502.135 kN **> 372.52T , Hence O.K.**

ALONG LOCAL Y - DIRECTION :

SQy = -0.526 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 550 , L/C No. - 406)*
Vu = 362.94 kN

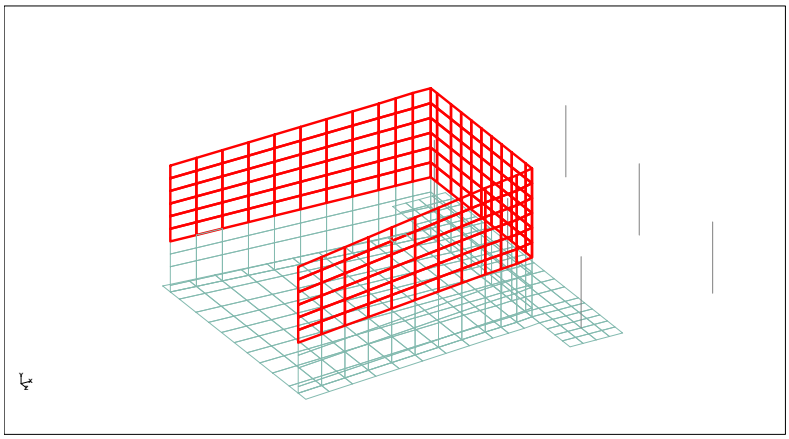
Pt provided = 0.003
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.003X362.94X10^3X690/344.107/10^3)X1000X690/10^3
= 665.261 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.003)X1000X690/10^3
= 688.325 KN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X690/10^3
= 1183.808 kN
HENCE, Vc = 665.261 kN
Φt*Vc = 0.75X665.261 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 498.946 kN **> 362.94T , Hence O.K.**

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
Date	03-11-2020	12-12-2020	03-11-2023	Date	03-11-2020	12-12-2020	03-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK	Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

6.3.5 Design of Ramp Side wall (550thk)

1) Staad Geometry



2) Design of Wall :

Wall Thickness =	550	mm	Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Width =	1000	mm	Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Clear Cover =	75	mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Compressive Strength of Concrete , fc' =	35	N/mm^2	Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	

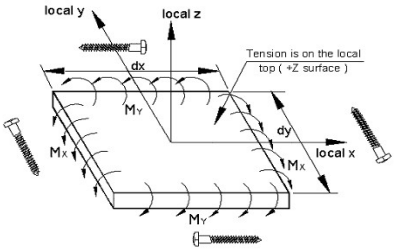
STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : FW_BM_700)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	62.197	kN-m	Corresponding Mxy	-22.856	kN-m	*(Plate No. - 106 , L/C No. - 406)*
Min. Mx	-83.577	kN-m	Corresponding Mxy	-142.534	kN-m	*(Plate No. - 148 , L/C No. - 406)*
Max. My	471.949	kN-m	Corresponding Mxy	-4.432	kN-m	*(Plate No. - 95 , L/C No. - 403)*
Min. My	-130.669	kN-m	Corresponding Mxy	-85.121	kN-m	*(Plate No. - 165 , L/C No. - 406)*

FOR INSIDE FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	62.197 + 22.856 =	85.053	kN-m
My1 =	My + abs(Mxy) =	471.949 + 4.432 =	476.381	kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	62.197 + 1.107 =	63.304	kN-m
My2 =	My+abs(Mxy^2/Mx) =	471.949 + 0.316 =	472.265	kN-m



DESIGN MOMENT CALCULATION FOR INSIDE FACE REINFORCEMENT :

Mxd = 85.053 kN-m
Myd = 476.381 kN-m

ALONG LOCAL X - DIRECTION : VERTICAL

Mu = 85.053 kN-m
Φ*Kn = Mu/bd^2 deff. = 437.5 mm
Mu/bd^2 = 85.053X10^6/(1000X437.5^2) = 0.444 N/mm^2 < 6.9 N/mm^2
Pt = 0.0012 Reqd. Ast = 0012X1000X437.5/100 = 5.25 cm^2
Provided **25** mm @ **150** C/C (Provide Ast = 32.72)cm^2

Total Provide Ast = 32.72 cm^2 **O.K.**
Total Provide Pt = 0.0075

ALONG LOCAL Y - DIRECTION : HORIZONTAL

Mu = 476.381 kN-m
Φ*Kn = Mu/bd^2 deff. = 462.5 mm
Mu/bd^2 = 76.381X10^6/(1000X462.5^2) = 2.227 N/mm^2 < 6.9 N/mm^2
Pt = 0.0062 Reqd. Ast = 0062X1000X437.5/100 = 28.68 cm^2
Provided **25** mm @ **150** C/C (Provide Ast = 32.72)cm^2

Total Provide Ast = 32.72 cm^2 **O.K.**
Total Provide Pt = 0.0071

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	03-11-2020	12-12-2020	03-11-2023				Date	03-11-2020	12-12-2020	03-11-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	RPK	RPK	RPK				Prepared By	RPK	RPK	RPK			
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

FOR OUTSIDE FACE REINFORCEMENT :

Mx1 = Mx - abs(Mxy) = -83.577 - 142.534 = -226.111 kN-m
My1 = My - abs(Mxy) = -130.669 - 85.121 = -215.79 kN-m
Mx2 = Mx-abs(Mxy^2/My) = -83.577 - 155.476 = -239.053 kN-m
My2 = My-abs(Mxy^2/Mx) = -130.669 - 86.694 = -43.975 kN-m

DESIGN MOMENT CALCULATION FOR OUTSIDE FACE REINFORCEMENT :

Mxd = -226.111 kN-m
Myd = -215.79 kN-m

ALONG LOCAL X - DIRECTION : VERTICAL

Mu = 226.111 kN-m
Φ*Kn = Mu/bd^2 deff. = 445 mm
Mu/bd^2 = 226.111X10^6/(1000X445^2) = 1.142 N/mm^2 < 6.9 N/mm^2
Pt = 0.0031 Reqd. Ast = 0.0031X1000X445/100 = 13.8 cm^2
Provided **20** mm @ **150** C/C (Provide Ast = 20.94)cm^2

Total Provide Ast = 20.94 cm^2 **O.K.**
Total Provide Pt = 0.0047

ALONG LOCAL Y - DIRECTION : HORIZONTAL

Mu = 215.79 kN-m
Φ*Kn = Mu/bd^2 deff. = 465 mm
Mu/bd^2 = 215.79X10^6/(1000X465^2) = 0.998 N/mm^2 < 6.9 N/mm^2
Pt = 0.0027 Reqd. Ast = 0.0027X1000X465/100 = 12.56 cm^2
Provided **20** mm @ **150** C/C (Provide Ast = 20.94)cm^2

Total Provide Ast = 20.94 cm^2 **O.K.**
Total Provide Pt = 0.0045

SHEAR CHECK

(Staad Group : FW_SF_700)
(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = 0.302 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 9 , L/C No. - 403)*
Vu = 134.39 kN

Provided Pt = 0.0047
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0047X134.39X10^3X445/226.111/10^6)X1000X445/10^3
= 430.629 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0047)X1000X445/10^3
= 456.78 kN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X445/10^3
= 763.47 kN
HENCE, Vc = 430.629 kN
Φt*Vc = 0.75X430.629 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 322.972 kN **> 134.39T , Hence O.K.**

ALONG LOCAL Y - DIRECTION :

SQy = -0.635 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 538 , L/C No. - 406)*
Vu = 295.275 kN

Pt provided = 0.0045
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0045X295.275X10^3X465/215.79/10^3)X1000X465/10^3
= 447.933 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0045)X1000X465/10^3
= 475.729 KN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X465/10^3
= 797.783 kN
HENCE, Vc = 447.933 kN
Φt*Vc = 0.75X447.933 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 335.95 kN **> 295.275T , Hence O.K.**

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01		
Date	03-11-2020	12-12-2020	03-11-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	RPK	RPK	RPK		
Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI		

Outotec / L&T Revision No.:	0	1	2		
Date	03-11-2020	12-12-2020	03-11-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	RPK	RPK	RPK		
Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI		

7. ANALYSIS & DESIGN OF RAMP PART 2

Wall is subjected to the following pressure.

1) Saturated soil pressure

2) Surcharge pressure

3) Water pressure Considered upto FGL as weep holes are provided in the wall water pressure not considered above Ground level

1) Saturated soil pressure

Level of Filling Top		916.100	m
Ground level		912.250	m
Bottom of footing		909.400	m
Unit weight of soil, γ =		18.0	KN/m^3
Considering ground water table is at +EL. =		0.00	m
Unit weight of water, γ_{water} =		10.0	KN/m^3
Submenrger weight of soil, γ_{sub} =		8.0	KN/m^3
Therefore,	γ_{sat}=	20.0	KN/m^3
Unit weight of concrete, $\gamma_{concrete}$ =		24	KN/m^3
Co-efficient of active earth pressure, k_a =		0.3	
Co-efficient of passive earth pressure, k_p =		3.33	
Soil Pressure at base heel side = $k_a.\gamma_{sat}.h$ =		(0.3 x 20 x 6.7) =	40.2 KN/m^2
Soil Pressure at base Toe side = $k_a.\gamma_{sat}.h$ =		(3.33 x 20 x 2.35) =	157 KN/m^2

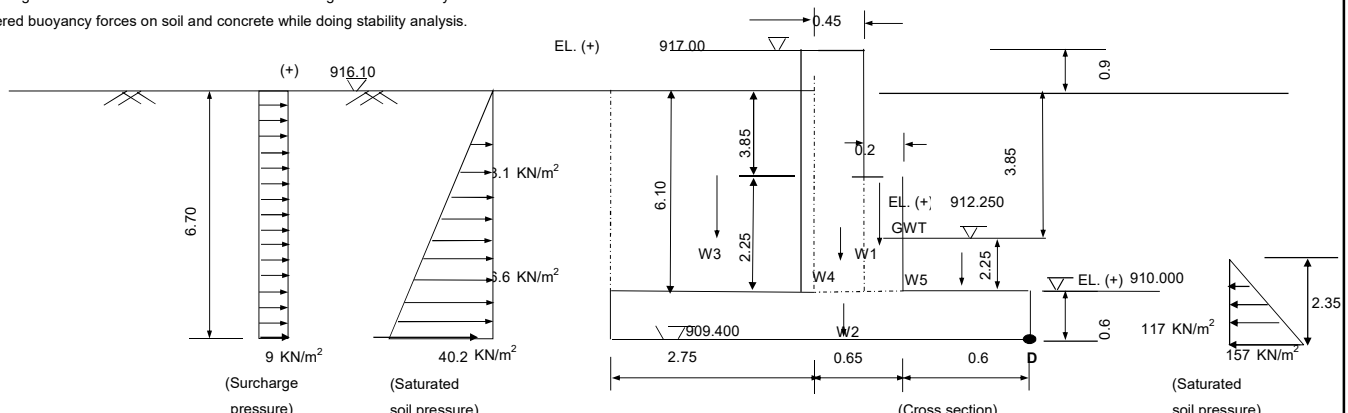
2) Surcharge Pressure:

Surcharge load on ground, q =	30.0	KN/m^2
Surcharge Pressure , $k_a.q$ =	(0.3 x 30) =	9 KN/m^2

3) Water Pressure:

For water pressure being same on both side of wall effect has been reglected for stability.

But we have considered buoyancy forces on soil and concrete while doing stability analysis.



**Buoyancy effect considered

LOAD MARKED	VERTICAL LOAD (KN)**	CENTRE OF GRAVITY DISTANCE OF LOAD FROM 'D'	RESTORING MOMENT (KN-m)
W1	6.3	0.7	4.40
W2	33.60	2	67.20
W3	335.50	2.625	880.70
W4	55.76	1.03	57.10
W5	27.000	0.30	8.10
W6	82.50	2.63	216.60
Total	**540.66**		**1234.10**
LOAD TYPE	**HORIZONTAL LOAD (KN)**	**CENTRE OF GRAVITY DISTANCE OF LOAD FROM 'D'**	**RESTORING MOMENT (KN-m)**
Passive EP toe	183.899	0.78	144.05
Total Restoring Moment			**1378.15**

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

LOAD TYPE	HORIZONTAL LOAD (KN)	CENTRE OF GRAVITY DISTANCE OF LOAD FROM 'D'	OVERTURNING MOMENT (KN-m)
1) S.P	60.300	3.35	202.0
2) S.S.P	134.670	2.23	300.8
Total	**194.97**		**502.77**

FACTOR OF SAFETY AGAINST OVERTURNING:

Factor of safety against overturning = (1378.15 / 502.77) = 2.741 > 1.50 **Hence safe**

FACTOR OF SAFETY AGAINST SLIDING:

Sliding force = 194.97 KN

Co-efficient of friction between soil and concrete, μ = 0.5

Resisting force Friction= 270.33 N Restoring Force Passive pressure = 183.90 KN

Total Resisting Force = 454.23 N

Factor of safety against sliding = 2.33 > 1.50 **Hence safe**

CHECK FOR ECCENTRICITY:

Total bottom width, b = 4 m

Resultant Moment = 875.38 KNm

z = (875.38 / 540.66) = 1.62 m

b/6 = (4 / 6) = 0.67 m

Eccentricity, e= z -(b/2) = (4 / 2) - 1.62 = 0.38 m < 0.67 m **Hence safe**

DESIGN OF STEM WALL:

Flexural reinforcement Fill Side

Wall thickness, D	=	**650** mm	Wall thickness, D	=	**450** mm
Diameter of bars	=	25 mm	Diameter of bars	=	20 mm
Clear cover	=	50 mm	Clear cover	=	50 mm
Effective cover, d'	=	50 + 12 + (25 / 2 = 74.5 mm	Effective cover, d'	=	50 + 12 + (20 / 2) = 72 mm
Effective depth, d	=	(650 - 74.5) = 575.5 mm	Effective depth, d	=	(450 - 72) = 378 mm
Width, b	=	1000 mm	Width, b	=	1000 mm
Grade of steel, f_{ck}	=	35 N/mm^2	Grade of steel, f_{ck}	=	35 N/mm^2
Grade of concrete, f_y	=	420 N/mm^2	Grade of concrete, f_y	=	420 N/mm^2
Maximum bending moment, M	=	**394.4** KN-m	Maximum bending moment, M	=	**123.8** KN-m
Maximum shear force	=	**166.5** KN	Maximum shear force	=	**79.1** KN
Load factor	=	1.7	Load factor	=	1.7
Factored Maximum bending moment, M	=	**671** KNm	Factored Maximum bending moment, M	=	**210** KNm
Factored Maximum shear force	=	**283** KN	Factored Maximum shear force	=	**134** KN

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Design for 650mm thickness **At Fill side as Vertical Main Reinforcement** *assuptions in clause 22.2 of SBC 304*

Mu lim

$= \emptyset\, 0.85\, f_c'\, \beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

$\frac{c}{d}\,max = \frac{0.003}{0.003 + 0.005} = 0.38$

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 1000 x 575.5 x 575.5 (1 - 0.8/2 x 0.375)

= 2261.3 KNm b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

$A_{st1} = \frac{0.85\, f_c'\, bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c'\, bd^2}}\right)$

= 0.85 x 35x 1000 x 575.5 x (1-√(1-(2.35 x 670.6 x 10^6)/(0.9 x 35 x 1000 x 575.5^2)))/420

= 3204.4 mm^2

required spacing = 153.1 mm

Spacing provided = 150 mm

Provide 25mm diameter bars at 150mm c/c.

Design for 450mm thickness **At Fill side as Vertical Main Reinforcement** *assuptions in clause 22.2 of SBC 304*

Mu lim

$= \emptyset\, 0.85\, f_c'\, \beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

$\frac{c}{d}\,max = \frac{0.003}{0.003 + 0.005} = 0.38$

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 1000 x 378 x 378 (1 - 0.8/2 x 0.375)

= 975.6 KNm b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

$A_{st1} = \frac{0.85\, f_c'\, bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c'\, bd^2}}\right)$

= 0.85 x 35x 1000 x 378 x (1-√(1-(2.35 x 210.5 x 10^6)/(0.9 x 35 x 1000 x 378^2)))/420

= 1513.5 mm^2

required spacing = 207.5 mm

Spacing provided = 150 mm

Provide 20mm diameter bars at 150mm c/c.

Area of steel provided, A_{st}= 3272.9 mm^2

P_t (provided) = 0.57 %

Flexural reinforcement Exposed Side **(At exposed side as Vertical main reinforcement)**

Design for 650mm thickness

Astmin(required) = $0.00125 b_w d$ Table 11.6.1 SBC 304

= 719.4 mm^2

Area of steel required, A_{st} = 719.4 mm^2

Provide diameter of bars = 16

Spacing required = 279.4 mm

Spacing provided = 150 mm

Provide 16mm diameter bars at 150mm c/c.

Area of steel provided, A_{st}= 1340.6 mm^2

P_t (provided)= 0.23 %

Design for 450mm thickness

Astmin(required) = $0.00125 b_w d$ Table 11.6.1 SBC 304

= 473 mm^2

Area of steel required, A_{st} = 473 mm^2

Provide diameter of bars = 12

Spacing required = 239 mm

Spacing provided = 150 mm

Provide 12mm diameter bars at 150mm c/c.

Area of steel provided, A_{st}= 754.1 mm^2

P_t (provided)= 0.20 %

Distribution reinforcement: **(At the soil face side and the exposure side as Horizontal distribution reinforcement)**

For 650mm Wall

P_t (required)= 0.00075 Table 11.6.1 SBC 304

Area of steel required, A_{st} = 431.625 mm^2

Provide diameter of bars = 12

Spacing required = 262.1 mm

Spacing provided = 150 mm

Provide 12mm diameter bars at 150mm c/c.

Area of steel provided, A_{st} = 754.1 mm^2

P_t (provided) = 0.13 %

For 450mm Wall

P_t (required)= 0.00075 Table 11.6.1 SBC 304

Area of steel required, A_{st} = 283.5 mm^2

Provide diameter of bars = 12

Spacing required = 399.0 mm

Spacing provided = 150 mm

Provide 12mm diameter bars at 150mm c/c.

Area of steel provided, A_{st} = 754.1 mm^2

P_t (provided) = 0.20 %

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Check for shear reinforcement:

For 650mm Wall

Allowable shear strength = 418.70 KN

Least i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ *Table 22.5.5.1 SBC 304*

ϕVc = 418.70 KN

Actual shear force = 283.10 KN **Hence Safe**

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$ *Table 22.5.5.1 SBC 304*

ϕVc = 450.29 KN

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$

= 740.52 KN

For 450mm Wall

Allowable shear strength = 274.98 KN

Least i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ *Table 22.5.5.1 SBC 304*

ϕVc = 274.98 KN

Actual shear force = 134.50 KN **Hence Safe**

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$ *Table 22.5.5.1 SBC 304*

ϕVc = 295.76 KN

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$

= 486.39 KN

DESIGN OF BASE SLAB (HEEL & TOE):

Centre of gravity of the vertical load from 'D' = (875.38 / 540.66) = 1.62 m

Centre of gravity of the base slab from 'D' = 2 m

Eccentricity, e = 0.380 m

Moment due to eccentricity, P x e = 205.40 KNm

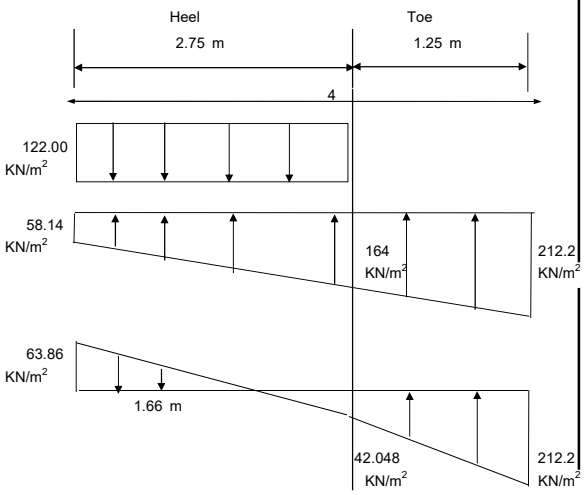
Vertical load, P = 540.66 KN

Base pressure= (P/A) ± (M/Z)

$\sigma_{(max)}$ = (P/A) + (M/Z) = 212.19 KN/m²

$\sigma_{(min)}$ = (P/A) - (M/Z) = 58.14 KN/m²

e = (M/P) = 0.380 < 0.67 **Hence ok**



Steel Reinforcement Calculation of Heel

Weight of soil on Heel = 122 KN/m²

$\sigma_{(min)}$ = 58.14 KN/m²

Moment at Stem, M = 116.34 KNm

Factored Moment at Stem face, M = 197.78 KNm Factor = 1.7

Shear Force at Stem face, SF = 52.95 KN

Factored Shear Force at Stem, SF = 90.01 KNm Factor = 1.7

Slab thickness, D = 600 mm

Diameter of bars = 16 mm

Clear cover = 75 mm

Effective cover, d' = 83 mm

Effective depth, d = 517 mm

Width, b = 1000 mm

Grade of steel, f_{ck} = 35 N/mm²

Grade of concrete, f_y = 420 N/mm²

assuptions in clause 22.2 of SBC 304

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

$\frac{c}{d}$ max = $\frac{0.003}{0.003 + 0.005}$ = 0.38

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 1000 x 517 x 517 (1 - 0.8/2 x 0.375)

= 1824.9 KNm

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

A_{st1} = $\frac{0.85\, f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

Min Ast = 465.30 mm²

= 0.85 x 35x 1000 x 517 x (1-√(1-(2.35 x 197.8 x 10^6)/(0.9 x 35 x 1000 x 517^2)))/420

= 1025.1 mm²

required spacing = 196.0 mm

Spacing provided = 150 mm

Provide 16mm diameter bars at 150mm c/c.

(Top transverse reinforcement)

Area of steel provided, A_{st} = 1340.6 mm²

P_t (provided)= 0.26 %

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Steel Reinforcement Calculation of Toe

Moment at Stem face, M = 79.45 KNm
Factored Moment at Stem face, M = 135.06 KNm Factor = 1.7
Shear Force at Stem face, SF = 127.12 KN
Factored Shear Force at Stem, SF = 216.10 KNm Factor = 1.7
Slab thickness, D = 600 mm
Diameter of bars = 16 mm
Clear cover = 75 mm
Effective cover, d' = 83 mm
Effective depth, d = 517 mm
Width, b = 1000 mm
Grade of steel, f_{ck} = 35 N/mm^2
Grade of concrete, f_y = 420 N/mm^2

assuptions in clause 22.2 of SBC 304

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.38

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 1000 x 517 x 517 (1 - 0.8/2 x 0.375)
= 1824.9 KNm

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Min Ast = 465 mm^2

A_{st1} = $\frac{0.85\, f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\, M_u}{\emptyset\, f_c'bd^2}}\right)$

= 0.85 x 35x 1000 x 517 x (1-√(1-(2.35 x 135.1 x 10^6)/(0.9 x 35 x 1000 x 517^2)))/420 0.43 0.27
= 696.9 mm^2 1.57
required spacing = 288.4 mm
Spacing provided = 150 mm

Provide 16mm diameter bars at 150mm c/c. **(Bottom transverse reinforcement)**

Area of steel provided, A_{st} = 1340.6 mm^2
P_t (provided)= 0.26 %

Distribution reinforcement:

P_t (required) = 0.09 % For each face Top and Bottom
Area of steel required, A_{st} = 465.3 mm^2
Provide diameter of bars = 12
Spacing required = 243 mm
Spacing provided = 150 mm

Provide 12mm diameter bars at 150mm c/c. **(Top and Bottom Longitudinal distribution steel)**

Area of steel provided, A_{st} = 754.1 mm^2
P_t (provided) = 0.15 %

Check for shear :

For heel

Allowable shear strength = 371.06 KN

Least i) V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ *Table 22.5.5.1 SBC 304*
φVc = 371.06 KN

Actual shear force = 90.01 KN **Hence Safe**

ii) V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$ *Table 22.5.5.1 SBC 304*
φVc = 384.13 KN

iii) V_c = $0.29\lambda\sqrt{f_c'}b_w d$
φVc = 665.25 KN

For Toe

Allowable shear strength = 381.17 KN

Least i) V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ *Table 22.5.5.1 SBC 304*
φVc = 381.17 KN

Actual shear force = 216.10 KN **Hence Safe**

ii) V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$ *Table 22.5.5.1 SBC 304*
φVc = 384.13 KN

iii) V_c = $0.29\lambda\sqrt{f_c'}b_w d$
φVc = 665.25 KN

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01		
Date	03-11-2020	12-12-2020	03-11-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	RPK	RPK	RPK		
Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI		

Outotec / L&T Revision No.:	0	1	2		
Date	03-11-2020	12-12-2020	03-11-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	RPK	RPK	RPK		
Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI		

8. ANALYSIS & DESIGN OF RAMP PART 3

Wall is subjected to the following pressure.

1) Saturated soil pressure

2) Surcharge pressure

3) Water pressure Considered upto FGL as weep holes are provided in the wall water pressure not considered above Ground level

1) Saturated soil pressure

Level of Filling Top		914.500 m	
Ground level		912.250 m	
Bottom of footing		909.400 m	
Unit weight of soil, γ =		18.0 KN/m^3	
Considering ground water table is at +EL. =		0.00 m	
Unit weight of water, γ_{water} =		10.0 KN/m^3	
Submenrger weight of soil, γ_{sub} =		8.0 KN/m^3	
Therefore,	γ_{sat}=	20.0 KN/m^3	
Unit weight of concrete, $\gamma_{concrete}$ =		24 KN/m^3	
Co-efficient of active earth pressure, k_a =		0.3	
Co-efficient of passive earth pressure, k_p =		3.33	
Soil Pressure at base heel side = $k_a.\gamma_{sat}.h$ =		(0.3 x 20 x 5.1)	= 30.6 KN/m^2
Soil Pressure at base Toe side = $k_a.\gamma_{sat}.h$ =		(3.33 x 20 x 2.35)	= 157 KN/m^2

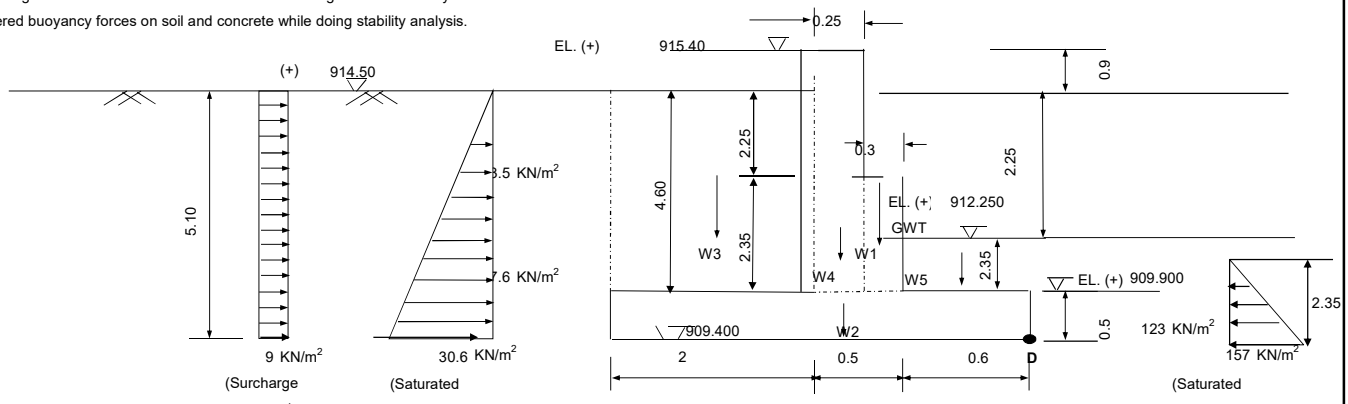
2) Surcharge Pressure:

Surcharge load on ground, q =	30.0 KN/m^2	
Surcharge Pressure , $k_a.q$ =	(0.3 x 30)	= 9 KN/m^2

3) Water Pressure:

For water pressure being same on both side of wall effect has been reglected for stability.

But we have considered buoyancy forces on soil and concrete while doing stability analysis.



**Buoyancy effect considered

LOAD MARKED	VERTICAL LOAD (KN)**	CENTRE OF GRAVITY DISTANCE OF LOAD FROM 'D'	RESTORING MOMENT (KN-m)
W1	8.225	0.725	6.00
W2	21.70	1.55	33.60
W3	184.00	2.1	386.40
W4	21.72	0.98	21.20
W5	28.200	0.30	8.50
W6	60.00	2.10	126.00
Total	**323.85**		**581.70**
LOAD TYPE	**HORIZONTAL LOAD (KN)**	**CENTRE OF GRAVITY DISTANCE OF LOAD FROM 'D'**	**RESTORING MOMENT (KN-m)**
Passive EP toe	183.899	0.78	144.05
Total Restoring Moment			**725.75**

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

LOAD TYPE	HORIZONTAL LOAD (KN)	CENTRE OF GRAVITY DISTANCE OF LOAD FROM 'D'	OVERTURNING MOMENT (KN-m)
1) S.P	45.900	2.55	117.0
2) S.S.P	78.030	1.70	132.7
Total	**123.93**		**249.70**

FACTOR OF SAFETY AGAINST OVERTURNING:

Factor of safety against overturning = (725.75 / 249.7) = 2.907 > 1.50 **Hence safe**

FACTOR OF SAFETY AGAINST SLIDING:

Sliding force = 123.93 KN

Co-efficient of friction between soil and concrete, μ = 0.5

Resisting force Friction= 161.93 N Restoring Force Passive pressure = 183.90 KN

Total Resisting Force = 345.83 N

Factor of safety against sliding = 2.79 > 1.50 **Hence safe**

CHECK FOR ECCENTRICITY:

Total bottom width, b = = 3.1 m

Resultant Moment = = 476.05 KNm

z = (476.05 / 323.85) = 1.47 m

b/6 = (3.1 / 6) = 0.52 m

Eccentricity, e= z -(b/2) = (3.1 / 2) - 1.47 = 0.08 m < 0.52 m **Hence safe**

DESIGN OF STEM WALL:

Flexural reinforcement Fill Side

Wall thickness, D	=	**500** mm	Wall thickness, D	=	**250** mm
Diameter of bars	=	20 mm	Diameter of bars	=	20 mm
Clear cover	=	50 mm	Clear cover	=	50 mm
Effective cover, d'	=	50 + 12 + (20 / 2) = 72 mm	Effective cover, d'	=	50 + 12 + (20 / 2) = 72 mm
Effective depth, d	=	(500 - 72) = 428 mm	Effective depth, d	=	(250 - 72) = 178 mm
Width, b	=	1000 mm	Width, b	=	1000 mm
Grade of steel, f_{ck}	=	35 N/mm^2	Grade of steel, f_{ck}	=	35 N/mm^2
Grade of concrete, f_y	=	420 N/mm^2	Grade of concrete, f_y	=	420 N/mm^2
Maximum bending moment, M	=	**192.6** KN-m	Maximum bending moment, M	=	**34.2** KN-m
Maximum shear force	=	**104.9** KN	Maximum shear force	=	**35.4** KN
Load factor	=	1.7	Load factor	=	1.7
Factored Maximum bending moment, M	=	**327** KNm	Factored Maximum bending moment, M	=	**58.1** KNm
Factored Maximum shear force	=	**178** KN	Factored Maximum shear force	=	**60.2** KN

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Design for 500mm thickness **At Fill side as Vertical Main Reinforcement** *assuptions in clause 22.2 of SBC 304*

Mu lim

$= \emptyset\, 0.85\, f_c'\, \beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

$\frac{c}{d}_{max} = \frac{0.003}{0.003 + 0.005} = 0.38$

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 1000 x 428 x 428 (1 - 0.8/2 x 0.375)

= 1250.7 KNm b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

$A_{st1} = \frac{0.85\, f_c'\, bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c'\, bd^2}}\right)$

= 0.85 x 35x 1000 x 428 x (1-√(1-(2.35 x 327.4 x 10^6)/(0.9 x 35 x 1000 x 428^2)))/420

= 2093.1 mm^2

required spacing = 150.0 mm

Spacing provided = 150 mm

Provide 20mm diameter bars at 150mm c/c.

Design for 250mm thickness **At Fill side as Vertical Main Reinforcement** *assuptions in clause 22.2 of SBC 304*

Mu lim

$= \emptyset\, 0.85\, f_c'\, \beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

$\frac{c}{d}_{max} = \frac{0.003}{0.003 + 0.005} = 0.38$

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 1000 x 178 x 178 (1 - 0.8/2 x 0.375)

= 216.3 KNm b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

$A_{st1} = \frac{0.85\, f_c'\, bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c'\, bd^2}}\right)$

= 0.85 x 35x 1000 x 178 x (1-√(1-(2.35 x 58.1 x 10^6)/(0.9 x 35 x 1000 x 178^2)))/420

= 894.0 mm^2

required spacing = 351.2 mm

Spacing provided = 150 mm

Provide 20mm diameter bars at 150mm c/c.

Area of steel provided, A_{st}= 2094.7 mm^2

P_t (provided) = 0.49 %

Flexural reinforcement Exposed Side **(At exposed side as Vertical main reinforcement)**

Design for 500mm thickness

Astmin(required) = $0.00125 b_w d$ Table 11.6.1 SBC 304

= 535 mm^2

Area of steel required, A_{st} = 535.0 mm^2

Provide diameter of bars = 12

Spacing required = 211.3 mm

Spacing provided = 150 mm

Provide 12mm diameter bars at 150mm c/c.

Area of steel provided, A_{st}= 754.1 mm^2

P_t (provided)= 0.18 %

Design for 250mm thickness

Astmin(required) = $0.00125 b_w d$ Table 11.6.1 SBC 304

= 223 mm^2

Area of steel required, A_{st} = 223 mm^2

Provide diameter of bars = 12

Spacing required = 508 mm

Spacing provided = 150 mm

Provide 12mm diameter bars at 150mm c/c.

Area of steel provided, A_{st}= 754.1 mm^2

P_t (provided)= 0.42 %

Distribution reinforcement: **(At the soil face side and the exposure side as Horizontal distribution reinforcement)**

For 500mm Wall

P_t (required)= 0.00075 Table 11.6.1 SBC 304

Area of steel required, A_{st} = 321 mm^2

Provide diameter of bars = 12

Spacing required = 352.4 mm

Spacing provided = 150 mm

Provide 12mm diameter bars at 150mm c/c.

Area of steel provided, A_{st} = 754.1 mm^2

P_t (provided) = 0.18 %

For 250mm Wall

P_t (required)= 0.00075 Table 11.6.1 SBC 304

Area of steel required, A_{st} = 133.5 mm^2

Provide diameter of bars = 12

Spacing required = 847.3 mm

Spacing provided = 150 mm

Provide 12mm diameter bars at 150mm c/c.

Area of steel provided, A_{st} = 754.1 mm^2

P_t (provided) = 0.42 %

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Check for shear reinforcement:

For 500mm Wall

Allowable shear strength = 310.08 KN

Least i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ *Table 22.5.5.1 SBC 304*

φVc = 310.08 KN

Actual shear force = 178.30 KN **Hence Safe**

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$ *Table 22.5.5.1 SBC 304*

φVc = 330.56 KN

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$

= 550.73 KN

For 250mm Wall

Allowable shear strength = 128.42 KN

Least i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ *Table 22.5.5.1 SBC 304*

φVc = 128.42 KN

Actual shear force = 60.24 KN **Hence Safe**

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$ *Table 22.5.5.1 SBC 304*

φVc = 137.47 KN

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$

= 229.04 KN

DESIGN OF BASE SLAB (HEEL & TOE):

Centre of gravity of the vertical load from 'D' = (476.05 / 323.85) = 1.47 m

Centre of gravity of the base slab from 'D' = 1.55 m

Eccentricity, e = 0.080 m

Moment due to eccentricity, P x e = 25.90 KNm

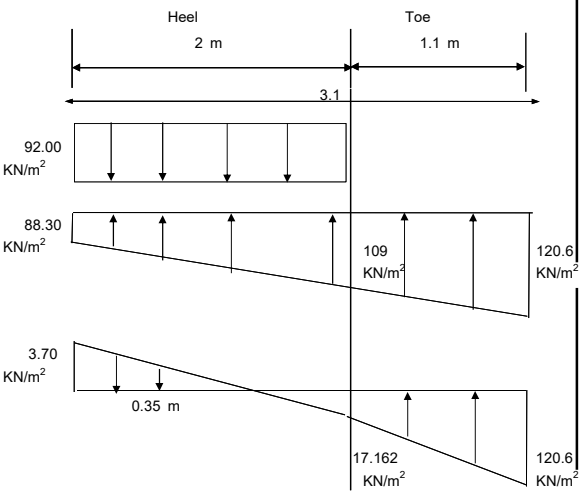
Vertical load, P = 323.85 KN

Base pressure= (P/A) ± (M/Z)

$\sigma_{(max)}$ = (P/A) + (M/Z) = 120.64 KN/m²

$\sigma_{(min)}$ = (P/A) - (M/Z) = 88.30 KN/m²

e = (M/P) = 0.080 < 0.52 **Hence ok**



Steel Reinforcement Calculation of Heel

Weight of soil on Heel = 92 KN/m²

$\sigma_{(min)}$ = 88.30 KN/m²

Moment at Stem, M = 1.24 KNm

Factored Moment at stem face, M = 2.10 KNm Factor = 1.7

Shear Force at Stem face, SF = 0.66 KN

Factored Shear Force at Stem, SF = 1.12 KNm Factor = 1.7

Slab thickness, D = 500 mm

Diameter of bars = 12 mm

Clear cover = 75 mm

Effective cover, d' = 81 mm

Effective depth, d = 419 mm

Width, b = 1000 mm

Grade of steel, f_{ck} = 35 N/mm²

Grade of concrete, f_y = 420 N/mm²

assuptions in clause 22.2 of SBC 304

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1\left(\frac{c}{d}\right)_{max} bd^2\left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

$\frac{c}{d}$ max = $\frac{0.003}{0.003 + 0.005}$ = 0.38

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 1000 x 419 x 419 (1 - 0.8/2 x 0.375)

= 1198.7 KNm

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

A_{st1} = $\frac{0.85\, f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

Min Ast = 377.10 mm²

= 0.85 x 35x 1000 x 419 x (1-√(1-(2.35 x 2.2 x 10^6)/(0.9 x 35 x 1000 x 419^2)))/420

= 13.3 mm²

required spacing = 299.8 mm

Spacing provided = 150 mm

Provide 12mm diameter bars at 150mm c/c. **(Top transverse reinforcement)**

Area of steel provided, A_{st} = 754.1 mm²

P_t (provided)= 0.18 %

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Steel Reinforcement Calculation of Toe

Moment at Stem face, M = 37.90 KNm
Factored Moment at Stem face, M = 64.42 KNm Factor = 1.7
Shear Force at Stem face, SF = 68.90 KN
Factored Shear Force at Stem, SF = 117.13 KNm Factor = 1.7
Slab thickness, D = 500 mm
Diameter of bars = 12 mm
Clear cover = 75 mm
Effective cover, d' = 81 mm
Effective depth, d = 419 mm
Width, b = 1000 mm
Grade of steel, f_{ck} = 35 N/mm^2
Grade of concrete, f_y = 420 N/mm^2

assuptions in clause 22.2 of SBC 304

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2\left(1-\frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

$\frac{c}{d}$ max = $\frac{0.003}{0.003+0.005}$ = 0.38

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 1000 x 419 x 419 (1 - 0.8/2 x 0.375)
= 1198.7 KNm

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

A_{st1} = $\frac{0.85\, f_c'bd}{f_y}\left(1-\sqrt{1-\frac{2.35\, M_u}{\emptyset\, f_c'bd^2}}\right)$

Min Ast = 377 mm^2

= 0.85 x 35x 1000 x 419 x (1-√(1-(2.35 x 64.5 x 10^6)/(0.9 x 35 x 1000 x 419^2)))/420
= 409.1 mm^2
required spacing = 276.3 mm
Spacing provided = 150 mm

Provide 12mm diameter bars at 150mm c/c. **(Bottom transverse reinforcement)**

Area of steel provided, A_{st} = 754.1 mm^2
P_t (provided)= 0.18 %

Distribution reinforcement:

P_t (required) = 0.09 % For each face Top and Bottom Table 24.4.3.2 SBC 304
Area of steel required, A_{st} = 377.1 mm^2
Provide diameter of bars = 12
Spacing required = 300 mm
Spacing provided = 150 mm

Provide 12mm diameter bars at 150mm c/c. **(Both ways as top reinforcement)**

Area of steel provided, A_{st} = 754.1 mm^2
P_t (provided) = 0.18 %

Check for shear :

For heel

Allowable shear strength = 299.60 KN
Actual shear force = 1.12 KN **Hence Safe**

Least i) $V_c = \left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ *Table 22.5.5.1 SBC 304*
$\emptyset Vc$ = 299.60 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$ *Table 22.5.5.1 SBC 304*
$\emptyset Vc$ = 307.08 KN

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$
$\emptyset Vc$ = 539.15 KN

For Toe

Allowable shear strength = 304.79 KN
Actual shear force = 117.13 KN **Hence Safe**

Least i) $V_c = \left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ *Table 22.5.5.1 SBC 304*
$\emptyset Vc$ = 304.79 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$ *Table 22.5.5.1 SBC 304*
$\emptyset Vc$ = 307.08 KN

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$
$\emptyset Vc$ = 539.15 KN

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

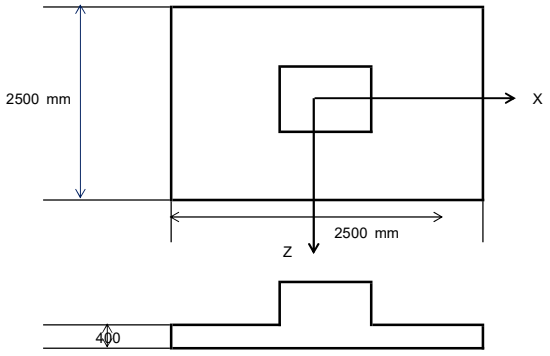
Ma'aden Revision No:	A01	C01	D01	Outotec / L&T	0	1	2
Date	03-11-2020	12-12-2020	03-11-2023	Date	03-11-2020	12-12-2020	03-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK	Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

9. ANALYSIS & DESIGN OF ERH SUPPORT FOOTING

Design of foundation F1 at Grid 3210/A'-3210/2

Finished Ground level	F.G.L	=	0.000	m	Weight of soil	=	259.75	KN
Bottom of foundation	B.O.F	=	-2.850	m	Weight of foundation	=	60.00	KN
Top of footing (staad support)	T.O.F	=	-2.450	m	Uplift	=	0.00	KN
Ground Water Level	G.W.L	=	-10.000	m	Surcharge	=	0	KN
Surcharge	Ws	=	0.000	KN/m^2	Pressure on top for tension	=	51.2	KN/m^2
Length of footing	Lz	=	2.50	m				
Width of footing	Lx	=	2.50	m				
Thickness of footing	D	=	0.40	m				
Depth of pedestal	Bx	=	0.60	m				
Width of pedestal	Bz	=	0.60	m				
Height of Load point from bottom of footing	H	=	0.200	m				
Section Modulus of Pedestal	Zxx	=	2.60	m^3				
Section Modulus of Pedestal	Zzz	=	2.60	m^3				
Allowable Net Bearing Capacity		=	500	KN/m^2				
Gross Bearing Capacity		=	551.3	KN/m^2				
Grade of concrete		=	35	N/mm^2				
Grade of steel		=	420	N/mm^2				
Clear cover		=	50	mm				
Top Reinfocement		=	12	mm				
Bottom Reinforcement		=	25	mm				
Co-efficient of friction between soil & concrete		=	0.3					
Unit wt of soil		=	18	KN/m^3				
Unit wt of concrete		=	24	KN/m^3				

*inputs



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$

β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301	2.9	142.585	0.0	0.00	0.0	-10.9	462.3	78.15	69.80	1	0.14
302	3.7	306.085	0.0	0.00	0	-13.9	625.8	105.46	94.81	1	0.19
303	8.2	184.721	0.5	1.92	0	-30.6	504.5	93.21	68.22	1	0.17
304	9.0	348.221	0.5	1.92	0.0	-33.6	668.0	120.52	93.23	1	0.22
305	2.0	186.049	-0.5	-1.92	0	-7.6	505.8	84.60	77.26	1	0.15
306	2.8	349.549	-0.5	-1.92	0	-10.6	669.3	111.91	102.27	1	0.20
307	31.6	247.343	0.0	-0.15	0.0	-118.7	567.1	136.36	45.11	1	0.25
308	-22.5	102.028	0.0	0.15	0	84.5	421.8	100.01	34.96	1	0.18
309	4.8	175.456	1.0	3.89	0	-17.8	495.2	87.57	70.89	1	0.16
310	4.3	173.915	-1.0	-3.89	0.0	-16.3	493.7	86.73	71.24	1	0.16
311	32.4	410.843	0.0	-0.15	0	-121.7	730.6	163.68	70.11	1	0.30
312	-21.7	265.528	0.0	0.15	0	81.5	585.3	125.02	62.27	1	0.23
313	5.6	338.956	1.0	3.89	0	-20.8	658.7	114.89	95.90	1	0.21
314	5.1	337.415	-1.0	-3.89	0	-19.3	657.2	114.05	96.25	1	0.21
315	37.9	182.428	-0.1	-0.21	0	-142.0	502.2	134.96	25.74	1	0.24
316	-34.4	-11.326	0.1	0.21	0	129.0	308.4	98.95	-0.25	1	0.18
317	2.0	86.578	1.4	5.18	0	-7.6	406.3	69.90	60.12	1	0.13
318	1.5	84.524	-1.4	-5.18	0	-5.5	404.3	68.78	60.58	1	0.12
319	38.3	280.528	-0.1	-0.21	0	-143.8	600.3	151.34	40.74	1	0.27
320	-33.9	86.774	0.1	0.21	0	127.2	406.5	113.95	16.14	1	0.21
321	2.5	184.678	1.4	5.18	0.0	-9.4	504.4	86.29	75.13	1	0.16
322	1.9	182.624	-1.4	-5.18	0	-7.3	502.4	85.17	75.59	1	0.15

Max P_{max} = 163.68 KN/m^2

Max P_{min} = -0.25 KN/m^2

Resultant F.O.S = 0.30

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	03-11-2020	12-12-2020	03-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Outotec / L&T	0	1	2
Date	03-11-2020	12-12-2020	03-11-2023
Revision Tex	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401	4.1	199.6	0.0	0.00	0.0	-15.23	37.79	26.09
402	5.2	428.5	0.0	0.00	0	-19.43	76.02	61.10
403	12.1	271.5	0.7	2.68	0	-45.34	61.87	24.99
404	3.5	273.3	-0.7	-2.68	0	-13.16	49.81	37.64
405	13.2	500.4	0.7	2.68	0	-49.54	100.11	60.00
406	4.6	502.2	-0.7	-2.68	0	-17.36	88.05	72.66
407	62.4	347.5	-0.1	-0.33	0	-233.94	145.56	-34.36
408	-53.2	37.5	0.1	0.33	0	199.59	82.77	-70.77
409	5.0	194.1	2.2	8.29	0	-18.83	41.48	20.65
410	4.1	190.9	-2.2	-8.29	0	-15.53	39.68	21.39
411	63.3	543.7	-0.1	-0.33	0	-237.54	178.34	-4.35
412	-52.3	233.7	0.1	0.33	0	195.99	112.78	-38.00
413	6.0	390.3	2.2	8.29	0	-22.43	74.25	50.66
414	5.1	387.1	-2.2	-8.29	0	-19.13	72.46	51.40
415	60.4	283.3	-0.1	-0.33	0	-226.56	132.46	-41.79
416	-55.2	-26.7	0.1	0.33	0	206.98	75.34	-83.88
417	3.1	130.0	2.2	8.29	0	-11.44	28.37	13.22
418	2.2	126.7	-2.2	-8.29	0	-8.14	26.58	13.96
419	61.1	430.5	-0.1	-0.33	0	-229.26	157.04	-19.28
420	-54.5	120.5	0.1	0.33	0	204.28	97.85	-59.29
421	3.8	277.1	2.2	8.29	0	-14.14	52.95	35.73
422	2.9	273.8	-2.2	-8.29	0	-10.84	51.16	36.47

Max Net factored B.P = 178.34 KN/m^2

DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 0.95 m

Moment about Z: Muz = 80.47 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 777.71 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 638.5 mm^2

Effective Depth Along X = 337.5 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

greater of or A_{stmin} = 0.0014bd

A_{stmin} = 607.5 mm^2

A_{req} = 638.5 mm^2

Required Spacing = **314.9** mm

Length of cantilever along Z = 0.95 m

Moment about X: Mux = 80.47 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 666.76 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 691.2 mm^2

Effective Depth Along X = 312.5 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

greater of or A_{stmin} = 0.0014bd

A_{stmin} = 562.5 mm^2

A_{req} = 691.2 mm^2

Required Spacing = **290.9** mm

Provide Bottom Reinforcement = **16 mm dia @ 200 c/c**

Ast prov = 1005.3 mm^2

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	03-11-2020	12-12-2020	03-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Outotec / L&T	0	1	2
Date	03-11-2020	12-12-2020	03-11-2023
Revision Tex	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Top Reinforcement

Length of cantilever along X = 0.95 m

Moment about Z: Muz = 23.09 KNm

$M_{uzlim} = \emptyset\, 0.85\, f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uzlim} = 807.95 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\, f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 178.0 mm^2

Effective Depth Along X = 344 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% more prov

greater of or

A_{stmin} = 50% of 0.0014bd bottom

Table 24.4.3.2 SBC 304

A_{stmin} = 309.6 mm^2

A_{req} = 178.0 mm^2

Required Spacing = **365.3** mm

Length of cantilever along Z = 0.95 m

Moment about X: Mux = 23.09 KNm

$M_{uxlim} = \emptyset\, 0.85\, f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uxlim} = 752.57 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\, f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 184.5 mm^2

Effective Depth Along Z = 332 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% more prov

greater of or

A_{stmin} = 50% of 0.0014bd bottom

Table 24.4.3.2 SBC 304

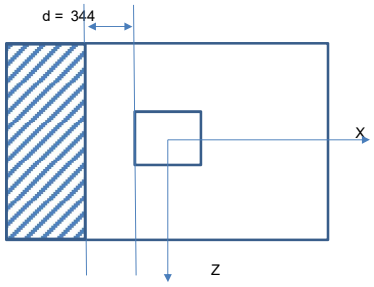
A_{stmin} = 298.8 mm^2

A_{req} = 184.5 mm^2

Required Spacing = **378.5** mm

Provide top Reinforcement = **12 mm dia @ 200 c/c** Ast prov = 565.5 mm^2

ONE WAY SHEAR CKECK



Along X Direction Cantilever L = 0.606 m

Reinforcement ratio ρ_w = 0.0029

Effective Depth Along X d = 344 mm

Shear Force at d distance V_u = 108.07 KN

Allowable shear strength φVc = 257.03 KN O.K

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ *Table 22.5.5.1 SBC 304*

λ = 1 Normal concrete

Mu = 32.75 KNm

φ = 0.75 for shear

φVc = 258.8 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$

φVc = 257.0 KN *Table 22.5.5.1 SBC 304*

iii) $V_c = 0.29\lambda\sqrt{f_c'} b_w d$

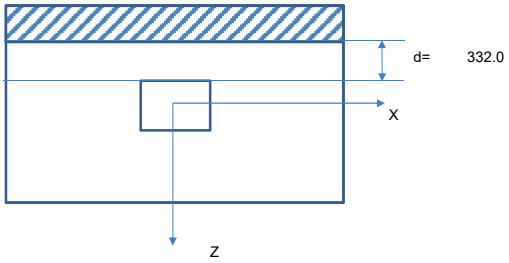
= 442.6 KN

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	03-11-2020	12-12-2020	03-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Outotec / L&T	0	1	2
Date	03-11-2020	12-12-2020	03-11-2023
Revision Tex	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

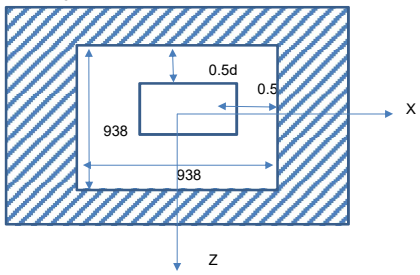
ONE WAY SHEAR CKECK



Along Z Direction	Cantilever	L	=	0.618	m	Least of	i)	V_c	= $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$
Reinforcement ratio		ρ_w	=	0.0030					*Table 22.5.5.1 SBC 304*
Effective Depth Along Z		d	=	332	mm			λ	= 1 Normal concrete
Shear Force at d distance		V_u	=	110.21	KN			Mu	= 34.06 KNm
Allowable shear strength		φVc	=	248.51	KN	O.K		φ	= 0.75 for shear
								φVc	= 249.5 KN
							ii)	V_c	= $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$
								φVc	= 248.5 KN
							iii)	V_c	= $0.29\lambda\sqrt{f_c'}b_w d$
								φVc	= 427.2 KN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area		=	6.25	m2
Punching area		=	0.88	m2
	λ	=	1	
	β	=	1	
	αs	=	40	
	b0	=	3752.0	mm

Allowable shear stress — least of — *Table 22.6.5.2 SBC 304*

v_c	= $0.33\lambda\sqrt{f_c'}$	=	1.95 N/mm²
v_c	= $0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$	=	3.02 N/mm²
v_c	= $0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$	=	2.75 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face	u	=	3.75	m		Vc	= $v_c b_0 d$
Depth at section	d	=	338	mm			= 2475.87 KN
Shear Force at section	Vu	=	957.7	KN		φ	= 0.75 for shear
Allowable shear strength	φVc	=	1856.90	KN	O.K	φV_n	= φVc
						φVc	= 1856.9 KN

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

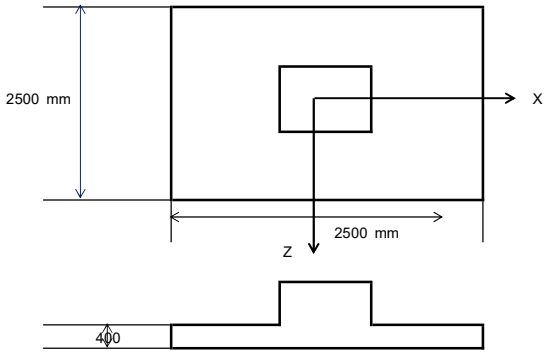
Ma'aden Revision No:	A01	C01	D01
Date	03-11-2020	12-12-2020	03-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Outotec / L&T	0	1	2
Date	03-11-2020	12-12-2020	03-11-2023
Revision Tex	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Design of foundation F1 at Grid 3210/A'-3210/3/2

Finished Ground level	F.G.L	=	0.000	m	Weight of soil	=	259.75	KN
Bottom of foundation	B.O.F	=	-2.850	m	Weight of foundation	=	60.00	KN
Top of footing (staad support)	T.O.F	=	-2.450	m	Uplift	=	0.00	KN
Ground Water Level	G.W.L	=	-10.000	m	Surcharge	=	0	KN
Surcharge	Ws	=	0.000	KN/m^2	Pressure on top for tension	=	51.2	KN/m^2
Length of footing	Lz	=	2.50	m				
Width of footing	Lx	=	2.50	m				
Thickness of footing	D	=	0.40	m				
Depth of pedestal	Bx	=	0.60	m				
Width of pedestal	Bz	=	0.60	m				
Height of Load point from bottom of footing	H	=	0.200	m				
Section Modulus of Pedestal	Zxx	=	2.60	m^3				
Section Modulus of Pedestal	Zzz	=	2.60	m^3				
Allowable Net Bearing Capacity		=	500	KN/m^2				
Gross Bearing Capacity		=	551.3	KN/m^2				
Grade of concrete		=	35	N/mm^2				
Grade of steel		=	420	N/mm^2				
Clear cover		=	50	mm				
Top Reinfocement		=	12	mm				
Bottom Reinforcement		=	25	mm				
Co-efficient of friction between soil & concrete		=	0.3					
Unit wt of soil		=	18	KN/m^3				
Unit wt of concrete		=	24	KN/m^3				

*inputs



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$

β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301	-2.0	217.685	38.7	145.13	0.0	7.5	537.4	144.60	27.38	1	0.26
302	-4.0	485.285	42.1	157.88	0	15.0	805.0	195.19	62.42	1	0.35
303	-3.1	274.426	51.6	193.56	0	11.5	594.2	173.82	16.31	1	0.32
304	-5.1	542.026	55.0	206.31	0.0	19.0	861.8	224.41	51.35	1	0.41
305	-7.5	272.944	45.0	168.69	0	28.2	592.7	170.45	19.22	1	0.31
306	-9.5	540.544	48.4	181.44	0	35.7	860.3	221.04	54.26	1	0.40
307	8.3	261.991	45.8	171.57	0.0	-31.2	581.7	170.94	15.22	1	0.31
308	-17.3	257.38	46.0	172.68	0	64.8	577.1	183.52	1.17	1	0.33
309	-4.6	298.025	64.1	240.48	0	17.2	617.8	197.80	-0.11	1	0.36
310	-4.4	221.345	27.7	103.77	0.0	16.3	541.1	132.70	40.45	1	0.24
311	6.3	529.591	49.2	184.32	0	-23.7	849.3	215.77	56.02	1	0.39
312	-19.3	524.98	49.4	185.43	0	72.3	844.7	234.11	36.21	1	0.42
313	-6.6	565.625	67.5	253.23	0	24.7	885.4	248.39	34.93	1	0.45
314	-6.4	488.945	31.1	116.52	0	23.8	808.7	183.29	75.49	1	0.33
315	15.9	133.685	23.0	86.33	0	-59.5	453.4	128.53	16.57	1	0.23
316	-18.3	127.537	23.4	87.82	0	68.5	447.3	131.58	11.55	1	0.24
317	-1.4	181.731	47.5	178.21	0	5.1	501.5	150.62	9.85	1	0.27
318	-1.0	79.491	-1.1	-4.06	0	3.9	399.2	66.94	60.82	1	0.12
319	14.7	294.245	25.1	93.98	0	-55.0	614.0	155.43	41.05	1	0.28
320	-19.5	288.097	25.5	95.47	0	73.0	607.8	161.93	32.58	1	0.29
321	-2.6	342.291	49.6	185.86	0.0	9.6	662.0	180.98	30.87	1	0.33
322	-2.2	240.051	1.0	3.59	0	8.4	559.8	94.18	84.96	1	0.17

Max P_{max} = 248.39 KN/m^2

Max P_{min} = -0.11 KN/m^2

Resultant F.O.S = 0.45

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	03-11-2020	12-12-2020	03-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Outotec / L&T	0	1	2
Date	03-11-2020	12-12-2020	03-11-2023
Revision Tex	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401	-2.8	304.8	54.2	203.18	0.0	10.50	130.81	-33.29
402	-5.6	679.4	58.9	221.03	0	21.00	201.64	15.77
403	-5.3	401.0	75.1	281.78	0	19.86	179.99	-51.67
404	-11.5	398.9	65.9	246.97	0	43.22	175.26	-47.61
405	-8.1	775.6	79.9	299.63	0	30.36	250.82	-2.61
406	-14.3	773.6	70.6	264.82	0	53.72	246.09	1.45
407	23.2	294.1	50.9	190.96	0	-87.15	153.86	-59.73
408	-31.3	284.3	51.6	193.34	0	117.52	164.86	-73.88
409	-4.3	371.0	90.1	337.97	0	16.12	195.33	-76.61
410	-3.8	207.4	12.4	46.33	0	14.25	56.45	9.93
411	20.8	615.3	55.0	206.26	0	-78.15	207.65	-10.77
412	-33.7	605.4	55.6	208.64	0	126.52	225.57	-31.84
413	-6.7	692.1	94.2	353.27	0	25.12	256.04	-34.56
414	-6.2	528.6	16.4	61.63	0	23.25	117.16	51.97
415	25.5	200.8	34.5	129.42	0	-95.59	118.54	-54.27
416	-29.1	191.0	35.1	131.81	0	109.09	123.06	-61.94
417	-2.1	277.7	73.7	276.43	0	7.69	153.53	-64.67
418	-1.6	114.1	-4.1	-15.21	0	5.81	26.33	10.19
419	23.7	441.7	37.6	140.89	0	-88.84	158.88	-17.55
420	-30.9	431.8	38.2	143.28	0	115.84	168.60	-30.41
421	-3.9	518.5	76.8	287.91	0	14.44	199.07	-33.13
422	-3.4	355.0	-1.0	-3.73	0	12.56	63.05	50.54

Max Net factored B.P = 256.04 KN/m^2

DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 0.95 m

Moment about Z: Muz = 115.54 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 777.71 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 922.3 mm^2

Effective Depth Along X = 337.5 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

greater of

or $A_{stmin} = 0.0014bd$

A_{stmin} = 607.5 mm^2

A_{req} = 922.3 mm^2

Required Spacing = **218.0** mm

Length of cantilever along Z = 0.95 m

Moment about X: Mux = 115.54 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 666.76 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 999.5 mm^2

Effective Depth Along X = 312.5 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

greater of

or $A_{stmin} = 0.0014bd$

A_{stmin} = 562.5 mm^2

A_{req} = 999.5 mm^2

Required Spacing = **201.2** mm

Provide Bottom Reinforcement = **16 mm dia @ 200 c/c**

Ast prov = 1005.3 mm^2

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	03-11-2020	12-12-2020	03-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Outotec / L&T	0	1	2
Date	03-11-2020	12-12-2020	03-11-2023
Revision Tex	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Top Reinforcement

Length of cantilever along X = 0.95 m

Moment about Z: Muz = 23.09 KNm

$M_{uzlim} = \emptyset\, 0.85\, f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uzlim} = 807.95 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\, f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 178.0 mm^2

Effective Depth Along X = 344 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% more prov

greater of or

A_{stmin} = 50% of 0.0014bd bottom

Table 24.4.3.2 SBC 304

A_{stmin} = 309.6 mm^2

A_{req} = 178.0 mm^2

Required Spacing = **365.3** mm

Length of cantilever along Z = 0.95 m

Moment about X: Mux = 23.09 KNm

$M_{uxlim} = \emptyset\, 0.85\, f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uxlim} = 752.57 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\, f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 184.5 mm^2

Effective Depth Along Z = 332 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% more prov

greater of or

A_{stmin} = 50% of 0.0014bd bottom

Table 24.4.3.2 SBC 304

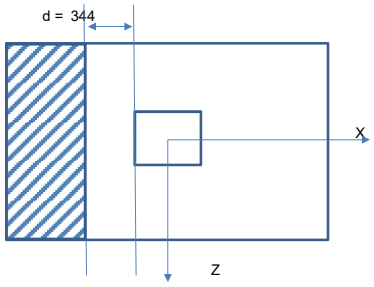
A_{stmin} = 298.8 mm^2

A_{req} = 184.5 mm^2

Required Spacing = **378.5** mm

Provide top Reinforcement = **12 mm dia @ 200 c/c** Ast prov = 565.5 mm^2

ONE WAY SHEAR CKECK



Along X Direction Cantilever L = 0.606 m

Reinforcement ratio ρ_w = 0.0029

Effective Depth Along X d = 344 mm

Shear Force at d distance V_u = 155.16 KN

Allowable shear strength φVc = 257.03 KN O.K

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ *Table 22.5.5.1 SBC 304*

λ = 1 Normal concrete

Mu = 47.01 KNm

φ = 0.75 for shear

φVc = 258.8 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$

φVc = 257.0 KN *Table 22.5.5.1 SBC 304*

iii) $V_c = 0.29\lambda\sqrt{f_c'} b_w d$

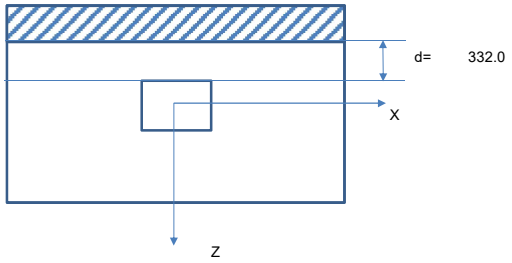
= 442.6 KN

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	03-11-2020	12-12-2020	03-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Outotec / L&T	0	1	2
Date	03-11-2020	12-12-2020	03-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

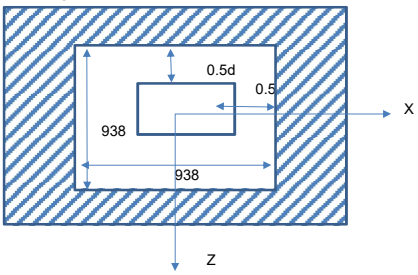
ONE WAY SHEAR CKECK



Along Z Direction	Cantilever	L	=	0.618	m	Least of	i)	V_c	= $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$
Reinforcement ratio		ρ_w	=	0.0030					*Table 22.5.5.1 SBC 304*
Effective Depth Along Z		d	=	332	mm			λ	= 1 Normal concrete
Shear Force at d distance		V_u	=	158.24	KN			Mu	= 48.89 KNm
Allowable shear strength		φVc	=	248.51	KN	O.K		φ	= 0.75 for shear
								φVc	= 249.5 KN
							ii)	V_c	= $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$
								φVc	= 248.5 KN
							iii)	V_c	= $0.29\lambda\sqrt{f_c'}b_w d$
								φVc	= 427.2 KN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area	=	6.25	m2
Punching area	=	0.88	m2
λ	=	1	
β	=	1	
αs	=	40	
b0	=	3752.0	mm

Allowable shear stress — least of — *Table 22.6.5.2 SBC 304*

v_c	= $0.33\lambda\sqrt{f_c'}$	=	1.95	N/mm²
v_c	= $0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$	=	3.02	N/mm²
v_c	= $0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$	=	2.75	N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face	u	=	3.75	m		Vc	= $v_c b_0 d$
Depth at section	d	=	338	mm			= 2475.87 KN
Shear Force at section	Vu	=	1375.0	KN		φ	= 0.75 for shear
Allowable shear strength	φVc	=	1856.90	KN	O.K	φV_n	= φVc
						φVc	= 1856.9 KN

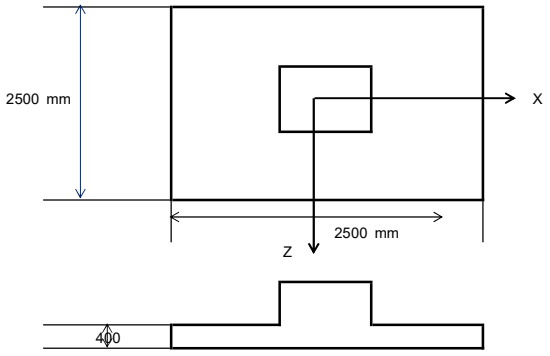
Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T	0	1	2
Date	03-11-2020	12-12-2020	03-11-2023	Date	03-11-2020	12-12-2020	03-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK	Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

Design of foundation F1 at Grid 3210/A'-3210/3

Finished Ground level	F.G.L	=	0.000	m	Weight of soil	=	259.75 KN
Bottom of foundation	B.O.F	=	-2.850	m	Weight of foundation	=	60.00 KN
Top of footing (staad support)	T.O.F	=	-2.450	m	Uplift	=	0.00 KN
Ground Water Level	G.W.L	=	-10.000	m	Surcharge	=	0 KN
Surcharge	Ws	=	0.000	KN/m^2	Pressure on top for tension	=	51.2 KN/m^2
Length of footing	Lz	=	2.50	m			
Width of footing	Lx	=	2.50	m			
Thickness of footing	D	=	0.40	m			
Depth of pedestal	Bx	=	0.60	m			
Width of pedestal	Bz	=	0.60	m			
Height of Load point from bottom of footing	H	=	0.200	m			
Section Modulus of Pedestal	Zxx	=	2.60	m^3			
Section Modulus of Pedestal	Zzz	=	2.60	m^3			
Allowable Net Bearing Capacity		=	500	KN/m^2			
Gross Bearing Capacity		=	551.3	KN/m^2			
Grade of concrete		=	35	N/mm^2			
Grade of steel		=	420	N/mm^2			
Clear cover		=	50	mm			
Top Reinfocement		=	12	mm			
Bottom Reinforcement		=	25	mm			
Co-efficient of friction between soil & concrete		=	0.3				
Unit wt of soil		=	18	KN/m^3			
Unit wt of concrete		=	24	KN/m^3			

*inputs



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$

β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301	-0.2	130.685	-1.7	-6.38	0.0	0.8	450.4	74.81	69.33	1	0.14
302	3.4	330.985	-13.8	-51.75	0	-12.8	650.7	128.89	79.35	1	0.23
303	0.4	162.308	-12.2	-45.57	0	-1.3	482.1	95.14	59.12	1	0.17
304	4.0	362.608	-24.3	-90.95	0.0	-14.8	682.4	149.79	68.56	1	0.27
305	-6.6	162.462	-3.4	-12.93	0	24.6	482.2	91.55	62.75	1	0.17
306	-3.0	362.762	-15.5	-58.30	0	11.1	682.5	135.84	82.56	1	0.25
307	26.1	225.685	-5.8	-21.79	0.0	-97.8	545.4	133.18	41.36	1	0.24
308	-30.8	83.236	-6.7	-25.28	0	115.6	403.0	118.56	10.39	1	0.22
309	-2.5	115.351	12.2	45.87	0	9.4	435.1	90.86	48.38	1	0.16
310	-2.2	193.57	-24.8	-92.93	0.0	8.4	513.3	121.03	43.23	1	0.22
311	29.7	425.985	-17.9	-67.16	0	-111.3	745.7	187.83	50.80	1	0.34
312	-27.2	283.536	-18.8	-70.65	0	102.1	603.3	162.85	30.20	1	0.30
313	1.1	315.651	0.1	0.50	0	-4.1	635.4	103.41	99.92	1	0.19
314	1.4	393.87	-36.9	-138.31	0	-5.1	713.6	169.26	59.10	1	0.31
315	37.8	173.377	-0.4	-1.50	0	-141.8	493.1	133.92	23.88	1	0.24
316	-38.0	-16.555	-1.6	-6.15	0	142.7	303.2	105.66	-8.64	1	0.19
317	-0.3	26.265	23.7	88.71	0	1.2	346.0	89.87	20.85	1	0.16
318	0.1	130.557	-25.7	-96.36	0	-0.3	450.3	109.15	34.94	1	0.20
319	40.0	293.557	-7.7	-28.73	0	-149.9	613.3	166.71	29.55	1	0.30
320	-35.9	103.625	-8.9	-33.38	0	134.6	423.4	132.23	3.25	1	0.24
321	1.8	146.445	16.4	61.49	0.0	-6.9	466.2	100.86	48.32	1	0.18
322	2.2	250.737	-33.0	-123.59	0	-8.4	570.5	141.95	40.61	1	0.26

Max P_{max} = 187.83 KN/m^2

Max P_{min} = -8.64 KN/m^2

Resultant F.O.S = 0.34

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	03-11-2020	12-12-2020	03-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Outotec / L&T	0	1	2
Date	03-11-2020	12-12-2020	03-11-2023
Revision Tex	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401	-0.3	183.0	-2.4	-8.93	0.0	1.05	33.10	25.44
402	4.8	463.4	-19.3	-72.45	0	-17.85	108.82	39.47
403	-0.4	236.7	-18.8	-70.67	0	1.41	65.56	10.20
404	-10.0	237.0	-6.7	-24.96	0	37.66	61.96	13.87
405	4.7	517.2	-35.8	-134.19	0	-17.49	140.99	24.50
406	-5.0	517.4	-23.6	-88.48	0	18.76	123.96	41.60
407	59.0	324.6	-4.1	-15.37	0	-221.22	142.79	-38.91
408	-62.4	20.7	-6.1	-22.81	0	233.89	101.89	-95.26
409	-2.0	89.2	34.4	128.97	0	7.48	66.68	-38.12
410	-1.4	256.1	-44.6	-167.14	0	5.19	107.15	-25.20
411	63.3	565.0	-18.6	-69.82	0	-237.42	208.38	-27.58
412	-58.1	261.1	-20.6	-77.26	0	217.69	155.04	-71.49
413	2.3	329.6	19.9	74.52	0	-8.72	84.70	20.77
414	2.9	496.5	-59.1	-221.59	0	-11.01	168.75	-9.88
415	60.5	269.6	-0.5	-2.02	0	-226.88	131.03	-44.77
416	-60.9	-34.3	-2.5	-9.46	0	228.23	85.78	-96.77
417	-0.5	34.2	38.0	142.32	0	1.82	60.82	-49.88
418	0.1	201.1	-41.0	-153.79	0	-0.47	91.41	-27.07
419	63.7	449.8	-11.4	-42.85	0	-239.03	180.22	-36.27
420	-57.6	145.9	-13.4	-50.30	0	216.08	125.64	-78.94
421	2.8	214.5	27.1	101.48	0	-10.33	77.25	-8.62
422	3.4	381.3	-51.9	-194.63	0	-12.62	140.60	-18.57

Max Net factored B.P = 208.38 KN/m^2

DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 0.95 m

Moment about Z: Muz = 94.03 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$ = 777.71 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$ = 747.8 mm^2

Effective Depth Along X = 337.5 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

greater of

or A_{stmin} = 0.0014bd

A_{stmin} = 607.5 mm^2

A_{req} = 747.8 mm^2

Required Spacing = **268.9** mm

Length of cantilever along Z = 0.95 m

Moment about X: Mux = 94.03 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$ = 666.76 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$ = 809.8 mm^2

Effective Depth Along X = 312.5 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

greater of

or A_{stmin} = 0.0014bd

A_{stmin} = 562.5 mm^2

A_{req} = 809.8 mm^2

Required Spacing = **248.3** mm

Provide Bottom Reinforcement = **16 mm dia @ 200 c/c**

Ast prov = 1005.3 mm^2

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	03-11-2020	12-12-2020	03-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Outotec / L&T	0	1	2
Date	03-11-2020	12-12-2020	03-11-2023
Revision Tex	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Top Reinforcement

Length of cantilever along X = 0.95 m

Moment about Z: Muz = 23.09 KNm

$M_{uzlim} = \emptyset\, 0.85\, f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uzlim} = 807.95 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\, f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 178.0 mm^2

Effective Depth Along X = 344 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% more prov

greater of or

A_{stmin} = 50% of 0.0014bd bottom

Table 24.4.3.2 SBC 304

A_{stmin} = 309.6 mm^2

A_{req} = 178.0 mm^2

Required Spacing = **365.3** mm

Length of cantilever along Z = 0.95 m

Moment about X: Mux = 23.09 KNm

$M_{uxlim} = \emptyset\, 0.85\, f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uxlim} = 752.57 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\, f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 184.5 mm^2

Effective Depth Along Z = 332 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% more prov

greater of or

A_{stmin} = 50% of 0.0014bd bottom

Table 24.4.3.2 SBC 304

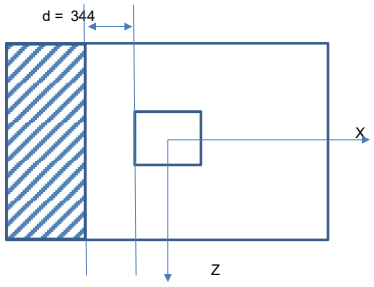
A_{stmin} = 298.8 mm^2

A_{req} = 184.5 mm^2

Required Spacing = **378.5** mm

Provide top Reinforcement = **12 mm dia @ 200 c/c** Ast prov = 565.5 mm^2

ONE WAY SHEAR CKECK



Along X Direction Cantilever: L = 0.606 m

Reinforcement ratio: ρ_w = 0.0029

Effective Depth Along X: d = 344 mm

Shear Force at d distance: V_u = 126.28 KN

Allowable shear strength: φVc = 257.03 KN O.K

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ *Table 22.5.5.1 SBC 304*

λ = 1 Normal concrete

Mu = 38.26 KNm

φ = 0.75 for shear

φVc = 258.8 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$

φVc = 257.0 KN *Table 22.5.5.1 SBC 304*

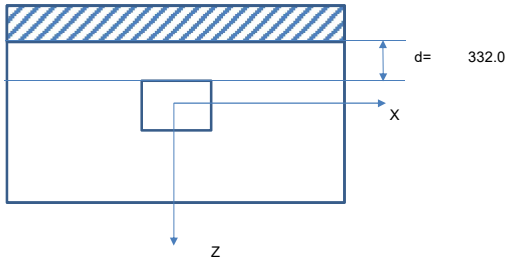
iii) $V_c = 0.29\lambda\sqrt{f_c'} b_w d$

= 442.6 KN

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T	0	1	2
Date	03-11-2020	12-12-2020	03-11-2023	Date	03-11-2020	12-12-2020	03-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK	Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

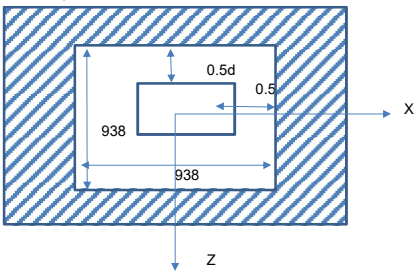
ONE WAY SHEAR CKECK



Along Z Direction	Cantilever	L	=	0.618	m		Least of	i)	V_c	= $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$
Reinforcement ratio		ρ_w	=	0.0030						*Table 22.5.5.1 SBC 304*
Effective Depth Along Z		d	=	332	mm				λ	= 1 Normal concrete
Shear Force at d distance		V_u	=	128.78	KN				Mu	= 39.79 KNm
Allowable shear strength		φVc	=	248.51	KN	O.K			φ	= 0.75 for shear
									φVc	= 249.5 KN
								ii)	V_c	= $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$
									φVc	= 248.5 KN
								iii)	V_c	= $0.29\lambda\sqrt{f_c'}b_w d$
									φVc	= 427.2 KN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area		=	6.25	m2
Punching area		=	0.88	m2
	λ	=	1	
	β	=	1	
	αs	=	40	
	b0	=	3752.0	mm

Allowable shear stress — least of — *Table 22.6.5.2 SBC 304*

v_c	= $0.33\lambda\sqrt{f_c'}$	=	1.95	N/mm²
v_c	= $0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$	=	3.02	N/mm²
v_c	= $0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$	=	2.75	N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

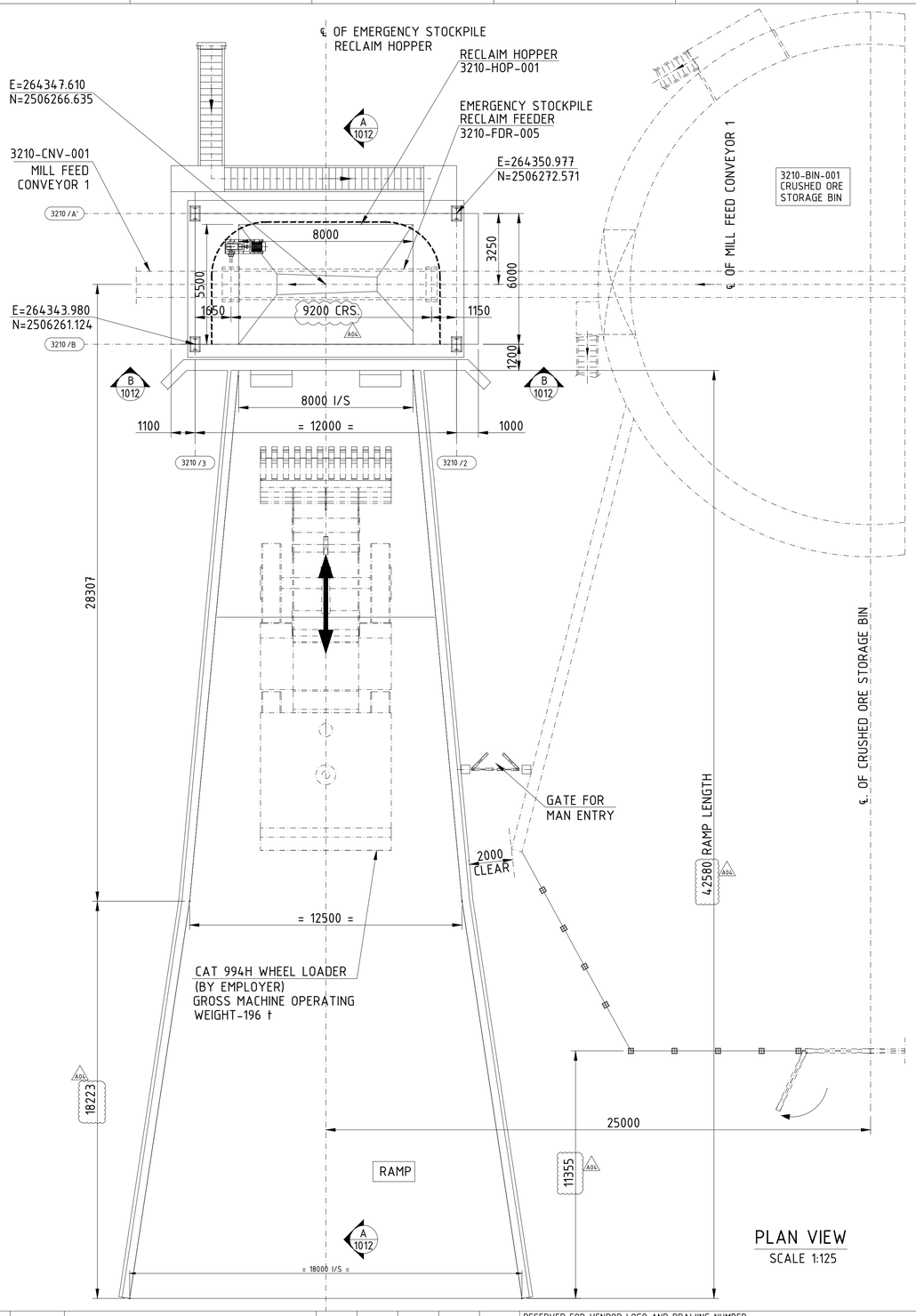
Perimeter at 0.5d from column face	u	=	3.75	m		Vc	= $v_c b_0 d$
Depth at section	d	=	338	mm			= 2475.87 KN
Shear Force at section	Vu	=	1119.0	KN		φ	= 0.75 for shear
Allowable shear strength	φVc	=	1856.90	KN	O.K	φV_n	= φVc
						φVc	= 1856.9 KN

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

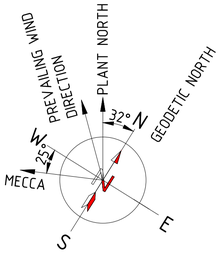
Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	03-11-2020	12-12-2020	03-11-2023				Date	03-11-2020	12-12-2020	03-11-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	RPK	RPK	RPK				Prepared By	RPK	RPK	RPK			
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

ATTACHMENT 1 **MECHANICAL GENERAL ARRANGEMENT DRAWING**

DRAWING NUMBER: MD-216-3210-EG-ME-LPX-1008
REV: A04

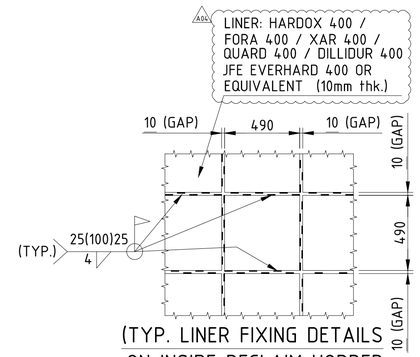


PLAN VIEW
SCALE 1:125



ABBREVIATION			
CONV.:	CONVEYOR	CRS.:	CENTERS
TBA:	TO BE ANNOUNCED	DS:	DUST SUPPRESSION
N.A.:	NOT APPLICABLE	VGTU.:	VERTICAL GRAVITY TAKE-UP
GP:	GROUND PORTAL	MKD:	MARKED
SARII:	SELF ALIGNING RETURN TRAINER INVERTED IDLER	DIA:	DIAMETER
		SYN:	SYNCHRONOUS
T.P.:	TANGENT POINT	thk.:	THICKNESS
I.P.:	INTERSECTION POINT	RCC:	REINFORCED CEMENT CONCRETE
W.P.:	WORKING POINT	B/W:	BELT WIDTH
STD.:	STANDARD	MAX:	MAXIMUM
STR.:	STRINGER	deg.:	DEGREE
SEC:	SECTION	U/S.:	UNDERSIDE
℄:	CENTER LINE	B.O.M.R:	BOTTOM OF MONORAIL
DN:	DOWNWARD	T.O.W.W.:	TOP OF WALKWAY
UP:	UPWARD	TR:	TRESTLE
TP:	TRANSFER POINT	TYP.:	TYPICAL
BRG.:	BEARING	OPNG.:	OPENING

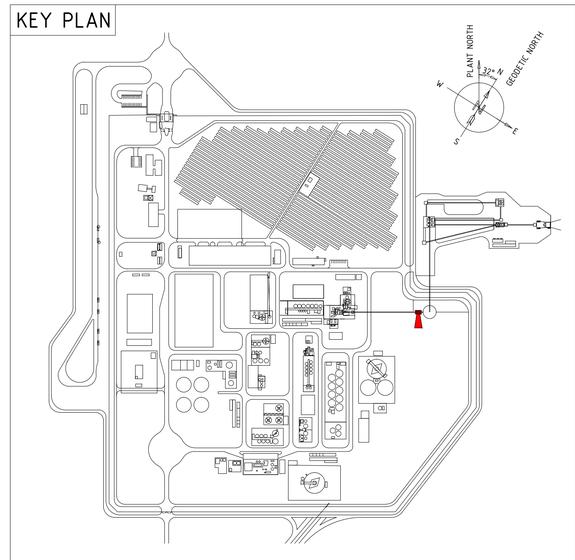
NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)912.450 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.250 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN INDICATIVE.



(TYP. LINER FIXING DETAILS ON INSIDE RECLAIM HOPPER
SCALE 1:20

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN



REV	DESCRIPTION	DRN	CHK	ENG	APP	EMP	DATE
A04	ISSUED FOR DETAIL DESIGN & REVISED AS MARKED △A04	SM	BRD	BD	BGR		25/06/2020
A03	ISSUED FOR APPROVAL & REVISED AS MARKED △A03						02/06/2020
A02	ISSUED FOR APPROVAL & REVISED AS MARKED △A02	HKD	BRD	BD	BGR		07/05/2020
A01	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR		16/03/2020

DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3210-EG-ME-ELV-1014	RECLAIM FEEDER 3210-FDR-005 - ELEVATION
MD-216-3210-EG-ME-ELV-1012	EMERGENCY STOCKPILE RECLAIM HOPPER - ELEVATION
MD-216-3210-EG-ME-LPX-1009	EMERGENCY STOCKPILE RECLAIM HOPPER - PLAN

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

This drawing is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Outotec

The content of these documents are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without prior written permission of Outotec is prohibited.

MA'ADEN معادن
Saudi Arabian Mining Company شركة معادن العربية السعودية

This document and the information contained herein is Ma'aden proprietary know-how. This document or any part thereof should not be reproduced, stored in a retrieval system or transmitted by any means, electronic or otherwise, or divulged without prior written permission of Ma'aden.

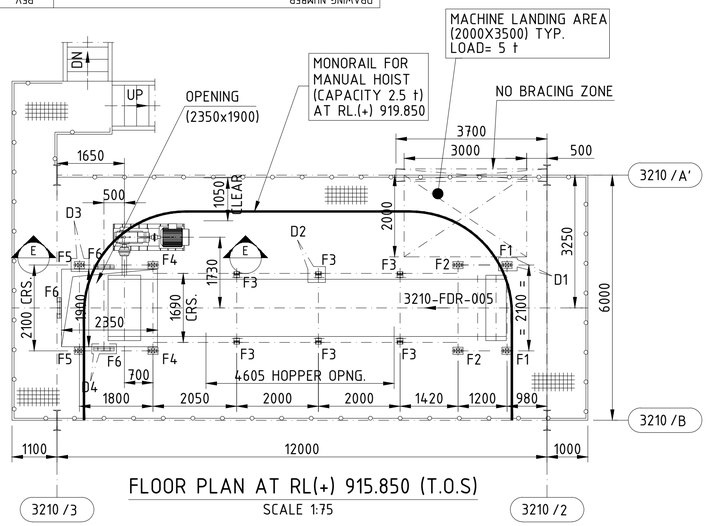
LARSEN & TOUBRO DRAWING NUMBER: 019045_100_H_G_16127_001 — REVISION: 3
OUTOTEC DRAWING NUMBER: 900707__GAU01__US01__MTL01__32108 — DOCUMENT ID: OU100669122 — REVISION: 3

DRAWN	HKD	08/03/2020
CHECKED	BRD	10/03/2020
ENGINEER	BD	12/03/2020
APPROVED	BGR	16/03/2020
EMPLOYER		
SCALE	1:125	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
GRINDING & CLASSIFICATION
EMERGENCY STOCKPILE RECLAIM HOPPER - PLAN

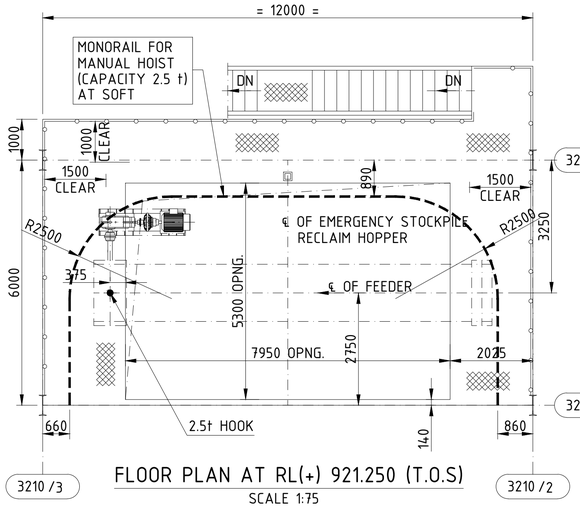
DRAWING NUMBER: MD-216-3210-EG-ME-LPX-1008 — REV: A04

DRAWING NUMBER MD-216-3210-EG-ME-LPX-1009 REV A03



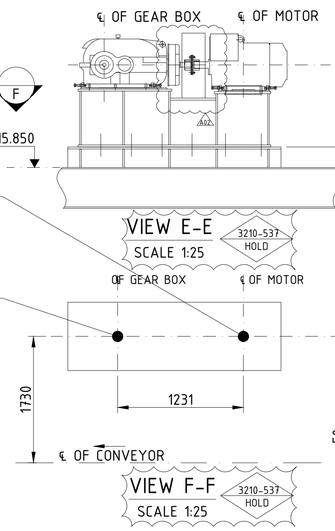
FLOOR PLAN AT RL(+) 915.850 (T.O.S)

SCALE 1:75



FLOOR PLAN AT RL(+) 921.250 (T.O.S)

SCALE 1:75



VIEW E-E

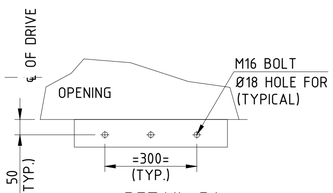
SCALE 1:25

3210-537 HOLD

VIEW F-F

SCALE 1:25

3210-537 HOLD



DETAIL-D4

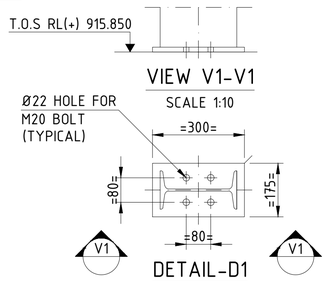
(TYPICAL)

SCALE 1:10

NOTE:

1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)912.450 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.250 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN INDICATIVE.

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

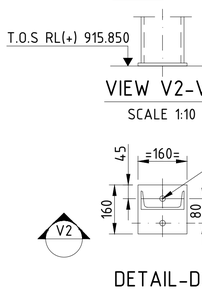


VIEW V1-V1

SCALE 1:10

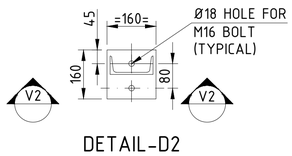
DETAIL-D1

SCALE 1:10



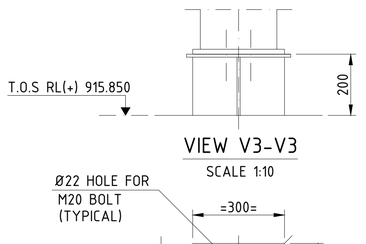
VIEW V2-V2

SCALE 1:10



DETAIL-D2

SCALE 1:10

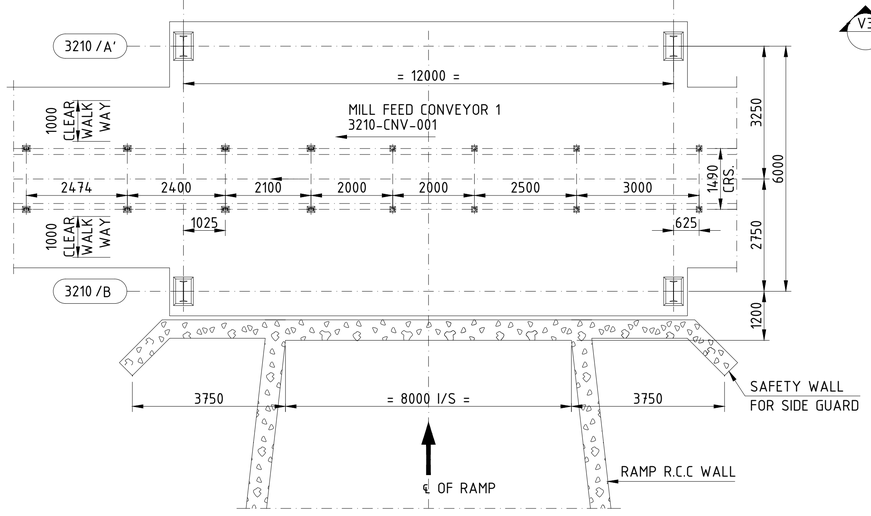


VIEW V3-V3

SCALE 1:10

DETAIL-D3

SCALE 1:10



FLOOR PLAN AT RL(+) 912.450 (F.F.L)

SCALE 1:75

TABLE-1 : LOAD FOR CONVEYOR SHORT POST

FOR CONV. 3210-FDR-005

ZONE	POINT	VERTICAL LOAD IN "t"		HORIZONTAL LOAD IN "t"	
		DEAD LOAD	LIVE LOAD	ALONG	ACROSS
IMPACT ZONE	F3	(+)0.90	(+)1.40	(±)0.15	(±)0.1

TABLE-2: LOAD FOR SCREW TAKE-UP FRAME

FOR BELT FEEDER : 3210-FDR-005

POINTS	ELEVATIONS IN "m"	VERTICAL LOAD (IN "t")	HORIZONTAL LOAD (IN "t")	
			ALONG THE CONV.	ACROSS THE CONV.
F1	RL(+) 915.850	(-)1.8	(+)3.0	(±)0.30
F2		(+)4.9	(+)3.5	(±)0.50

TABLE-3 : LOAD FOR DRIVE PULLEY FRAME

FOR CONV. 3210-FDR-005

POINTS	ELEVATIONS IN "m"	LOAD CASE I			LOAD CASE II		
		V (IN "t")	H ALONG (IN "t")	H ACROSS (IN "t")	V (IN "t")	H ALONG (IN "t")	H ACROSS (IN "t")
F4	RL(+) 915.850	(+)5.1	(-)4.0	(±)0.6	(+)4.3	(-)6.0	(±)0.5
F5		(+)1.9	(-)5.0	(±)0.2	(-)1.1	(-)4.1	(±)0.2

NOTE:
CASE I : CONVEYOR NORMAL RUNNING + MATERIAL LOAD + DEAD LOAD
CASE II : CONVEYOR STARTING + CHUTE EMPTY + DEAD LOAD

TABLE-4: LOAD FOR CHUTE SADDLE

POINTS	ELEVATIONS IN "m"	VERTICAL LOAD (IN "t")	
		CHUTE EMPTY	CHUTE BLOCKED
F6	RL(+) 915.850	(+)1.5	(+)3.5

SIGN CONVENTION

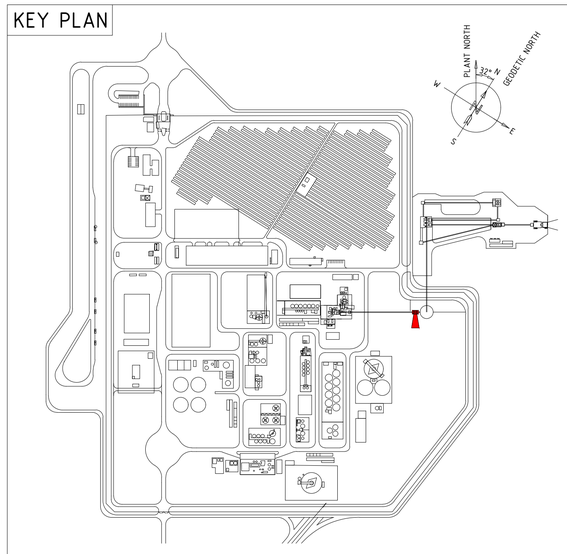
UPWARD (-)
DOWNWARD (+)
TOWARDS DRIVE PULLEY=(+)
TOWARDS TAIL PULLEY=(-)
REVERSIBLE LOAD=(±)

Note: Wind & seismic load to be considered during civil & structural design

ABBREVIATION

CONV.:	CONVEYOR	CRS.:	CENTERS
TBA:	TO BE ANNOUNCED	DS:	DUST SUPPRESSION
N.A.:	NOT APPLICABLE	VGTU.:	VERTICAL GRAVITY TAKE-UP
GP:	GROUND PORTAL	MKD:	MARKED
SARII:	SELF ALIGNING RETURN TRAINER INVERTED IDLER	DIA:	DIAMETER
T.P.:	TANGENT POINT	SYN:	SYNCHRONOUS
I.P.:	INTERSECTION POINT	thk.:	THICKNESS
W.P.:	WORKING POINT	RCC:	REINFORCED CEMENT CONCRETE
STD.:	STANDARD	B/W:	BELT WIDTH
STR.:	STRINGER	MAX:	MAXIMUM
SEC:	SECTION	deg.:	DEGREE
℄:	CENTER LINE	U/S.:	UNDERSIDE
DN:	DOWNWARD	B.O.M.R:	BOTTOM OF MONORAIL
UP:	UPWARD	T.O.W.W.:	TOP OF WALKWAY
TP:	TRANSFER POINT	TR:	TRESTLE
BRG.:	BEARING	TYP.:	TYPICAL
		OPNG.:	OPENING

KEY PLAN



REV	DESCRIPTION	DRN	CHK	ENG	APP	EMP	DATE
A03	ISSUED FOR DETAIL DESIGN	SM	BRD	BD	BGR		02/06/2020
A02	ISSUED FOR APPROVAL & REVISED AS PER EMPLOYER	HKD	BRD	BD	BGR		07/05/2020
	COMMENTS DT. 08.04.2020 & REVISED AS MARKED A02						
A01	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR		16/03/2020

DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3210-EG-ME-ELV-1014	RECLAIM FEEDER 3210-FDR-005 - ELEVATION
MD-216-3210-EG-ME-ELV-1012	EMERGENCY STOCKPILE RECLAIM HOPPER - ELEVATION
MD-216-3210-EG-ME-LPX-1008	EMERGENCY STOCKPILE RECLAIM HOPPER - PLAN

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

This drawing is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Outotec

The content of these documents are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without prior written permission of Outotec is prohibited.

MA'ADEN معادن
Saudi Arabian Mining Company شركة معادن العربية السعودية

This document and the information contained herein is Ma'aden proprietary know-how. This document or any part thereof should not be reproduced, stored in a retrieval system or transmitted by any means, electronic or otherwise, or divulged without prior written permission of Ma'aden.

LARSEN & TOUBRO DRAWING NUMBER		REVISION
019045_100_H_G_16127_004		2
OUTOTEC DRAWING NUMBER	DOCUMENT ID	REVISION
900707__GAU01__US01__MTL01__32109	OU100669123	2

DRAWN	HKD	08/03/2020
CHECKED	BRD	10/03/2020
ENGINEER	BD	12/03/2020
APPROVED	BGR	16/03/2020
EMPLOYER		
SCALE	1:25, 1:75	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
GRINDING & CLASSIFICATION
EMERGENCY STOCKPILE RECLAIM HOPPER - PLAN

DRAWING NUMBER: MD-216-3210-EG-ME-LPX-1009 REV: A03

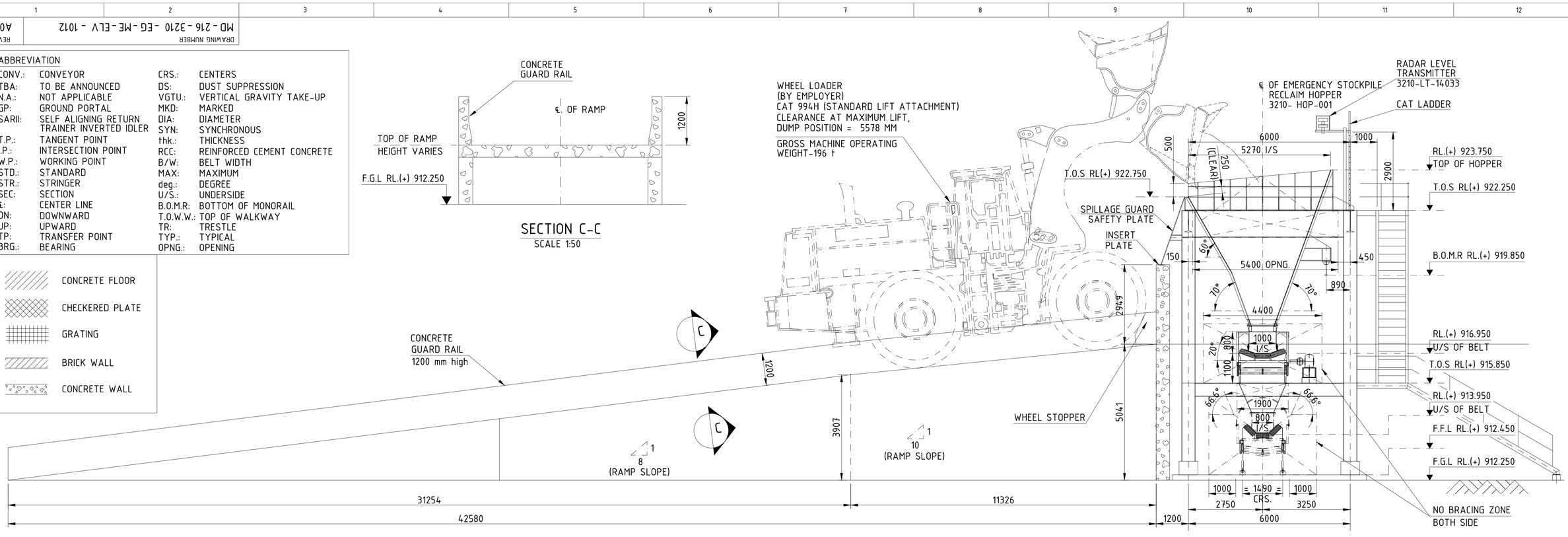
DRAWING NUMBER: MD-216-3210-EG-ME-ELV-1012 | REV: A04

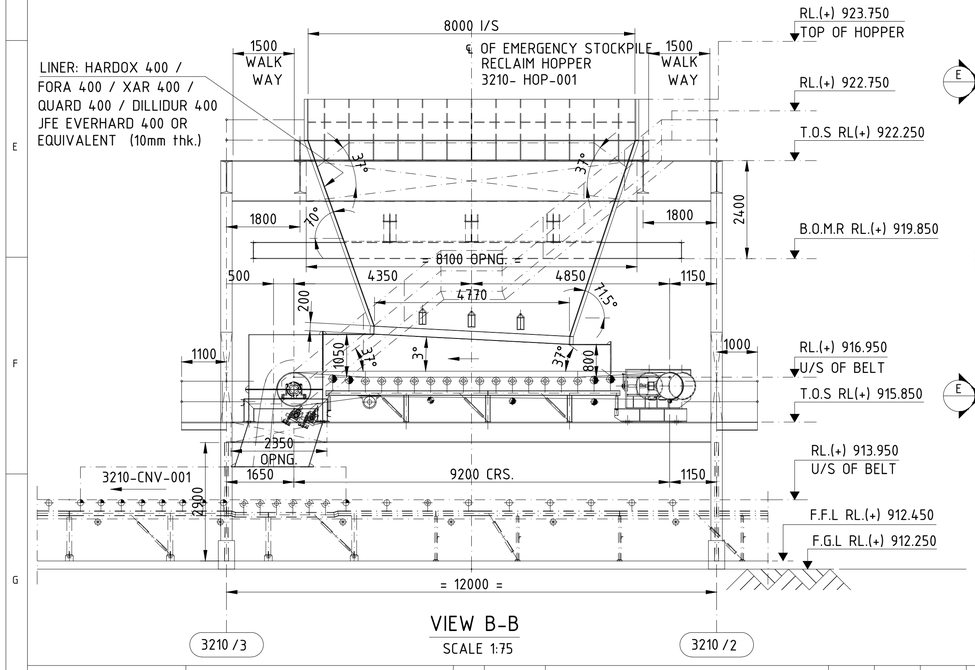
ABBREVIATION

CONV.:	CONVEYOR	CRS.:	CENTERS
TBA:	TO BE ANNOUNCED	DS:	DUST SUPPRESSION
N.A.:	NOT APPLICABLE	VGTU.:	VERTICAL GRAVITY TAKE-UP
GP:	GROUND PORTAL	MKD:	MARKED
SARII:	SELF ALIGNING RETURN TRAINER INVERTED IDLER	DIA:	DIAMETER
		SYN:	SYNCHRONOUS
T.P.:	TANGENT POINT	thk.:	THICKNESS
I.P.:	INTERSECTION POINT	RCC:	REINFORCED CEMENT CONCRETE
W.P.:	WORKING POINT	B/W:	BELT WIDTH
STD.:	STANDARD	MAX:	MAXIMUM
STR.:	STRINGER	deg.:	DEGREE
SEC:	SECTION	U/S.:	UNDERSIDE
℄:	CENTER LINE	B.O.M.R:	BOTTOM OF MONORAIL
DN:	DOWNWARD	T.O.W.W.:	TOP OF WALKWAY
UP:	UPWARD	TR:	TRESTLE
TP:	TRANSFER POINT	TYP.:	TYPICAL
BRG.:	BEARING	OPNG.:	OPENING

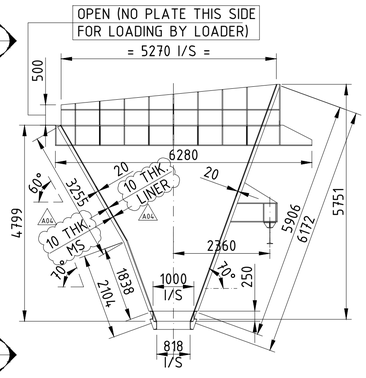
CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

SECTION C-C
SCALE 1:50



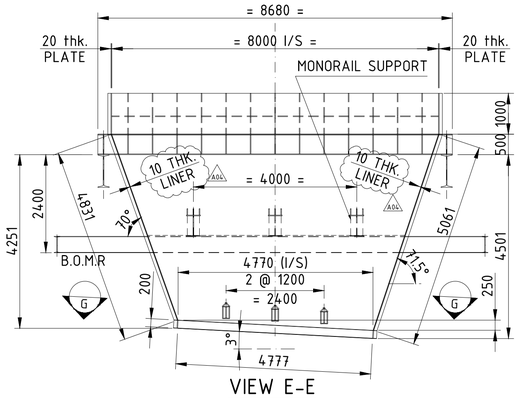


VIEW B-B
SCALE 1:75

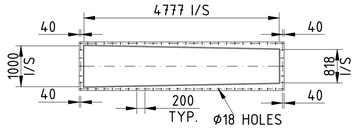


DETAIL OF EMERGENCY STOCKPILE RECLAIM HOPPER 3210- HOP-001
SCALE 1:75

VIEW A-A
SCALE 1:75

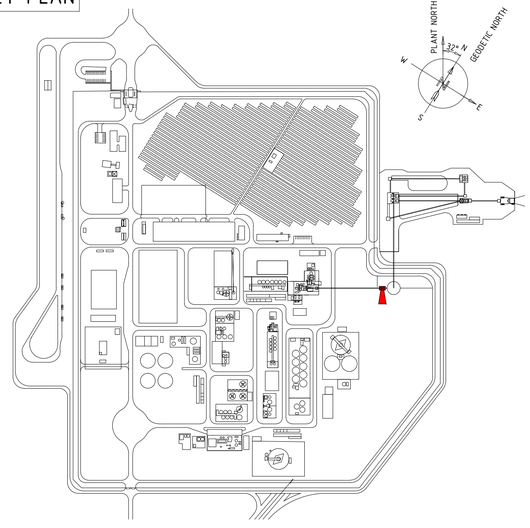


VIEW E-E
SCALE 1:75



VIEW G-G
SCALE 1:75

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A04	25/06/2020	ISSUED FOR DETAIL DESIGN & REVISED AS MARKED A04	HKD	BRD	BD	BGR	
		A03	02/06/2020	ISSUED FOR APPROVAL & REVISED AS PER EMPLOYER	HKD	BRD	BD	BGR	
				COMMENTS DT. 29.05.2020 & REVISED AS MARKED A02					
MD-216-3210-EG-ME-ELV-1014	RECLAIM FEEDER 3210-FDR-005 - ELEVATION	A02	07/05/2020	ISSUED FOR APPROVAL & REVISED AS PER EMPLOYER	HKD	BRD	BD	BGR	
MD-216-3210-EG-ME-LPX-1009	EMERGENCY STOCKPILE RECLAIM HOPPER - PLAN			COMMENTS DT. 08.04.2020 & REVISED AS MARKED A02					
MD-216-3210-EG-ME-LPX-1008	EMERGENCY STOCKPILE RECLAIM HOPPER - PLAN	A01	16/03/2020	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

This drawing is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Outotec

The content of these documents are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without prior written permission of Outotec is prohibited.

MA'ADEN معادن
Saudi Arabian Mining Company شركة معادن العربية السعودية

This document and the information contained herein is Ma'aden proprietary know-how. This document or any part thereof should not be reproduced, stored in a retrieval system or transmitted by any means, electronic or otherwise, or divulged without prior written permission of Ma'aden.

LARSEN & TOUBRO DRAWING NUMBER: O19045_100_H_G_16127_002 | REVISION: 3
OUTOTEC DRAWING NUMBER: 900707__GAU01__US01__MTL01__32117 | DOCUMENT ID: OU100669136 | REVISION: 3

DRAWN	HKD	08/03/2020
CHECKED	BRD	10/03/2020
ENGINEER	BD	12/03/2020
APPROVED	BGR	16/03/2020
EMPLOYER		
SCALE	1:75	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
GRINDING & CLASSIFICATION
EMERGENCY STOCKPILE RECLAIM HOPPER - ELEVATION

DRAWING NUMBER: MD-216-3210-EG-ME-ELV-1012 | REV: A04

DRAWING NUMBER: MD-216-3210-EG-ME-ELV-1014 | REV: A03

ABBREVIATION

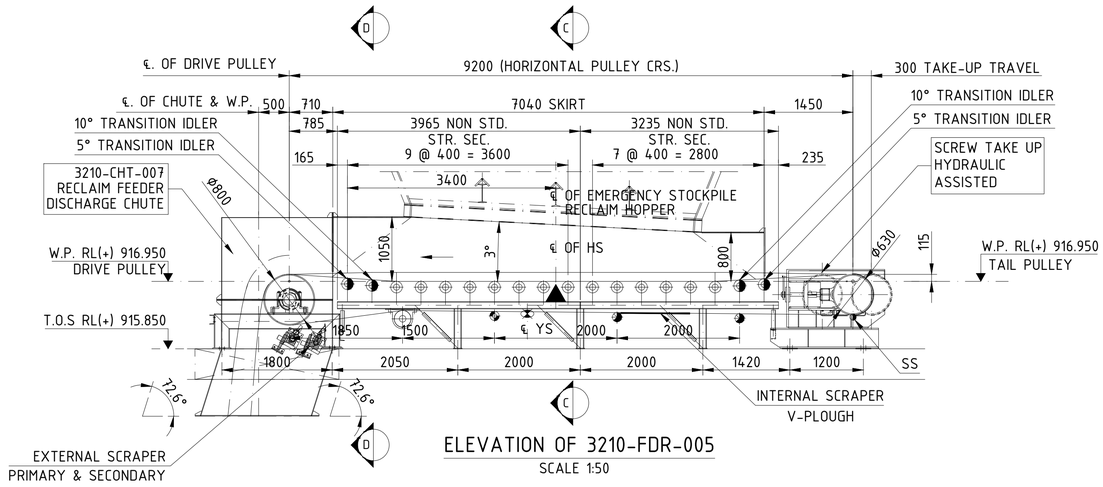
CONV.:	CONVEYOR	CRS.:	CENTERS
TBA:	TO BE ANNOUNCED	DS:	DUST SUPPRESSION
N.A.:	NOT APPLICABLE	VGTU.:	VERTICAL GRAVITY TAKE-UP
GP:	GROUND PORTAL	MKD:	MARKED
SARII:	SELF ALIGNING RETURN TRAINER INVERTED IDLER	DIA:	DIAMETER
		SYN:	SYNCHRONOUS
T.P.:	TANGENT POINT	thk.:	THICKNESS
I.P.:	INTERSECTION POINT	RCC:	REINFORCED CEMENT CONCRETE
W.P.:	WORKING POINT	B/W:	BELT WIDTH
STD.:	STANDARD	MAX:	MAXIMUM
STR.:	STRINGER	deg.:	DEGREE
SEC:	SECTION	U/S.:	UNDERSIDE
℄:	CENTER LINE	B.O.M.R:	BOTTOM OF MONORAIL
DN:	DOWNWARD	T.O.W.W.:	TOP OF WALKWAY
UP:	UPWARD	TR:	TRESTLE
TP:	TRANSFER POINT	TYP.:	TYPICAL
BRG.:	BEARING	OPNG.:	OPENING

CONVEYOR CHARACTERISTICS

	PARTICULARS	SPECIFICATION	QUANTITY
MISCELLANEOUS EQUIPMENT	TAKE-UP / TRAVEL	SCREW / 300 mm	
	TAKE-UP LOAD	N.A.	
	SCRAPER AT DISCHARGE PULLEY	EXTERNAL / MAIN & PRE CLEANER	1 NO. + 1 NO.
	SCRAPER AT TAKE-UP	N.A.	
	SCRAPER AT TAIL PULLEY	INTERNAL V-PLOUGH	1 NO.
	BELT SWAY SWITCH (ZS)	@ 50 m INTERVAL	
	PULL CORD SWITCH (HS)	@ 30 m INTERVAL	1 PAIRS
	ZERO SPEED SWITCH (SS)	AT TAIL PULLEY	1 NO.
	BELT RIP SWITCH (YS)	AFTER FEEDING ZONE	1 NO.
	DECK PLATE	3.15 mm thk. CONTINUOUS	1 LOT.
	BELT HOOD	N.A.	

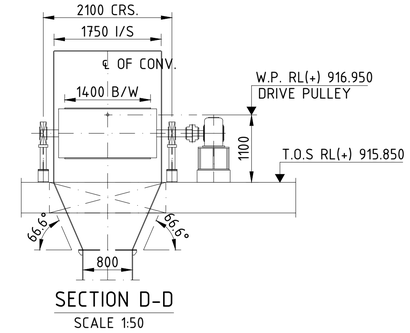
CONVEYOR CHARACTERISTICS

		PARTICULARS	SPECIFICATION	QUANTITY
CONVEYOR		EQUIPMENT TAG NO.	3210-FDR-005	1 NO.
		EQUIPMENT DESCRIPTION	EMERGENCYHOPPER DISCHARGE FEEDER	
		LOCATION	EMERGENCY STOCKPILE RECLAIM HOPPER	
		CAPACITY RATED / DESIGN (t/h)	500 / 600	
		BELT WIDTH (mm)	1400	
		BELT SPEED (m/sec)	0.686	
		TRIPPER FIXED / MOVABLE	N.A.	
		CONVEYOR LENGTH / LIFT (mm)	9200 / 0	
MATERIAL		MATERIAL	GOLD ORE : FRESH ORE / OXIDE ORE	
		BULK DENSITY (t/m³)	1.6	
		MAX. LUMP SIZE (mm)	(-)10.0	
		MOISTURE CONTENT (%)	2	
		SURCHARGE ANGLE (deg.)	20	
BELTING		MAX. RUNNING TENSION (T1 kN)	79.786	
		SLACK SIDE TENSION (T2 kN)	58.684	
		TAIL TENSION (T0 kN)	61.952	
		BELT DESIGNATION	EP-1000 / 4 (HEAVY DUTY)	
		TOP / BOTTOM COVER THICKNESS (mm)	8 / 3	
		COVER GRADE / FABRIC	DIN-X / EP	
PULLEYS		DISCHARGE WITH DRIVE (MKD. x DIA.)	D-2 x 800 mm (A03)	1 NO.
		DISCHARGE WITHOUT DRIVE (MKD. x DIA.)	N.A.	
		SNUB (MKD. x DIA.)	N.A.	
		TAIL (MKD. x DIA.)	ND-6 x 630 mm	1 NO.
		TAKE-UP (MKD. x DIA.)	N.A.	
		DEFLECTOR ROLLER (MKD. x DIA.)	DR-3 x 273 mm	1 NO.
		TRIPPER DISCHARGE (MKD. x DIA.)	N.A.	
		TRIPPER BEND (MKD. x DIA.)	N.A.	
		BEND (MKD. x DIA.)	N.A.	
		LAGGING (mm)	DRIVE: 16thk CERAMIC	
			NON-DRIVE: 16thk PLAIN RUBBER	
BEARING/SLEEVE PLUMMER BLOCK		DISCHARGE WITH DRIVE	TBA	2 SETS
		DISCHARGE WITHOUT DRIVE	N.A.	
		SNUB	TBA	
		TAIL	TBA	2 SETS
		TAKE-UP	TBA	2 SETS
		DEFLECTOR ROLLER	TBA	2 SETS
		BEND	TBA	
		TRIPPER DISCHARGE	N.A.	
		TRIPPER BEND	N.A.	
IDLER		TROUGHING (DIA. x deg.)		
		TROUGHING RETRACTABLE (DIA. x deg.)		
		TROUGHING TRAINER (DIA. x deg.)		
		RETURN (DIA. x deg.)		
		SELF ALIGNING RETURN TRAINER INVERTED IDLER (DIA. x deg.)		
		IMPACT RETRACTABLE (DIA. x deg.)	190 x 20°	14 NOS.
		TRANSITION (ADJUSTABLE) (DIA. x deg.)	152.4 x 10°/ 5°	(2+2) NOS.
		RETURN SELF CLEANING (DIA. x deg.)	190 x 0°	1 NO.
		NON MAGNETIC TROUGHING (DIA. x deg.)		
		WEIGHING IDLER (DIA. x deg.)		
		SPECIAL MACHINED IDLER (DIA. x deg.)		
		RETURN FLAT IDLER (DIA. x deg.)	152.4 x 0°	2 NOS.
DRIVE	MOTOR	RATING kW/rpm (SYN)	45 / 1800 VVF DRIVE OPERATED	1 NO.
		FRAME SIZE	TBA	
		TYPE	SQUIRREL CAGE	
		SUPPLY	400V, 60Hz	
	REDUCER	TYPE	HELICAL	1 NO.
		MODEL NO./COOLING/RATIO	TBA	
		RATIO	112	
		HOLD BACK	YES	
	BRAKE	TYPE	N.A.	
		MODEL NO.	N.A.	
		DRAM DIAMETER	N.A.	
	COUPLING HIGH SPEED	TYPE	RESILIENT COUPLING	1 NO.
		MODEL NO.	TBA	
	COUPLING LOW SPEED	TYPE	GEAR	1 NO.
		MODEL NO.	TBA	



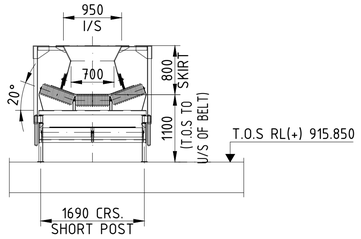
ELEVATION OF 3210-FDR-005

SCALE 1:50



SECTION D-D

SCALE 1:50



SECTION C-C

SCALE 1:50

NOTE:

1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)912.450 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.250 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN INDICATIVE.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3210-EG-ME-ELV-1012	EMERGENCY STOCKPILE RECLAIM HOPPER - ELEVATION	A03	02/06/2020	ISSUED FOR DETAIL DESIGN & REVISED AS MARKED (A03)	SM	BRD	BD	BGR	
MD-216-3210-EG-ME-LPX-1009	EMERGENCY STOCKPILE RECLAIM HOPPER - PLAN	A02	07/05/2020	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR	
MD-216-3210-EG-ME-LPX-1008	EMERGENCY STOCKPILE RECLAIM HOPPER - PLAN	A01	16/03/2020	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

This drawing is the property of *LARSEN & TOUBRO* and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Outotec

The content of these documents are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without prior written permission of Outotec is prohibited.

MA'ADEN معادن
Saudi Arabian Mining Company شركة معادن العربية السعودية

This document and the information contained herein is Ma'aden proprietary know-how. This document or any part thereof should not be reproduced, stored in a retrieval system or transmitted by any means, electronic or otherwise, or divulged without prior written permission of Ma'aden.

LARSEN & TOUBRO DRAWING NUMBER	019045_100_H_G_16127_006	REVISION 2
OUTOTEC DRAWING NUMBER	900707 _GAU01 _US01 _MTL01_ 32119	DOCUMENT ID OU100669138 / REVISION 2

DRAWN	HKD	08/03/2020
CHECKED	BRD	10/03/2020
ENGINEER	BD	12/03/2020
APPROVED	BGR	16/03/2020
EMPLOYER		
SCALE	1:10, 1:50	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
GRINDING & CLASSIFICATION
RECLAIM FEEDER 3210-FDR-005 - ELEVATION

DRAWING NUMBER: MD-216-3210-EG-ME-ELV-1014 | REV: A03

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	03-11-2020	12-12-2020	03-11-2023				Date	03-11-2020	12-12-2020	03-11-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	RPK	RPK	RPK				Prepared By	RPK	RPK	RPK		
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 2 **STAAD INPUT & OUTPUT FILE OF RAMP PART-1**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 20-Oct-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 12.5 3.55 -0.8; 2 12.5 3.55 10.8; 3 18.5 3.55 -0.8; 4 18.5 3.55 10.8;
17 12.5 3.55 5; 21 18.5 3.55 5; 100 0 0 0; 101 11.3 0 1; 102 0 0 10.1;
103 11.3 0 9; 104 11.3 7.4 1; 105 11.3 7.4 9; 106 0 6.3 0; 107 0 6.3 10.1;
108 0 5.67 0; 109 1.13 5.769 0.1; 110 1.13 6.41 0.1; 111 0 5.04 0;
112 1.13 5.128 0.1; 113 0 4.41 0; 114 1.13 4.487 0.1; 115 0 3.78 0;
116 1.13 3.846 0.1; 117 0 3.15 0; 118 1.13 3.205 0.1; 119 0 2.52 0;
120 1.13 2.564 0.1; 121 0 1.89 0; 122 1.13 1.923 0.1; 123 0 1.26 0;
124 1.13 1.282 0.1; 125 0 0.33 0; 126 1.13 0.341 0.1; 127 1.13 0 0.1;
128 2.26 5.868 0.2; 129 2.26 6.52 0.2; 130 2.26 5.216 0.2; 131 2.26 4.564 0.2;
132 2.26 3.912 0.2; 133 2.26 3.26 0.2; 134 2.26 2.608 0.2; 135 2.26 1.956 0.2;
136 2.26 1.304 0.2; 137 2.26 0.352 0.2; 138 2.26 0 0.2; 139 3.39 5.967 0.3;
140 3.39 6.63 0.3; 141 3.39 5.304 0.3; 142 3.39 4.641 0.3; 143 3.39 3.978 0.3;
144 3.39 3.315 0.3; 145 3.39 2.652 0.3; 146 3.39 1.989 0.3; 147 3.39 1.326 0.3;
148 3.39 0.363 0.3; 149 3.39 0 0.3; 150 4.52 6.066 0.4; 151 4.52 6.74 0.4;
152 4.52 5.392 0.4; 153 4.52 4.718 0.4; 154 4.52 4.044 0.4; 155 4.52 3.37 0.4;
156 4.52 2.696 0.4; 157 4.52 2.022 0.4; 158 4.52 1.348 0.4; 159 4.52 0.374 0.4;
160 4.52 0 0.4; 161 5.65 6.165 0.5; 162 5.65 6.85 0.5; 163 5.65 5.48 0.5;
164 5.65 4.795 0.5; 165 5.65 4.11 0.5; 166 5.65 3.425 0.5; 167 5.65 2.74 0.5;
168 5.65 2.055 0.5; 169 5.65 1.37 0.5; 170 5.65 0.385 0.5; 171 5.65 0 0.5;
172 6.78 6.264 0.6; 173 6.78 6.96 0.6; 174 6.78 5.568 0.6; 175 6.78 4.872 0.6;
176 6.78 4.176 0.6; 177 6.78 3.48 0.6; 178 6.78 2.784 0.6; 179 6.78 2.088 0.6;
180 6.78 1.392 0.6; 181 6.78 0.396 0.6; 182 6.78 0 0.6; 183 7.91 6.363 0.7;
184 7.91 7.07 0.7; 185 7.91 5.656 0.7; 186 7.91 4.949 0.7; 187 7.91 4.242 0.7;
188 7.91 3.535 0.7; 189 7.91 2.828 0.7; 190 7.91 2.121 0.7; 191 7.91 1.414 0.7;
192 7.91 0.407 0.7; 193 7.91 0 0.7; 194 9.04 6.462 0.8; 195 9.04 7.18 0.8;
196 9.04 5.744 0.8; 197 9.04 5.026 0.8; 198 9.04 4.308 0.8; 199 9.04 3.59 0.8;
200 9.04 2.872 0.8; 201 9.04 2.154 0.8; 202 9.04 1.436 0.8; 203 9.04 0.418 0.8;
204 9.04 0 0.8; 205 10.54 6.561 0.9; 206 10.54 7.33 0.9; 207 10.54 5.832 0.9;
208 10.54 5.103 0.9; 209 10.54 4.374 0.9; 210 10.54 3.645 0.9;
211 10.54 2.916 0.9; 212 10.54 2.187 0.9; 213 10.54 1.458 0.9;
214 10.54 0.429 0.9; 215 10.54 0 0.9; 216 11.3 6.66 1; 217 11.3 5.92 1;
218 11.3 5.18 1; 219 11.3 4.44 1; 220 11.3 3.7 1; 221 11.3 2.96 1;
222 11.3 2.22 1; 223 11.3 1.48 1; 224 11.3 0.44 1; 225 11.3 6.66 9;
226 10.54 6.561 9.11; 227 10.54 7.33 9.11; 228 11.3 5.92 9;
229 10.54 5.832 9.11; 230 11.3 5.18 9; 231 10.54 5.103 9.11; 232 11.3 4.44 9;
233 10.54 4.374 9.11; 234 11.3 3.7 9; 235 10.54 3.645 9.11; 236 11.3 2.96 9;
237 10.54 2.916 9.11; 238 11.3 2.22 9; 239 10.54 2.187 9.11; 240 11.3 1.48 9;
241 10.54 1.458 9.11; 242 11.3 0.44 9; 243 10.54 0.429 9.11; 244 10.54 0 9.11;
245 9.04 6.462 9.22; 246 9.04 7.18 9.22; 247 9.04 5.744 9.22;
248 9.04 5.026 9.22; 249 9.04 4.308 9.22; 250 9.04 3.59 9.22;
251 9.04 2.872 9.22; 252 9.04 2.154 9.22; 253 9.04 1.436 9.22;
254 9.04 0.418 9.22; 255 9.04 0 9.22; 256 7.91 6.363 9.33; 257 7.91 7.07 9.33;
258 7.91 5.656 9.33; 259 7.91 4.949 9.33; 260 7.91 4.242 9.33;
261 7.91 3.535 9.33; 262 7.91 2.828 9.33; 263 7.91 2.121 9.33;
264 7.91 1.414 9.33; 265 7.91 0.407 9.33; 266 7.91 0 9.33; 267 6.78 6.264 9.44;
268 6.78 6.96 9.44; 269 6.78 5.568 9.44; 270 6.78 4.872 9.44;
271 6.78 4.176 9.44; 272 6.78 3.48 9.44; 273 6.78 2.784 9.44;
```

274 6.78 2.088 9.44; 275 6.78 1.392 9.44; 276 6.78 0.396 9.44; 277 6.78 0 9.44;
278 5.65 6.165 9.55; 279 5.65 6.85 9.55; 280 5.65 5.48 9.55;
281 5.65 4.795 9.55; 282 5.65 4.11 9.55; 283 5.65 3.425 9.55;
284 5.65 2.74 9.55; 285 5.65 2.055 9.55; 286 5.65 1.37 9.55;
287 5.65 0.385 9.55; 288 5.65 0 9.55; 289 4.52 6.066 9.66; 290 4.52 6.74 9.66;
291 4.52 5.392 9.66; 292 4.52 4.718 9.66; 293 4.52 4.044 9.66;
294 4.52 3.37 9.66; 295 4.52 2.696 9.66; 296 4.52 2.022 9.66;
297 4.52 1.348 9.66; 298 4.52 0.374 9.66; 299 4.52 0 9.66; 300 3.39 5.967 9.77;
301 3.39 6.63 9.77; 302 3.39 5.304 9.77; 303 3.39 4.641 9.77;
304 3.39 3.978 9.77; 305 3.39 3.315 9.77; 306 3.39 2.652 9.77;
307 3.39 1.989 9.77; 308 3.39 1.326 9.77; 309 3.39 0.363 9.77; 310 3.39 0 9.77;
311 2.26 5.868 9.88; 312 2.26 6.52 9.88; 313 2.26 5.216 9.88;
314 2.26 4.564 9.88; 315 2.26 3.912 9.88; 316 2.26 3.26 9.88;
317 2.26 2.608 9.88; 318 2.26 1.956 9.88; 319 2.26 1.304 9.88;
320 2.26 0.352 9.88; 321 2.26 0 9.88; 322 1.13 5.769 9.99; 323 1.13 6.41 9.99;
324 1.13 5.128 9.99; 325 1.13 4.487 9.99; 326 1.13 3.846 9.99;
327 1.13 3.205 9.99; 328 1.13 2.564 9.99; 329 1.13 1.923 9.99;
330 1.13 1.282 9.99; 331 1.13 0.341 9.99; 332 1.13 0 9.99;
333 -1.30773e-06 5.67 10.1; 334 -1.26839e-06 5.04 10.1;
335 -1.22905e-06 4.41 10.1; 336 -1.18971e-06 3.78 10.1;
337 -1.15037e-06 3.15 10.1; 338 -1.11103e-06 2.52 10.1;
339 -1.07169e-06 1.89 10.1; 340 -1.03235e-06 1.26 10.1;
341 -9.93013e-07 0.33 10.1; 342 0 0 0.71; 343 1.13 0 0.789; 344 0 0 2.02;
345 1.13 0 2.078; 346 0 0 3.03; 347 1.13 0 3.067; 348 0 0 4.04;
349 1.13 0 4.056; 350 0 0 5.05; 351 1.13 0 5.045; 352 0 0 6.06;
353 1.13 0 6.034; 354 0 0 7.07; 355 1.13 0 7.023; 356 0 0 8.08;
357 1.13 0 8.012; 358 0 0 9.39; 359 1.13 0 9.301; 360 2.26 0 0.868;
361 2.26 0 2.136; 362 2.26 0 3.104; 363 2.26 0 4.072; 364 2.26 0 5.04;
365 2.26 0 6.008; 366 2.26 0 6.976; 367 2.26 0 7.944; 368 2.26 0 9.212;
369 3.39 0 0.947; 370 3.39 0 2.194; 371 3.39 0 3.141; 372 3.39 0 4.088;
373 3.39 0 5.035; 374 3.39 0 5.982; 375 3.39 0 6.929; 376 3.39 0 7.876;
377 3.39 0 9.123; 378 4.52 0 1.026; 379 4.52 0 2.252; 380 4.52 0 3.178;
381 4.52 0 4.104; 382 4.52 0 5.03; 383 4.52 0 5.956; 384 4.52 0 6.882;
385 4.52 0 7.808; 386 4.52 0 9.034; 387 5.65 0 1.105; 388 5.65 0 2.31;
389 5.65 0 3.215; 390 5.65 0 4.12; 391 5.65 0 5.025; 392 5.65 0 5.93;
393 5.65 0 6.835; 394 5.65 0 7.74; 395 5.65 0 8.945; 396 6.78 0 1.184;
397 6.78 0 2.368; 398 6.78 0 3.252; 399 6.78 0 4.136; 400 6.78 0 5.02;
401 6.78 0 5.904; 402 6.78 0 6.788; 403 6.78 0 7.672; 404 6.78 0 8.856;
405 7.91 0 1.263; 406 7.91 0 2.426; 407 7.91 0 3.289; 408 7.91 0 4.152;
409 7.91 0 5.015; 410 7.91 0 5.878; 411 7.91 0 6.741; 412 7.91 0 7.604;
413 7.91 0 8.767; 414 9.04 0 1.342; 415 9.04 0 2.484; 416 9.04 0 3.326;
417 9.04 0 4.168; 418 9.04 0 5.01; 419 9.04 0 5.852; 420 9.04 0 6.694;
421 9.04 0 7.536; 422 9.04 0 8.678; 423 10.54 0 1.421; 424 10.54 0 2.542;
425 10.54 0 3.363; 426 10.54 0 4.184; 427 10.54 0 5.005; 428 10.54 0 5.826;
429 10.54 0 6.647; 430 10.54 0 7.468; 431 10.54 0 8.589; 432 11.3 0 1.5;
433 11.3 0 2.6; 434 11.3 0 3.4; 435 11.3 0 4.2; 436 11.3 0 5; 437 11.3 0 5.8;
438 11.3 0 6.6; 439 11.3 0 7.4; 440 11.3 0 8.5; 441 11.3 6.66 1.5;
442 11.3 7.4 1.5; 443 11.3 5.92 1.5; 444 11.3 5.18 1.5; 445 11.3 4.44 1.5;
446 11.3 3.7 1.5; 447 11.3 2.96 1.5; 448 11.3 2.22 1.5; 449 11.3 1.48 1.5;
450 11.3 0.44 1.5; 451 11.3 6.66 2.6; 452 11.3 7.4 2.6; 453 11.3 5.92 2.6;
454 11.3 5.18 2.6; 455 11.3 4.44 2.6; 456 11.3 3.7 2.6; 457 11.3 2.96 2.6;
458 11.3 2.22 2.6; 459 11.3 1.48 2.6; 460 11.3 0.44 2.6; 461 11.3 6.66 3.4;
462 11.3 7.4 3.4; 463 11.3 5.92 3.4; 464 11.3 5.18 3.4; 465 11.3 4.44 3.4;
466 11.3 3.7 3.4; 467 11.3 2.96 3.4; 468 11.3 2.22 3.4; 469 11.3 1.48 3.4;

```
470 11.3 0.44 3.4; 471 11.3 6.66 4.2; 472 11.3 7.4 4.2; 473 11.3 5.92 4.2;
474 11.3 5.18 4.2; 475 11.3 4.44 4.2; 476 11.3 3.7 4.2; 477 11.3 2.96 4.2;
478 11.3 2.22 4.2; 479 11.3 1.48 4.2; 480 11.3 0.44 4.2; 481 11.3 6.66 5;
482 11.3 7.4 5; 483 11.3 5.92 5; 484 11.3 5.18 5; 485 11.3 4.44 5;
486 11.3 3.7 5; 487 11.3 2.96 5; 488 11.3 2.22 5; 489 11.3 1.48 5;
490 11.3 0.44 5; 491 11.3 6.66 5.8; 492 11.3 7.4 5.8; 493 11.3 5.92 5.8;
494 11.3 5.18 5.8; 495 11.3 4.44 5.8; 496 11.3 3.7 5.8; 497 11.3 2.96 5.8;
498 11.3 2.22 5.8; 499 11.3 1.48 5.8; 500 11.3 0.44 5.8; 501 11.3 6.66 6.6;
502 11.3 7.4 6.6; 503 11.3 5.92 6.6; 504 11.3 5.18 6.6; 505 11.3 4.44 6.6;
506 11.3 3.7 6.6; 507 11.3 2.96 6.6; 508 11.3 2.22 6.6; 509 11.3 1.48 6.6;
510 11.3 0.44 6.6; 511 11.3 6.66 7.4; 512 11.3 7.4 7.4; 513 11.3 5.92 7.4;
514 11.3 5.18 7.4; 515 11.3 4.44 7.4; 516 11.3 3.7 7.4; 517 11.3 2.96 7.4;
518 11.3 2.22 7.4; 519 11.3 1.48 7.4; 520 11.3 0.44 7.4; 521 11.3 6.66 8.5;
522 11.3 7.4 8.5; 523 11.3 5.92 8.5; 524 11.3 5.18 8.5; 525 11.3 4.44 8.5;
526 11.3 3.7 8.5; 527 11.3 2.96 8.5; 528 11.3 2.22 8.5; 529 11.3 1.48 8.5;
530 11.3 0.44 8.5; 531 0 0 10.7; 532 11.3 0 9.6; 533 0 0 -0.6; 534 11.3 0 0.4;
535 0 0 9.8; 536 1.13 0 10.59; 537 2.26 0 10.48; 538 3.39 0 10.37;
539 4.52 0 10.26; 540 5.65 0 10.15; 541 6.78 0 10.04; 542 7.91 0 9.93;
543 9.04 0 9.82; 544 10.54 0 9.71; 545 1.13 0 -0.5; 546 2.26 0 -0.4;
547 3.39 0 -0.3; 548 4.52 0 -0.2; 549 5.65 0 -0.1; 550 6.78 0 0;
551 7.91 0 0.1; 552 9.04 0 0.2; 553 10.54 0 0.3; 554 11.3 0 -2; 555 11.3 0 12;
556 13.7 0 -2; 557 13.7 0 12; 558 13.7 0 1; 559 13.7 0 9; 560 13.7 0 1.5;
561 13.7 0 2.6; 562 13.7 0 3.4; 563 13.7 0 4.2; 564 13.7 0 5; 565 13.7 0 5.8;
566 13.7 0 6.6; 567 13.7 0 7.4; 568 13.7 0 8.5; 569 13.7 0 9.6; 570 13.7 0 0.4;
571 11.9 0 1.5; 572 11.9 0 1; 573 11.9 0 2.6; 574 11.9 0 3.4; 575 11.9 0 4.2;
576 11.9 0 5; 577 11.9 0 5.8; 578 11.9 0 6.6; 579 11.9 0 7.4; 580 11.9 0 8.5;
581 11.9 0 9; 582 12.5 0 1.5; 583 12.5 0 1; 584 12.5 0 2.6; 585 12.5 0 3.4;
586 12.5 0 4.2; 587 12.5 0 5; 588 12.5 0 5.8; 589 12.5 0 6.6; 590 12.5 0 7.4;
591 12.5 0 8.5; 592 12.5 0 9; 593 13.1 0 1.5; 594 13.1 0 1; 595 13.1 0 2.6;
596 13.1 0 3.4; 597 13.1 0 4.2; 598 13.1 0 5; 599 13.1 0 5.8; 600 13.1 0 6.6;
601 13.1 0 7.4; 602 13.1 0 8.5; 603 13.1 0 9; 604 11.9 0 9.6; 605 12.5 0 9.6;
606 13.1 0 9.6; 607 11.9 0 0.4; 608 12.5 0 0.4; 609 13.1 0 0.4;
610 11.3 0 -1.4; 611 11.9 0 -1.4; 612 11.9 0 -2; 613 11.3 0 -0.8;
614 11.9 0 -0.8; 615 11.3 0 -0.2; 616 11.9 0 -0.2; 617 12.5 0 -1.4;
618 12.5 0 -2; 619 12.5 0 -0.8; 620 12.5 0 -0.2; 621 13.1 0 -1.4;
622 13.1 0 -2; 623 13.1 0 -0.8; 624 13.1 0 -0.2; 625 13.7 0 -1.4;
626 13.7 0 -0.8; 627 13.7 0 -0.2; 628 11.3 0 10.2; 629 11.9 0 10.2;
630 11.3 0 10.8; 631 11.9 0 10.8; 632 11.3 0 11.4; 633 11.9 0 11.4;
634 11.9 0 12; 635 12.5 0 10.2; 636 12.5 0 10.8; 637 12.5 0 11.4;
638 12.5 0 12; 639 13.1 0 10.2; 640 13.1 0 10.8; 641 13.1 0 11.4;
642 13.1 0 12; 643 13.7 0 10.2; 644 13.7 0 10.8; 645 13.7 0 11.4;
649 18.5 0 10.8; 651 18.5 0 5; 653 18.5 0 -0.8; 654 9.79 6.5115 0.85;
655 9.79 7.255 0.85; 656 9.79 5.788 0.85; 657 9.79 5.0645 0.85;
658 9.79 4.341 0.85; 659 9.79 3.6175 0.85; 660 9.79 2.894 0.85;
661 9.79 2.1705 0.85; 662 9.79 1.447 0.85; 663 9.79 0.4235 0.85;
664 9.79 0 0.85; 665 9.79 6.5115 9.165; 666 9.79 7.255 9.165;
667 9.79 5.788 9.165; 668 9.79 5.0645 9.165; 669 9.79 4.341 9.165;
670 9.79 3.6175 9.165; 671 9.79 2.894 9.165; 672 9.79 2.1705 9.165;
673 9.79 1.447 9.165; 674 9.79 0.4235 9.165; 675 9.79 0 9.165;
676 9.79 0 1.3815; 677 9.79 0 2.513; 678 9.79 0 3.3445; 679 9.79 0 4.176;
680 9.79 0 5.0075; 681 9.79 0 5.839; 682 9.79 0 6.6705; 683 9.79 0 7.502;
684 9.79 0 8.6335; 685 9.79 0 9.765; 686 9.79 0 0.25;
MEMBER INCIDENCES
512 636 2; 513 587 17; 514 619 1; 515 649 4; 516 651 21; 517 653 3;
```

ELEMENT INCIDENCES SHELL
5 106 108 109 110; 6 108 111 112 109; 7 111 113 114 112; 8 113 115 116 114;
9 115 117 118 116; 10 117 119 120 118; 11 119 121 122 120; 12 121 123 124 122;
13 123 125 126 124; 14 125 100 127 126; 15 110 109 128 129; 16 109 112 130 128;
17 112 114 131 130; 18 114 116 132 131; 19 116 118 133 132; 20 118 120 134 133;
21 120 122 135 134; 22 122 124 136 135; 23 124 126 137 136; 24 126 127 138 137;
25 129 128 139 140; 26 128 130 141 139; 27 130 131 142 141; 28 131 132 143 142;
29 132 133 144 143; 30 133 134 145 144; 31 134 135 146 145; 32 135 136 147 146;
33 136 137 148 147; 34 137 138 149 148; 35 140 139 150 151; 36 139 141 152 150;
37 141 142 153 152; 38 142 143 154 153; 39 143 144 155 154; 40 144 145 156 155;
41 145 146 157 156; 42 146 147 158 157; 43 147 148 159 158; 44 148 149 160 159;
45 151 150 161 162; 46 150 152 163 161; 47 152 153 164 163; 48 153 154 165 164;
49 154 155 166 165; 50 155 156 167 166; 51 156 157 168 167; 52 157 158 169 168;
53 158 159 170 169; 54 159 160 171 170; 55 162 161 172 173; 56 161 163 174 172;
57 163 164 175 174; 58 164 165 176 175; 59 165 166 177 176; 60 166 167 178 177;
61 167 168 179 178; 62 168 169 180 179; 63 169 170 181 180; 64 170 171 182 181;
65 173 172 183 184; 66 172 174 185 183; 67 174 175 186 185; 68 175 176 187 186;
69 176 177 188 187; 70 177 178 189 188; 71 178 179 190 189; 72 179 180 191 190;
73 180 181 192 191; 74 181 182 193 192; 75 184 183 194 195; 76 183 185 196 194;
77 185 186 197 196; 78 186 187 198 197; 79 187 188 199 198; 80 188 189 200 199;
81 189 190 201 200; 82 190 191 202 201; 83 191 192 203 202; 84 192 193 204 203;
95 206 205 216 104; 96 205 207 217 216; 97 207 208 218 217; 98 208 209 219 218;
99 209 210 220 219; 100 210 211 221 220; 101 211 212 222 221;
102 212 213 223 222; 103 213 214 224 223; 104 214 215 101 224;
105 105 225 226 227; 106 225 228 229 226; 107 228 230 231 229;
108 230 232 233 231; 109 232 234 235 233; 110 234 236 237 235;
111 236 238 239 237; 112 238 240 241 239; 113 240 242 243 241;
114 242 103 244 243; 125 246 245 256 257; 126 245 247 258 256;
127 247 248 259 258; 128 248 249 260 259; 129 249 250 261 260;
130 250 251 262 261; 131 251 252 263 262; 132 252 253 264 263;
133 253 254 265 264; 134 254 255 266 265; 135 257 256 267 268;
136 256 258 269 267; 137 258 259 270 269; 138 259 260 271 270;
139 260 261 272 271; 140 261 262 273 272; 141 262 263 274 273;
142 263 264 275 274; 143 264 265 276 275; 144 265 266 277 276;
145 268 267 278 279; 146 267 269 280 278; 147 269 270 281 280;
148 270 271 282 281; 149 271 272 283 282; 150 272 273 284 283;
151 273 274 285 284; 152 274 275 286 285; 153 275 276 287 286;
154 276 277 288 287; 155 279 278 289 290; 156 278 280 291 289;
157 280 281 292 291; 158 281 282 293 292; 159 282 283 294 293;
160 283 284 295 294; 161 284 285 296 295; 162 285 286 297 296;
163 286 287 298 297; 164 287 288 299 298; 165 290 289 300 301;
166 289 291 302 300; 167 291 292 303 302; 168 292 293 304 303;
169 293 294 305 304; 170 294 295 306 305; 171 295 296 307 306;
172 296 297 308 307; 173 297 298 309 308; 174 298 299 310 309;
175 301 300 311 312; 176 300 302 313 311; 177 302 303 314 313;
178 303 304 315 314; 179 304 305 316 315; 180 305 306 317 316;
181 306 307 318 317; 182 307 308 319 318; 183 308 309 320 319;
184 309 310 321 320; 185 312 311 322 323; 186 311 313 324 322;
187 313 314 325 324; 188 314 315 326 325; 189 315 316 327 326;
190 316 317 328 327; 191 317 318 329 328; 192 318 319 330 329;
193 319 320 331 330; 194 320 321 332 331; 195 323 322 333 107;
196 322 324 334 333; 197 324 325 335 334; 198 325 326 336 335;
199 326 327 337 336; 200 327 328 338 337; 201 328 329 339 338;
202 329 330 340 339; 203 330 331 341 340; 204 331 332 102 341;

205 100 342 343 127; 206 342 344 345 343; 207 344 346 347 345;
208 346 348 349 347; 209 348 350 351 349; 210 350 352 353 351;
211 352 354 355 353; 212 354 356 357 355; 213 356 358 359 357;
214 358 102 332 359; 215 127 343 360 138; 216 343 345 361 360;
217 345 347 362 361; 218 347 349 363 362; 219 349 351 364 363;
220 351 353 365 364; 221 353 355 366 365; 222 355 357 367 366;
223 357 359 368 367; 224 359 332 321 368; 225 138 360 369 149;
226 360 361 370 369; 227 361 362 371 370; 228 362 363 372 371;
229 363 364 373 372; 230 364 365 374 373; 231 365 366 375 374;
232 366 367 376 375; 233 367 368 377 376; 234 368 321 310 377;
235 149 369 378 160; 236 369 370 379 378; 237 370 371 380 379;
238 371 372 381 380; 239 372 373 382 381; 240 373 374 383 382;
241 374 375 384 383; 242 375 376 385 384; 243 376 377 386 385;
244 377 310 299 386; 245 160 378 387 171; 246 378 379 388 387;
247 379 380 389 388; 248 380 381 390 389; 249 381 382 391 390;
250 382 383 392 391; 251 383 384 393 392; 252 384 385 394 393;
253 385 386 395 394; 254 386 299 288 395; 255 171 387 396 182;
256 387 388 397 396; 257 388 389 398 397; 258 389 390 399 398;
259 390 391 400 399; 260 391 392 401 400; 261 392 393 402 401;
262 393 394 403 402; 263 394 395 404 403; 264 395 288 277 404;
265 182 396 405 193; 266 396 397 406 405; 267 397 398 407 406;
268 398 399 408 407; 269 399 400 409 408; 270 400 401 410 409;
271 401 402 411 410; 272 402 403 412 411; 273 403 404 413 412;
274 404 277 266 413; 275 193 405 414 204; 276 405 406 415 414;
277 406 407 416 415; 278 407 408 417 416; 279 408 409 418 417;
280 409 410 419 418; 281 410 411 420 419; 282 411 412 421 420;
283 412 413 422 421; 284 413 266 255 422; 295 215 423 432 101;
296 423 424 433 432; 297 424 425 434 433; 298 425 426 435 434;
299 426 427 436 435; 300 427 428 437 436; 301 428 429 438 437;
302 429 430 439 438; 303 430 431 440 439; 304 431 244 103 440;
305 104 216 441 442; 306 216 217 443 441; 307 217 218 444 443;
308 218 219 445 444; 309 219 220 446 445; 310 220 221 447 446;
311 221 222 448 447; 312 222 223 449 448; 313 223 224 450 449;
314 224 101 432 450; 315 442 441 451 452; 316 441 443 453 451;
317 443 444 454 453; 318 444 445 455 454; 319 445 446 456 455;
320 446 447 457 456; 321 447 448 458 457; 322 448 449 459 458;
323 449 450 460 459; 324 450 432 433 460; 325 452 451 461 462;
326 451 453 463 461; 327 453 454 464 463; 328 454 455 465 464;
329 455 456 466 465; 330 456 457 467 466; 331 457 458 468 467;
332 458 459 469 468; 333 459 460 470 469; 334 460 433 434 470;
335 462 461 471 472; 336 461 463 473 471; 337 463 464 474 473;
338 464 465 475 474; 339 465 466 476 475; 340 466 467 477 476;
341 467 468 478 477; 342 468 469 479 478; 343 469 470 480 479;
344 470 434 435 480; 345 472 471 481 482; 346 471 473 483 481;
347 473 474 484 483; 348 474 475 485 484; 349 475 476 486 485;
350 476 477 487 486; 351 477 478 488 487; 352 478 479 489 488;
353 479 480 490 489; 354 480 435 436 490; 355 482 481 491 492;
356 481 483 493 491; 357 483 484 494 493; 358 484 485 495 494;
359 485 486 496 495; 360 486 487 497 496; 361 487 488 498 497;
362 488 489 499 498; 363 489 490 500 499; 364 490 436 437 500;
365 492 491 501 502; 366 491 493 503 501; 367 493 494 504 503;
368 494 495 505 504; 369 495 496 506 505; 370 496 497 507 506;
371 497 498 508 507; 372 498 499 509 508; 373 499 500 510 509;
374 500 437 438 510; 375 502 501 511 512; 376 501 503 513 511;

377 503 504 514 513; 378 504 505 515 514; 379 505 506 516 515;
380 506 507 517 516; 381 507 508 518 517; 382 508 509 519 518;
383 509 510 520 519; 384 510 438 439 520; 385 512 511 521 522;
386 511 513 523 521; 387 513 514 524 523; 388 514 515 525 524;
389 515 516 526 525; 390 516 517 527 526; 391 517 518 528 527;
392 518 519 529 528; 393 519 520 530 529; 394 520 439 440 530;
395 522 521 225 105; 396 521 523 228 225; 397 523 524 230 228;
398 524 525 232 230; 399 525 526 234 232; 400 526 527 236 234;
401 527 528 238 236; 402 528 529 240 238; 403 529 530 242 240;
404 530 440 103 242; 407 102 531 536 332; 408 332 536 537 321;
409 321 537 538 310; 410 310 538 539 299; 411 299 539 540 288;
412 288 540 541 277; 413 277 541 542 266; 414 266 542 543 255;
416 244 544 532 103; 417 533 100 127 545; 418 545 127 138 546;
419 546 138 149 547; 420 547 149 160 548; 421 548 160 171 549;
422 549 171 182 550; 423 550 182 193 551; 424 551 193 204 552;
426 553 215 101 534; 428 101 432 571 572; 429 432 433 573 571;
430 433 434 574 573; 431 434 435 575 574; 432 435 436 576 575;
433 436 437 577 576; 434 437 438 578 577; 435 438 439 579 578;
436 439 440 580 579; 437 440 103 581 580; 438 572 571 582 583;
439 571 573 584 582; 440 573 574 585 584; 441 574 575 586 585;
442 575 576 587 586; 443 576 577 588 587; 444 577 578 589 588;
445 578 579 590 589; 446 579 580 591 590; 447 580 581 592 591;
448 583 582 593 594; 449 582 584 595 593; 450 584 585 596 595;
451 585 586 597 596; 452 586 587 598 597; 453 587 588 599 598;
454 588 589 600 599; 455 589 590 601 600; 456 590 591 602 601;
457 591 592 603 602; 458 594 593 560 558; 459 593 595 561 560;
460 595 596 562 561; 461 596 597 563 562; 462 597 598 564 563;
463 598 599 565 564; 464 599 600 566 565; 465 600 601 567 566;
466 601 602 568 567; 467 602 603 559 568; 472 103 532 604 581;
473 581 604 605 592; 474 592 605 606 603; 475 603 606 569 559;
476 534 101 572 607; 477 607 572 583 608; 478 608 583 594 609;
479 609 594 558 570; 480 554 610 611 612; 481 610 613 614 611;
482 613 615 616 614; 483 615 534 607 616; 484 612 611 617 618;
485 611 614 619 617; 486 614 616 620 619; 487 616 607 608 620;
488 618 617 621 622; 489 617 619 623 621; 490 619 620 624 623;
491 620 608 609 624; 492 622 621 625 556; 493 621 623 626 625;
494 623 624 627 626; 495 624 609 570 627; 496 532 628 629 604;
497 628 630 631 629; 498 630 632 633 631; 499 632 555 634 633;
500 604 629 635 605; 501 629 631 636 635; 502 631 633 637 636;
503 633 634 638 637; 504 605 635 639 606; 505 635 636 640 639;
506 636 637 641 640; 507 637 638 642 641; 508 606 639 643 569;
509 639 640 644 643; 510 640 641 645 644; 511 641 642 557 645;
518 195 194 654 655; 519 655 654 205 206; 520 194 196 656 654;
521 654 656 207 205; 522 196 197 657 656; 523 656 657 208 207;
524 197 198 658 657; 525 657 658 209 208; 526 198 199 659 658;
527 658 659 210 209; 528 199 200 660 659; 529 659 660 211 210;
530 200 201 661 660; 531 660 661 212 211; 532 201 202 662 661;
533 661 662 213 212; 534 202 203 663 662; 535 662 663 214 213;
536 203 204 664 663; 537 663 664 215 214; 538 227 226 665 666;
539 666 665 245 246; 540 226 229 667 665; 541 665 667 247 245;
542 229 231 668 667; 543 667 668 248 247; 544 231 233 669 668;
545 668 669 249 248; 546 233 235 670 669; 547 669 670 250 249;
548 235 237 671 670; 549 670 671 251 250; 550 237 239 672 671;
551 671 672 252 251; 552 239 241 673 672; 553 672 673 253 252;

```
554 241 243 674 673; 555 673 674 254 253; 556 243 244 675 674;
557 674 675 255 254; 558 204 414 676 664; 559 664 676 423 215;
560 414 415 677 676; 561 676 677 424 423; 562 415 416 678 677;
563 677 678 425 424; 564 416 417 679 678; 565 678 679 426 425;
566 417 418 680 679; 567 679 680 427 426; 568 418 419 681 680;
569 680 681 428 427; 570 419 420 682 681; 571 681 682 429 428;
572 420 421 683 682; 573 682 683 430 429; 574 421 422 684 683;
575 683 684 431 430; 576 422 255 675 684; 577 684 675 244 431;
578 255 543 685 675; 579 675 685 544 244; 580 552 204 664 686;
581 686 664 215 553;
START GROUP DEFINITION
ELEMENT
_FW_BM_700 311 TO 314 321 TO 324 331 TO 334 341 TO 344 351 TO 354 361 TO 364 -
371 TO 374 381 TO 384 391 TO 394 401 TO 404
_FW_BM_550 305 TO 310 315 TO 320 325 TO 330 335 TO 340 345 TO 350 355 TO 360 -
365 TO 370 375 TO 380 385 TO 390 395 TO 400
_SW_BM_700 11 TO 14 21 TO 24 31 TO 34 41 TO 44 51 TO 54 61 TO 64 71 TO 74 -
81 TO 84 101 TO 104 111 TO 114 131 TO 134 141 TO 144 151 TO 154 161 TO 164 -
171 TO 174 181 TO 184 191 TO 194 201 TO 204 530 TO 537 550 TO 557
_SW_BM_550 5 TO 10 15 TO 20 25 TO 30 35 TO 40 45 TO 50 55 TO 60 65 TO 70 75 -
76 TO 80 95 TO 100 105 TO 110 125 TO 130 135 TO 140 145 TO 150 155 TO 160 -
165 TO 170 175 TO 180 185 TO 190 195 TO 200 518 TO 529 538 TO 549
_FW_SF_700 321 TO 323 331 TO 333 341 TO 343 351 TO 353 361 TO 363 371 TO 373 -
381 TO 383 391 TO 393
_FW_SF_550 315 TO 320 325 TO 330 335 TO 340 345 TO 350 355 TO 360 365 TO 370 -
375 TO 380 385 TO 390
_SW_SF_700 10 TO 13 20 TO 23 30 TO 33 40 TO 43 50 TO 53 60 TO 63 70 TO 73 -
80 TO 83 130 TO 133 140 TO 143 150 TO 153 160 TO 163 170 TO 173 180 TO 183 -
190 TO 193 200 TO 203 528 TO 535 548 TO 555
_SW_SF_550 5 TO 9 15 TO 19 25 TO 29 35 TO 39 45 TO 49 55 TO 59 65 TO 69 75 -
76 TO 79 125 TO 129 135 TO 139 145 TO 149 155 TO 159 165 TO 169 175 TO 179 -
185 TO 189 195 TO 199 518 TO 527 538 TO 547
_RAFT_SHEAR 206 TO 213 216 TO 223 226 TO 233 236 TO 243 246 TO 253 -
256 TO 263 266 TO 273 276 TO 283 438 TO 441 444 TO 451 454 TO 467 -
473 TO 475 477 TO 484 487 488 491 TO 500 503 504 507 TO 511 560 TO 575
JOINT
MEMBER
END GROUP DEFINITION
ELEMENT PROPERTY
205 TO 284 295 TO 304 407 TO 414 416 TO 424 426 428 TO 467 472 TO 511 558 -
559 TO 581 THICKNESS 0.75
5 TO 10 15 TO 20 25 TO 30 35 TO 40 45 TO 50 55 TO 60 65 TO 70 75 TO 80 95 -
96 TO 100 105 TO 110 125 TO 130 135 TO 140 145 TO 150 155 TO 160 165 TO 170 -
175 TO 180 185 TO 190 195 TO 200 305 TO 310 315 TO 320 325 TO 330 -
335 TO 340 345 TO 350 355 TO 360 365 TO 370 375 TO 380 385 TO 390 -
395 TO 400 518 TO 529 538 TO 549 THICKNESS 0.55
11 TO 14 21 TO 24 31 TO 34 41 TO 44 51 TO 54 61 TO 64 71 TO 74 81 TO 84 101 -
102 TO 104 111 TO 114 131 TO 134 141 TO 144 151 TO 154 161 TO 164 171 TO 174 -
181 TO 184 191 TO 194 201 TO 204 311 TO 314 321 TO 324 331 TO 334 -
341 TO 344 351 TO 354 361 TO 364 371 TO 374 381 TO 384 391 TO 394 -
401 TO 404 530 TO 537 550 TO 557 THICKNESS 0.75
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+07
```

```
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-05
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
513 516 PRIS YD 0.5 ZD 0.7
MEMBER PROPERTY
512 514 515 517 PRIS YD 0.7 ZD 0.5
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
649 651 653 FIXED
205 TO 284 295 TO 304 407 TO 414 416 TO 424 426 428 TO 467 472 TO 511 558 -
559 TO 581 PLATE MAT DIRECT Y SUBGRADE 12000 PRINT
LOAD 1 DEAD LOAD
SELFWEIGHT Y -1
JOINT LOAD
1 FX 0.9 FY -42 FZ -10.1
2 FX 0.3 FY -34.8 FZ 0
3 FX -0.9 FY -56 FZ -7.3
4 FX 1.5 FY -50.3 FZ 0
17 FX 0 FY -67 FZ 8.3
21 FX -1.8 FY -83.9 FZ 9.1
LOAD 2 LIVE LOAD
JOINT LOAD
1 FX 2.8 FY -14.2 FZ -11
2 FX 1.1 FY -6.7 FZ 0
3 FX -2.9 FY -31.7 FZ -6.1
4 FX 2.2 FY -42.8 FZ 0
17 FX 0.1 FY -41.3 FZ 7.6
21 FX -3.3 FY -56 FZ 9.6
ELEMENT LOAD
8 TO 14 18 TO 24 28 TO 34 38 TO 44 48 TO 54 58 TO 64 68 TO 74 78 TO 84 98 -
99 TO 104 108 TO 114 128 TO 134 138 TO 144 148 TO 154 158 TO 164 168 TO 174 -
178 TO 184 188 TO 194 198 TO 204 308 TO 314 318 TO 324 328 TO 334 -
338 TO 344 348 TO 354 358 TO 364 368 TO 374 378 TO 384 388 TO 394 -
398 TO 404 524 TO 537 544 TO 557 PR -16.6
205 TO 284 295 TO 304 558 TO 577 PR GY -39.3
JOINT LOAD
442 452 462 472 482 492 502 512 522 MZ -150
LOAD 3 SDL1
JOINT LOAD
1 FX -1.7 FY -38 FZ 4.5
2 FX -0.2 FY -51.5 FZ 0
3 FX 0.7 FY -33.7 FZ 5.6
4 FX 1.4 FY -51.3 FZ 0
17 FX 0 FY -151.7 FZ 32.4
21 FX -0.2 FY -92.8 FZ 29.6
LOAD 4 SDL2
JOINT LOAD
1 FX -5.1 FY -255.3 FZ -10.9
```

2 FX 0.7 FY -201.2 FZ -0.1
3 FX 4.3 FY -234 FZ -6.5
4 FX 2.2 FY -214.8 FZ 0
17 FX 0.1 FY -638.6 FZ 34.3
21 FX -2.2 FY -360.4 FZ 33
LOAD 5 WIND LOAD (+X)
JOINT LOAD
1 FX 36.403 FY 94.924 FZ -0.206
2 FX 35.971 FY 96.835 FZ 0.028
3 FX 37.926 FY -94.966 FZ 0.62
4 FX 36.128 FY -96.877 FZ -0.055
17 FX 3.206 FY 3.158 FZ -0.188
21 FX 17.056 FY -3.074 FZ -0.199
LOAD 6 WIND LOAD (-X)
JOINT LOAD
1 FX -36.403 FY -94.924 FZ 0.206
2 FX -35.971 FY -96.835 FZ -0.028
3 FX -37.926 FY 94.966 FZ -0.62
4 FX -36.128 FY 96.877 FZ 0.055
17 FX -3.206 FY -3.158 FZ 0.188
21 FX -17.056 FY 3.074 FZ 0.199
LOAD 7 WIND LOAD (+Z)
JOINT LOAD
1 FX -0.145 FY 62.227 FZ 28.649
2 FX 0.214 FY 0.398 FZ 1.257
3 FX -0.191 FY 52.146 FZ 24.676
4 FX 0.275 FY -1.027 FZ 1.382
17 FX 0.004 FY -62.625 FZ 29.244
21 FX -0.156 FY -51.12 FZ 24.303
LOAD 8 WIND LOAD (-Z)
JOINT LOAD
1 FX 0.145 FY -62.227 FZ -28.649
2 FX -0.214 FY -0.398 FZ -1.257
3 FX 0.191 FY -52.146 FZ -24.676
4 FX -0.275 FY 1.027 FZ -1.382
17 FX -0.004 FY 62.625 FZ -29.244
21 FX 0.156 FY 51.12 FZ -24.303
LOAD 9 TEMP LOAD 1
JOINT LOAD
1 FX -4.19 FY 0.952 FZ -4.008
2 FX -4.329 FY 0.366 FZ 0.527
3 FX 3.452 FY 0.077 FZ -4.353
4 FX 3.064 FY 0.664 FZ 0.511
17 FX -0.223 FY -1.318 FZ 4.008
21 FX 2.225 FY -0.741 FZ 3.315
LOAD 10 TEMP LOAD 2
JOINT LOAD
1 FX 4.19 FY -0.952 FZ 4.008
2 FX 4.329 FY -0.366 FZ -0.527
3 FX -3.452 FY -0.077 FZ 4.353
4 FX -3.064 FY -0.664 FZ -0.511
17 FX 0.223 FY 1.318 FZ -4.008
21 FX -2.225 FY 0.741 FZ -3.315
LOAD 11 EARTH PRESSURE

```
ELEMENT LOAD
5 6 15 16 25 26 35 36 45 46 55 56 65 66 75 76 95 96 105 106 125 126 135 136 -
145 146 155 156 165 166 175 176 185 186 195 196 305 306 315 316 325 326 335 -
336 345 346 355 356 365 366 375 376 385 386 395 396 518 TO 521 538 TO 540 -
541 PR -7.4
7 8 17 18 27 28 37 38 47 48 57 58 67 68 77 78 97 98 107 108 127 128 137 138 -
147 148 157 158 167 168 177 178 187 188 197 198 307 308 317 318 327 328 337 -
338 347 348 357 358 367 368 377 378 387 388 397 398 522 TO 525 542 TO 544 -
545 PR -22.2
9 10 19 20 29 30 39 40 49 50 59 60 69 70 79 80 99 100 109 110 129 130 139 -
140 149 150 159 160 169 170 179 180 189 190 199 200 309 310 319 320 329 330 -
339 340 349 350 359 360 369 370 379 380 389 390 399 400 526 TO 529 -
546 TO 549 PR -37
11 12 21 22 31 32 41 42 51 52 61 62 71 72 81 82 101 102 111 112 131 132 141 -
142 151 152 161 162 171 172 181 182 191 192 201 202 311 312 321 322 331 332 -
341 342 351 352 361 362 371 372 381 382 391 392 401 402 530 TO 533 -
550 TO 553 PR -51.8
13 14 23 24 33 34 43 44 53 54 63 64 73 74 83 84 103 104 113 114 133 134 143 -
144 153 154 163 164 173 174 183 184 193 194 203 204 313 314 323 324 333 334 -
343 344 353 354 363 364 373 374 383 384 393 394 403 404 534 TO 537 -
554 TO 557 PR -66.6
11 12 21 22 31 32 41 42 51 52 61 62 71 72 81 82 101 102 111 112 131 132 141 -
142 151 152 161 162 171 172 181 182 191 192 201 202 311 312 321 322 331 332 -
341 342 351 352 361 362 371 372 381 382 391 392 401 402 530 TO 533 -
550 TO 553 PR 7.4
13 14 23 24 33 34 43 44 53 54 63 64 73 74 83 84 103 104 113 114 133 134 143 -
144 153 154 163 164 173 174 183 184 193 194 203 204 313 314 323 324 333 334 -
343 344 353 354 363 364 373 374 383 384 393 394 403 404 534 TO 537 -
554 TO 557 PR 22
**
ELEMENT LOAD
205 TO 214 407 417 PR GY -127
215 TO 224 408 418 PR GY -129
225 TO 234 409 419 PR GY -132
235 TO 244 410 420 PR GY -134
245 TO 254 411 421 PR GY -136
255 TO 264 412 422 PR GY -138
265 TO 274 413 423 PR GY -140
275 TO 284 414 424 PR GY -143
558 TO 581 PR GY -145
295 TO 304 416 426 PR GY -147
407 TO 414 416 TO 424 426 428 TO 467 472 TO 511 578 TO 581 PR GY -57
LOAD 301 DL+SDL1
REPEAT LOAD
1 1.0 3 1.0
LOAD 302 DL+SDL2
REPEAT LOAD
1 1.0 4 1.0
LOAD 303 DL + SDL1 + TL1 + LL + EP
REPEAT LOAD
1 1.0 3 1.0 9 1.0 2 1.0 11 1.0
LOAD 304 DL + SDL1 + TL2 + LL + EP
REPEAT LOAD
1 1.0 4 1.0 9 1.0 2 1.0 11 1.0
```

```
LOAD 305 DL + SDL2 + TL1 + LL + EP
REPEAT LOAD
1 1.0 3 1.0 10 1.0 2 1.0 11 1.0
LOAD 306 DL + SDL2 + TL2 + LL + EP
REPEAT LOAD
1 1.0 4 1.0 10 1.0 2 1.0 11 1.0
LOAD 307 DL + SDL1 + EP + 0.75LL + 0.75WL(+X)
REPEAT LOAD
1 1.0 3 1.0 11 1.0 2 0.75 5 0.75
LOAD 308 DL + SDL1 + EP + 0.75LL + 0.75WL(-X)
REPEAT LOAD
1 1.0 3 1.0 11 1.0 2 0.75 6 0.75
LOAD 309 DL + SDL1 + EP + 0.75LL + 0.75WL(+Z)
REPEAT LOAD
1 1.0 3 1.0 11 1.0 2 0.75 7 0.75
LOAD 310 DL + SDL1 + EP + 0.75LL + 0.75WL(-Z)
REPEAT LOAD
1 1.0 3 1.0 11 1.0 2 0.75 8 0.75
LOAD 311 DL + SDL2 + EP + 0.75LL + 0.75WL(+X)
REPEAT LOAD
1 1.0 4 1.0 11 1.0 2 0.75 5 0.75
LOAD 312 DL + SDL2 + EP + 0.75LL + 0.75WL(-X)
REPEAT LOAD
1 1.0 4 1.0 11 1.0 2 0.75 6 0.75
LOAD 313 DL + SDL2 + EP + 0.75LL + 0.75WL(+Z)
REPEAT LOAD
1 1.0 4 1.0 11 1.0 2 0.75 7 0.75
LOAD 314 DL + SDL2 + EP + 0.75LL + 0.75WL(-Z)
REPEAT LOAD
1 1.0 4 1.0 11 1.0 2 0.75 8 0.75
LOAD 315 0.6 (DL + SDL1) + WL(+X) + EP
REPEAT LOAD
1 0.6 3 0.6 5 1.0 11 1.0
LOAD 316 0.6 (DL + SDL1) + WL(-X) + EP
REPEAT LOAD
1 0.6 3 0.6 6 1.0 11 1.0
LOAD 317 0.6 (DL + SDL1) + WL(+Z) + EP
REPEAT LOAD
1 0.6 3 0.6 7 1.0 11 1.0
LOAD 318 0.6 (DL + SDL1) + WL(-Z) + EP
REPEAT LOAD
1 0.6 3 0.6 8 1.0 11 1.0
LOAD 319 0.6 (DL + SDL2) + WL(+X) + EP
REPEAT LOAD
1 0.6 4 0.6 5 1.0 11 1.0
LOAD 320 0.6 (DL + SDL2) + WL(-X) + EP
REPEAT LOAD
1 0.6 4 0.6 6 1.0 11 1.0
LOAD 321 0.6 (DL + SDL2) + WL(+Z) + EP
REPEAT LOAD
1 0.6 4 0.6 7 1.0 11 1.0
LOAD 322 0.6 (DL + SDL2) + WL(-Z) + EP
REPEAT LOAD
1 0.6 4 0.6 8 1.0 11 1.0
```

```
****
*Load Combination for Concrete design
LOAD 401 1.4 (DL+SDL1)
REPEAT LOAD
1 1.4 3 1.4
LOAD 402 1.4 (DL+SDL2)
REPEAT LOAD
1 1.4 4 1.4
LOAD 403 1.4 (DL + SDL1 + TL1) + 1.7 LL + 1.7 EP
REPEAT LOAD
1 1.4 3 1.4 9 1.4 2 1.7 11 1.7
LOAD 404 1.4 (DL + SDL1 + TL2) + 1.7 LL  + 1.7 EP
REPEAT LOAD
1 1.4 3 1.4 10 1.4 2 1.7 11 1.7
LOAD 405 1.4 (DL + SDL2 + TL1) + 1.7 LL  + 1.7 EP
REPEAT LOAD
1 1.4 4 1.4 9 1.4 2 1.7 11 1.7
LOAD 406 1.4 (DL + SDL2 + TL2) + 1.7 LL  + 1.7 EP
REPEAT LOAD
1 1.4 4 1.4 10 1.4 2 1.7 11 1.7
LOAD 407 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 5 1.6
LOAD 408 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 6 1.6
LOAD 409 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 7 1.6
LOAD 410 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 8 1.6
LOAD 411 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 5 1.6
LOAD 412 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 6 1.6
LOAD 413 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 7 1.6
LOAD 414 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 8 1.6
LOAD 415 0.9 (DL + SDL1) + 1.6 WL(+X) + 1.6 EP
REPEAT LOAD
1 0.9 3 0.9 5 1.6 11 1.6
LOAD 416 0.9 (DL + SDL1) + 1.6 WL(-X) + 1.6 EP
REPEAT LOAD
1 0.9 3 0.9 6 1.6 11 1.6
LOAD 417 0.9 (DL + SDL1) + 1.6 WL(+Z) + 1.6 EP
REPEAT LOAD
1 0.9 3 0.9 7 1.6 11 1.6
LOAD 418 0.9 (DL + SDL1) + 1.6 WL(-Z) + 1.6 EP
```

```
REPEAT LOAD
1 0.9 3 0.9 8 1.6 11 1.6
LOAD 419 0.9 (DL + SDL2) + 1.6 WL(+X) + 1.6 EP
REPEAT LOAD
1 0.9 4 0.9 5 1.6 11 1.6
LOAD 420 0.9 (DL + SDL2) + 1.6 WL(-X) + 1.6 EP
REPEAT LOAD
1 0.9 4 0.9 6 1.6 11 1.6
LOAD 421 0.9 (DL + SDL2) + 1.6 WL(+Z) + 1.6 EP
REPEAT LOAD
1 0.9 4 0.9 7 1.6 11 1.6
LOAD 422 0.9 (DL + SDL2) + 1.6 WL(-Z) + 1.6 EP
REPEAT LOAD
1 0.9 4 0.9 8 1.6 11 1.6
PERFORM ANALYSIS
FINISH
```

PAGE NO. 1

```
**************************************************
*                                                *
*           STAAD.Pro CONNECT Edition            *
*           Version  22.00.00.15                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    OCT 30, 2020                *
*           Time=    22:20:19                    *
*                                                *
*  Licensed to:                                  *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\F Drive\Maaden Gold\ERH hopper\Civil\Staad\RAMP Part-1.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 20-OCT-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1 12.5 3.55 -0.8; 2 12.5 3.55 10.8; 3 18.5 3.55 -0.8; 4 18.5 3.55 10.8
     9. 17 12.5 3.55 5; 21 18.5 3.55 5; 100 0 0 0; 101 11.3 0 1; 102 0 0 10.1
    10. 103 11.3 0 9; 104 11.3 7.4 1; 105 11.3 7.4 9; 106 0 6.3 0; 107 0 6.3 10.1
    11. 108 0 5.67 0; 109 1.13 5.769 0.1; 110 1.13 6.41 0.1; 111 0 5.04 0
    12. 112 1.13 5.128 0.1; 113 0 4.41 0; 114 1.13 4.487 0.1; 115 0 3.78 0
    13. 116 1.13 3.846 0.1; 117 0 3.15 0; 118 1.13 3.205 0.1; 119 0 2.52 0
    14. 120 1.13 2.564 0.1; 121 0 1.89 0; 122 1.13 1.923 0.1; 123 0 1.26 0
    15. 124 1.13 1.282 0.1; 125 0 0.33 0; 126 1.13 0.341 0.1; 127 1.13 0 0.1
    16. 128 2.26 5.868 0.2; 129 2.26 6.52 0.2; 130 2.26 5.216 0.2; 131 2.26 4.564 0.2
    17. 132 2.26 3.912 0.2; 133 2.26 3.26 0.2; 134 2.26 2.608 0.2; 135 2.26 1.956 0.2
    18. 136 2.26 1.304 0.2; 137 2.26 0.352 0.2; 138 2.26 0 0.2; 139 3.39 5.967 0.3
    19. 140 3.39 6.63 0.3; 141 3.39 5.304 0.3; 142 3.39 4.641 0.3; 143 3.39 3.978 0.3
    20. 144 3.39 3.315 0.3; 145 3.39 2.652 0.3; 146 3.39 1.989 0.3; 147 3.39 1.326 0.3
    21. 148 3.39 0.363 0.3; 149 3.39 0 0.3; 150 4.52 6.066 0.4; 151 4.52 6.74 0.4
    22. 152 4.52 5.392 0.4; 153 4.52 4.718 0.4; 154 4.52 4.044 0.4; 155 4.52 3.37 0.4
    23. 156 4.52 2.696 0.4; 157 4.52 2.022 0.4; 158 4.52 1.348 0.4; 159 4.52 0.374 0.4
    24. 160 4.52 0 0.4; 161 5.65 6.165 0.5; 162 5.65 6.85 0.5; 163 5.65 5.48 0.5
    25. 164 5.65 4.795 0.5; 165 5.65 4.11 0.5; 166 5.65 3.425 0.5; 167 5.65 2.74 0.5
    26. 168 5.65 2.055 0.5; 169 5.65 1.37 0.5; 170 5.65 0.385 0.5; 171 5.65 0 0.5
    27. 172 6.78 6.264 0.6; 173 6.78 6.96 0.6; 174 6.78 5.568 0.6; 175 6.78 4.872 0.6
    28. 176 6.78 4.176 0.6; 177 6.78 3.48 0.6; 178 6.78 2.784 0.6; 179 6.78 2.088 0.6
    29. 180 6.78 1.392 0.6; 181 6.78 0.396 0.6; 182 6.78 0 0.6; 183 7.91 6.363 0.7
    30. 184 7.91 7.07 0.7; 185 7.91 5.656 0.7; 186 7.91 4.949 0.7; 187 7.91 4.242 0.7
    31. 188 7.91 3.535 0.7; 189 7.91 2.828 0.7; 190 7.91 2.121 0.7; 191 7.91 1.414 0.7
    32. 192 7.91 0.407 0.7; 193 7.91 0 0.7; 194 9.04 6.462 0.8; 195 9.04 7.18 0.8
    33. 196 9.04 5.744 0.8; 197 9.04 5.026 0.8; 198 9.04 4.308 0.8; 199 9.04 3.59 0.8
    34. 200 9.04 2.872 0.8; 201 9.04 2.154 0.8; 202 9.04 1.436 0.8; 203 9.04 0.418 0.8
    35. 204 9.04 0 0.8; 205 10.54 6.561 0.9; 206 10.54 7.33 0.9; 207 10.54 5.832 0.9
    36. 208 10.54 5.103 0.9; 209 10.54 4.374 0.9; 210 10.54 3.645 0.9
    37. 211 10.54 2.916 0.9; 212 10.54 2.187 0.9; 213 10.54 1.458 0.9
    38. 214 10.54 0.429 0.9; 215 10.54 0 0.9; 216 11.3 6.66 1; 217 11.3 5.92 1
```

39. 218 11.3 5.18 1; 219 11.3 4.44 1; 220 11.3 3.7 1; 221 11.3 2.96 1
40. 222 11.3 2.22 1; 223 11.3 1.48 1; 224 11.3 0.44 1; 225 11.3 6.66 9
41. 226 10.54 6.561 9.11; 227 10.54 7.33 9.11; 228 11.3 5.92 9
42. 229 10.54 5.832 9.11; 230 11.3 5.18 9; 231 10.54 5.103 9.11; 232 11.3 4.44 9
43. 233 10.54 4.374 9.11; 234 11.3 3.7 9; 235 10.54 3.645 9.11; 236 11.3 2.96 9
44. 237 10.54 2.916 9.11; 238 11.3 2.22 9; 239 10.54 2.187 9.11; 240 11.3 1.48 9
45. 241 10.54 1.458 9.11; 242 11.3 0.44 9; 243 10.54 0.429 9.11; 244 10.54 0 9.11
46. 245 9.04 6.462 9.22; 246 9.04 7.18 9.22; 247 9.04 5.744 9.22
47. 248 9.04 5.026 9.22; 249 9.04 4.308 9.22; 250 9.04 3.59 9.22
48. 251 9.04 2.872 9.22; 252 9.04 2.154 9.22; 253 9.04 1.436 9.22
49. 254 9.04 0.418 9.22; 255 9.04 0 9.22; 256 7.91 6.363 9.33; 257 7.91 7.07 9.33
50. 258 7.91 5.656 9.33; 259 7.91 4.949 9.33; 260 7.91 4.242 9.33
51. 261 7.91 3.535 9.33; 262 7.91 2.828 9.33; 263 7.91 2.121 9.33
52. 264 7.91 1.414 9.33; 265 7.91 0.407 9.33; 266 7.91 0 9.33; 267 6.78 6.264 9.44
53. 268 6.78 6.96 9.44; 269 6.78 5.568 9.44; 270 6.78 4.872 9.44
54. 271 6.78 4.176 9.44; 272 6.78 3.48 9.44; 273 6.78 2.784 9.44
55. 274 6.78 2.088 9.44; 275 6.78 1.392 9.44; 276 6.78 0.396 9.44; 277 6.78 0 9.44
56. 278 5.65 6.165 9.55; 279 5.65 6.85 9.55; 280 5.65 5.48 9.55
57. 281 5.65 4.795 9.55; 282 5.65 4.11 9.55; 283 5.65 3.425 9.55
58. 284 5.65 2.74 9.55; 285 5.65 2.055 9.55; 286 5.65 1.37 9.55
59. 287 5.65 0.385 9.55; 288 5.65 0 9.55; 289 4.52 6.066 9.66; 290 4.52 6.74 9.66
60. 291 4.52 5.392 9.66; 292 4.52 4.718 9.66; 293 4.52 4.044 9.66
61. 294 4.52 3.37 9.66; 295 4.52 2.696 9.66; 296 4.52 2.022 9.66
62. 297 4.52 1.348 9.66; 298 4.52 0.374 9.66; 299 4.52 0 9.66; 300 3.39 5.967 9.77
63. 301 3.39 6.63 9.77; 302 3.39 5.304 9.77; 303 3.39 4.641 9.77
64. 304 3.39 3.978 9.77; 305 3.39 3.315 9.77; 306 3.39 2.652 9.77
65. 307 3.39 1.989 9.77; 308 3.39 1.326 9.77; 309 3.39 0.363 9.77; 310 3.39 0 9.77
66. 311 2.26 5.868 9.88; 312 2.26 6.52 9.88; 313 2.26 5.216 9.88
67. 314 2.26 4.564 9.88; 315 2.26 3.912 9.88; 316 2.26 3.26 9.88
68. 317 2.26 2.608 9.88; 318 2.26 1.956 9.88; 319 2.26 1.304 9.88
69. 320 2.26 0.352 9.88; 321 2.26 0 9.88; 322 1.13 5.769 9.99; 323 1.13 6.41 9.99
70. 324 1.13 5.128 9.99; 325 1.13 4.487 9.99; 326 1.13 3.846 9.99
71. 327 1.13 3.205 9.99; 328 1.13 2.564 9.99; 329 1.13 1.923 9.99
72. 330 1.13 1.282 9.99; 331 1.13 0.341 9.99; 332 1.13 0 9.99
73. 333 -1.30773E-06 5.67 10.1; 334 -1.26839E-06 5.04 10.1
74. 335 -1.22905E-06 4.41 10.1; 336 -1.18971E-06 3.78 10.1
75. 337 -1.15037E-06 3.15 10.1; 338 -1.11103E-06 2.52 10.1
76. 339 -1.07169E-06 1.89 10.1; 340 -1.03235E-06 1.26 10.1
77. 341 -9.93013E-07 0.33 10.1; 342 0 0 0.71; 343 1.13 0 0.789; 344 0 0 2.02
78. 345 1.13 0 2.078; 346 0 0 3.03; 347 1.13 0 3.067; 348 0 0 4.04
79. 349 1.13 0 4.056; 350 0 0 5.05; 351 1.13 0 5.045; 352 0 0 6.06
80. 353 1.13 0 6.034; 354 0 0 7.07; 355 1.13 0 7.023; 356 0 0 8.08
81. 357 1.13 0 8.012; 358 0 0 9.39; 359 1.13 0 9.301; 360 2.26 0 0.868
82. 361 2.26 0 2.136; 362 2.26 0 3.104; 363 2.26 0 4.072; 364 2.26 0 5.04
83. 365 2.26 0 6.008; 366 2.26 0 6.976; 367 2.26 0 7.944; 368 2.26 0 9.212
84. 369 3.39 0 0.947; 370 3.39 0 2.194; 371 3.39 0 3.141; 372 3.39 0 4.088
85. 373 3.39 0 5.035; 374 3.39 0 5.982; 375 3.39 0 6.929; 376 3.39 0 7.876
86. 377 3.39 0 9.123; 378 4.52 0 1.026; 379 4.52 0 2.252; 380 4.52 0 3.178
87. 381 4.52 0 4.104; 382 4.52 0 5.03; 383 4.52 0 5.956; 384 4.52 0 6.882
88. 385 4.52 0 7.808; 386 4.52 0 9.034; 387 5.65 0 1.105; 388 5.65 0 2.31
89. 389 5.65 0 3.215; 390 5.65 0 4.12; 391 5.65 0 5.025; 392 5.65 0 5.93
90. 393 5.65 0 6.835; 394 5.65 0 7.74; 395 5.65 0 8.945; 396 6.78 0 1.184
91. 397 6.78 0 2.368; 398 6.78 0 3.252; 399 6.78 0 4.136; 400 6.78 0 5.02
92. 401 6.78 0 5.904; 402 6.78 0 6.788; 403 6.78 0 7.672; 404 6.78 0 8.856
93. 405 7.91 0 1.263; 406 7.91 0 2.426; 407 7.91 0 3.289; 408 7.91 0 4.152
94. 409 7.91 0 5.015; 410 7.91 0 5.878; 411 7.91 0 6.741; 412 7.91 0 7.604

STAAD SPACE -- PAGE NO. 3

95. 413 7.91 0 8.767; 414 9.04 0 1.342; 415 9.04 0 2.484; 416 9.04 0 3.326
96. 417 9.04 0 4.168; 418 9.04 0 5.01; 419 9.04 0 5.852; 420 9.04 0 6.694
97. 421 9.04 0 7.536; 422 9.04 0 8.678; 423 10.54 0 1.421; 424 10.54 0 2.542
98. 425 10.54 0 3.363; 426 10.54 0 4.184; 427 10.54 0 5.005; 428 10.54 0 5.826
99. 429 10.54 0 6.647; 430 10.54 0 7.468; 431 10.54 0 8.589; 432 11.3 0 1.5
100. 433 11.3 0 2.6; 434 11.3 0 3.4; 435 11.3 0 4.2; 436 11.3 0 5; 437 11.3 0 5.8
101. 438 11.3 0 6.6; 439 11.3 0 7.4; 440 11.3 0 8.5; 441 11.3 6.66 1.5
102. 442 11.3 7.4 1.5; 443 11.3 5.92 1.5; 444 11.3 5.18 1.5; 445 11.3 4.44 1.5
103. 446 11.3 3.7 1.5; 447 11.3 2.96 1.5; 448 11.3 2.22 1.5; 449 11.3 1.48 1.5
104. 450 11.3 0.44 1.5; 451 11.3 6.66 2.6; 452 11.3 7.4 2.6; 453 11.3 5.92 2.6
105. 454 11.3 5.18 2.6; 455 11.3 4.44 2.6; 456 11.3 3.7 2.6; 457 11.3 2.96 2.6
106. 458 11.3 2.22 2.6; 459 11.3 1.48 2.6; 460 11.3 0.44 2.6; 461 11.3 6.66 3.4
107. 462 11.3 7.4 3.4; 463 11.3 5.92 3.4; 464 11.3 5.18 3.4; 465 11.3 4.44 3.4
108. 466 11.3 3.7 3.4; 467 11.3 2.96 3.4; 468 11.3 2.22 3.4; 469 11.3 1.48 3.4
109. 470 11.3 0.44 3.4; 471 11.3 6.66 4.2; 472 11.3 7.4 4.2; 473 11.3 5.92 4.2
110. 474 11.3 5.18 4.2; 475 11.3 4.44 4.2; 476 11.3 3.7 4.2; 477 11.3 2.96 4.2
111. 478 11.3 2.22 4.2; 479 11.3 1.48 4.2; 480 11.3 0.44 4.2; 481 11.3 6.66 5
112. 482 11.3 7.4 5; 483 11.3 5.92 5; 484 11.3 5.18 5; 485 11.3 4.44 5
113. 486 11.3 3.7 5; 487 11.3 2.96 5; 488 11.3 2.22 5; 489 11.3 1.48 5
114. 490 11.3 0.44 5; 491 11.3 6.66 5.8; 492 11.3 7.4 5.8; 493 11.3 5.92 5.8
115. 494 11.3 5.18 5.8; 495 11.3 4.44 5.8; 496 11.3 3.7 5.8; 497 11.3 2.96 5.8
116. 498 11.3 2.22 5.8; 499 11.3 1.48 5.8; 500 11.3 0.44 5.8; 501 11.3 6.66 6.6
117. 502 11.3 7.4 6.6; 503 11.3 5.92 6.6; 504 11.3 5.18 6.6; 505 11.3 4.44 6.6
118. 506 11.3 3.7 6.6; 507 11.3 2.96 6.6; 508 11.3 2.22 6.6; 509 11.3 1.48 6.6
119. 510 11.3 0.44 6.6; 511 11.3 6.66 7.4; 512 11.3 7.4 7.4; 513 11.3 5.92 7.4
120. 514 11.3 5.18 7.4; 515 11.3 4.44 7.4; 516 11.3 3.7 7.4; 517 11.3 2.96 7.4
121. 518 11.3 2.22 7.4; 519 11.3 1.48 7.4; 520 11.3 0.44 7.4; 521 11.3 6.66 8.5
122. 522 11.3 7.4 8.5; 523 11.3 5.92 8.5; 524 11.3 5.18 8.5; 525 11.3 4.44 8.5
123. 526 11.3 3.7 8.5; 527 11.3 2.96 8.5; 528 11.3 2.22 8.5; 529 11.3 1.48 8.5
124. 530 11.3 0.44 8.5; 531 0 0 10.7; 532 11.3 0 9.6; 533 0 0 -0.6; 534 11.3 0 0.4
125. 535 0 0 9.8; 536 1.13 0 10.59; 537 2.26 0 10.48; 538 3.39 0 10.37
126. 539 4.52 0 10.26; 540 5.65 0 10.15; 541 6.78 0 10.04; 542 7.91 0 9.93
127. 543 9.04 0 9.82; 544 10.54 0 9.71; 545 1.13 0 -0.5; 546 2.26 0 -0.4
128. 547 3.39 0 -0.3; 548 4.52 0 -0.2; 549 5.65 0 -0.1; 550 6.78 0 0
129. 551 7.91 0 0.1; 552 9.04 0 0.2; 553 10.54 0 0.3; 554 11.3 0 -2; 555 11.3 0 12
130. 556 13.7 0 -2; 557 13.7 0 12; 558 13.7 0 1; 559 13.7 0 9; 560 13.7 0 1.5
131. 561 13.7 0 2.6; 562 13.7 0 3.4; 563 13.7 0 4.2; 564 13.7 0 5; 565 13.7 0 5.8
132. 566 13.7 0 6.6; 567 13.7 0 7.4; 568 13.7 0 8.5; 569 13.7 0 9.6; 570 13.7 0 0.4
133. 571 11.9 0 1.5; 572 11.9 0 1; 573 11.9 0 2.6; 574 11.9 0 3.4; 575 11.9 0 4.2
134. 576 11.9 0 5; 577 11.9 0 5.8; 578 11.9 0 6.6; 579 11.9 0 7.4; 580 11.9 0 8.5
135. 581 11.9 0 9; 582 12.5 0 1.5; 583 12.5 0 1; 584 12.5 0 2.6; 585 12.5 0 3.4
136. 586 12.5 0 4.2; 587 12.5 0 5; 588 12.5 0 5.8; 589 12.5 0 6.6; 590 12.5 0 7.4
137. 591 12.5 0 8.5; 592 12.5 0 9; 593 13.1 0 1.5; 594 13.1 0 1; 595 13.1 0 2.6
138. 596 13.1 0 3.4; 597 13.1 0 4.2; 598 13.1 0 5; 599 13.1 0 5.8; 600 13.1 0 6.6
139. 601 13.1 0 7.4; 602 13.1 0 8.5; 603 13.1 0 9; 604 11.9 0 9.6; 605 12.5 0 9.6
140. 606 13.1 0 9.6; 607 11.9 0 0.4; 608 12.5 0 0.4; 609 13.1 0 0.4
141. 610 11.3 0 -1.4; 611 11.9 0 -1.4; 612 11.9 0 -2; 613 11.3 0 -0.8
142. 614 11.9 0 -0.8; 615 11.3 0 -0.2; 616 11.9 0 -0.2; 617 12.5 0 -1.4
143. 618 12.5 0 -2; 619 12.5 0 -0.8; 620 12.5 0 -0.2; 621 13.1 0 -1.4
144. 622 13.1 0 -2; 623 13.1 0 -0.8; 624 13.1 0 -0.2; 625 13.7 0 -1.4
145. 626 13.7 0 -0.8; 627 13.7 0 -0.2; 628 11.3 0 10.2; 629 11.9 0 10.2
146. 630 11.3 0 10.8; 631 11.9 0 10.8; 632 11.3 0 11.4; 633 11.9 0 11.4
147. 634 11.9 0 12; 635 12.5 0 10.2; 636 12.5 0 10.8; 637 12.5 0 11.4
148. 638 12.5 0 12; 639 13.1 0 10.2; 640 13.1 0 10.8; 641 13.1 0 11.4
149. 642 13.1 0 12; 643 13.7 0 10.2; 644 13.7 0 10.8; 645 13.7 0 11.4
150. 649 18.5 0 10.8; 651 18.5 0 5; 653 18.5 0 -0.8; 654 9.79 6.5115 0.85

```
151. 655 9.79 7.255 0.85; 656 9.79 5.788 0.85; 657 9.79 5.0645 0.85
152. 658 9.79 4.341 0.85; 659 9.79 3.6175 0.85; 660 9.79 2.894 0.85
153. 661 9.79 2.1705 0.85; 662 9.79 1.447 0.85; 663 9.79 0.4235 0.85
154. 664 9.79 0 0.85; 665 9.79 6.5115 9.165; 666 9.79 7.255 9.165
155. 667 9.79 5.788 9.165; 668 9.79 5.0645 9.165; 669 9.79 4.341 9.165
156. 670 9.79 3.6175 9.165; 671 9.79 2.894 9.165; 672 9.79 2.1705 9.165
157. 673 9.79 1.447 9.165; 674 9.79 0.4235 9.165; 675 9.79 0 9.165
158. 676 9.79 0 1.3815; 677 9.79 0 2.513; 678 9.79 0 3.3445; 679 9.79 0 4.176
159. 680 9.79 0 5.0075; 681 9.79 0 5.839; 682 9.79 0 6.6705; 683 9.79 0 7.502
160. 684 9.79 0 8.6335; 685 9.79 0 9.765; 686 9.79 0 0.25
161. MEMBER INCIDENCES
162. 512 636 2; 513 587 17; 514 619 1; 515 649 4; 516 651 21; 517 653 3
163. ELEMENT INCIDENCES SHELL
164. 5 106 108 109 110; 6 108 111 112 109; 7 111 113 114 112; 8 113 115 116 114
165. 9 115 117 118 116; 10 117 119 120 118; 11 119 121 122 120; 12 121 123 124 122
166. 13 123 125 126 124; 14 125 100 127 126; 15 110 109 128 129; 16 109 112 130 128
167. 17 112 114 131 130; 18 114 116 132 131; 19 116 118 133 132; 20 118 120 134 133
168. 21 120 122 135 134; 22 122 124 136 135; 23 124 126 137 136; 24 126 127 138 137
169. 25 129 128 139 140; 26 128 130 141 139; 27 130 131 142 141; 28 131 132 143 142
170. 29 132 133 144 143; 30 133 134 145 144; 31 134 135 146 145; 32 135 136 147 146
171. 33 136 137 148 147; 34 137 138 149 148; 35 140 139 150 151; 36 139 141 152 150
172. 37 141 142 153 152; 38 142 143 154 153; 39 143 144 155 154; 40 144 145 156 155
173. 41 145 146 157 156; 42 146 147 158 157; 43 147 148 159 158; 44 148 149 160 159
174. 45 151 150 161 162; 46 150 152 163 161; 47 152 153 164 163; 48 153 154 165 164
175. 49 154 155 166 165; 50 155 156 167 166; 51 156 157 168 167; 52 157 158 169 168
176. 53 158 159 170 169; 54 159 160 171 170; 55 162 161 172 173; 56 161 163 174 172
177. 57 163 164 175 174; 58 164 165 176 175; 59 165 166 177 176; 60 166 167 178 177
178. 61 167 168 179 178; 62 168 169 180 179; 63 169 170 181 180; 64 170 171 182 181
179. 65 173 172 183 184; 66 172 174 185 183; 67 174 175 186 185; 68 175 176 187 186
180. 69 176 177 188 187; 70 177 178 189 188; 71 178 179 190 189; 72 179 180 191 190
181. 73 180 181 192 191; 74 181 182 193 192; 75 184 183 194 195; 76 183 185 196 194
182. 77 185 186 197 196; 78 186 187 198 197; 79 187 188 199 198; 80 188 189 200 199
183. 81 189 190 201 200; 82 190 191 202 201; 83 191 192 203 202; 84 192 193 204 203
184. 95 206 205 216 104; 96 205 207 217 216; 97 207 208 218 217; 98 208 209 219 218
185. 99 209 210 220 219; 100 210 211 221 220; 101 211 212 222 221
186. 102 212 213 223 222; 103 213 214 224 223; 104 214 215 101 224
187. 105 105 225 226 227; 106 225 228 229 226; 107 228 230 231 229
188. 108 230 232 233 231; 109 232 234 235 233; 110 234 236 237 235
189. 111 236 238 239 237; 112 238 240 241 239; 113 240 242 243 241
190. 114 242 103 244 243; 125 246 245 256 257; 126 245 247 258 256
191. 127 247 248 259 258; 128 248 249 260 259; 129 249 250 261 260
192. 130 250 251 262 261; 131 251 252 263 262; 132 252 253 264 263
193. 133 253 254 265 264; 134 254 255 266 265; 135 257 256 267 268
194. 136 256 258 269 267; 137 258 259 270 269; 138 259 260 271 270
195. 139 260 261 272 271; 140 261 262 273 272; 141 262 263 274 273
196. 142 263 264 275 274; 143 264 265 276 275; 144 265 266 277 276
197. 145 268 267 278 279; 146 267 269 280 278; 147 269 270 281 280
198. 148 270 271 282 281; 149 271 272 283 282; 150 272 273 284 283
199. 151 273 274 285 284; 152 274 275 286 285; 153 275 276 287 286
200. 154 276 277 288 287; 155 279 278 289 290; 156 278 280 291 289
201. 157 280 281 292 291; 158 281 282 293 292; 159 282 283 294 293
202. 160 283 284 295 294; 161 284 285 296 295; 162 285 286 297 296
203. 163 286 287 298 297; 164 287 288 299 298; 165 290 289 300 301
204. 166 289 291 302 300; 167 291 292 303 302; 168 292 293 304 303
205. 169 293 294 305 304; 170 294 295 306 305; 171 295 296 307 306
206. 172 296 297 308 307; 173 297 298 309 308; 174 298 299 310 309
```

207. 175 301 300 311 312; 176 300 302 313 311; 177 302 303 314 313
208. 178 303 304 315 314; 179 304 305 316 315; 180 305 306 317 316
209. 181 306 307 318 317; 182 307 308 319 318; 183 308 309 320 319
210. 184 309 310 321 320; 185 312 311 322 323; 186 311 313 324 322
211. 187 313 314 325 324; 188 314 315 326 325; 189 315 316 327 326
212. 190 316 317 328 327; 191 317 318 329 328; 192 318 319 330 329
213. 193 319 320 331 330; 194 320 321 332 331; 195 323 322 333 107
214. 196 322 324 334 333; 197 324 325 335 334; 198 325 326 336 335
215. 199 326 327 337 336; 200 327 328 338 337; 201 328 329 339 338
216. 202 329 330 340 339; 203 330 331 341 340; 204 331 332 102 341
217. 205 100 342 343 127; 206 342 344 345 343; 207 344 346 347 345
218. 208 346 348 349 347; 209 348 350 351 349; 210 350 352 353 351
219. 211 352 354 355 353; 212 354 356 357 355; 213 356 358 359 357
220. 214 358 102 332 359; 215 127 343 360 138; 216 343 345 361 360
221. 217 345 347 362 361; 218 347 349 363 362; 219 349 351 364 363
222. 220 351 353 365 364; 221 353 355 366 365; 222 355 357 367 366
223. 223 357 359 368 367; 224 359 332 321 368; 225 138 360 369 149
224. 226 360 361 370 369; 227 361 362 371 370; 228 362 363 372 371
225. 229 363 364 373 372; 230 364 365 374 373; 231 365 366 375 374
226. 232 366 367 376 375; 233 367 368 377 376; 234 368 321 310 377
227. 235 149 369 378 160; 236 369 370 379 378; 237 370 371 380 379
228. 238 371 372 381 380; 239 372 373 382 381; 240 373 374 383 382
229. 241 374 375 384 383; 242 375 376 385 384; 243 376 377 386 385
230. 244 377 310 299 386; 245 160 378 387 171; 246 378 379 388 387
231. 247 379 380 389 388; 248 380 381 390 389; 249 381 382 391 390
232. 250 382 383 392 391; 251 383 384 393 392; 252 384 385 394 393
233. 253 385 386 395 394; 254 386 299 288 395; 255 171 387 396 182
234. 256 387 388 397 396; 257 388 389 398 397; 258 389 390 399 398
235. 259 390 391 400 399; 260 391 392 401 400; 261 392 393 402 401
236. 262 393 394 403 402; 263 394 395 404 403; 264 395 288 277 404
237. 265 182 396 405 193; 266 396 397 406 405; 267 397 398 407 406
238. 268 398 399 408 407; 269 399 400 409 408; 270 400 401 410 409
239. 271 401 402 411 410; 272 402 403 412 411; 273 403 404 413 412
240. 274 404 277 266 413; 275 193 405 414 204; 276 405 406 415 414
241. 277 406 407 416 415; 278 407 408 417 416; 279 408 409 418 417
242. 280 409 410 419 418; 281 410 411 420 419; 282 411 412 421 420
243. 283 412 413 422 421; 284 413 266 255 422; 295 215 423 432 101
244. 296 423 424 433 432; 297 424 425 434 433; 298 425 426 435 434
245. 299 426 427 436 435; 300 427 428 437 436; 301 428 429 438 437
246. 302 429 430 439 438; 303 430 431 440 439; 304 431 244 103 440
247. 305 104 216 441 442; 306 216 217 443 441; 307 217 218 444 443
248. 308 218 219 445 444; 309 219 220 446 445; 310 220 221 447 446
249. 311 221 222 448 447; 312 222 223 449 448; 313 223 224 450 449
250. 314 224 101 432 450; 315 442 441 451 452; 316 441 443 453 451
251. 317 443 444 454 453; 318 444 445 455 454; 319 445 446 456 455
252. 320 446 447 457 456; 321 447 448 458 457; 322 448 449 459 458
253. 323 449 450 460 459; 324 450 432 433 460; 325 452 451 461 462
254. 326 451 453 463 461; 327 453 454 464 463; 328 454 455 465 464
255. 329 455 456 466 465; 330 456 457 467 466; 331 457 458 468 467
256. 332 458 459 469 468; 333 459 460 470 469; 334 460 433 434 470
257. 335 462 461 471 472; 336 461 463 473 471; 337 463 464 474 473
258. 338 464 465 475 474; 339 465 466 476 475; 340 466 467 477 476
259. 341 467 468 478 477; 342 468 469 479 478; 343 469 470 480 479
260. 344 470 434 435 480; 345 472 471 481 482; 346 471 473 483 481
261. 347 473 474 484 483; 348 474 475 485 484; 349 475 476 486 485
262. 350 476 477 487 486; 351 477 478 488 487; 352 478 479 489 488

263. 353 479 480 490 489; 354 480 435 436 490; 355 482 481 491 492
264. 356 481 483 493 491; 357 483 484 494 493; 358 484 485 495 494
265. 359 485 486 496 495; 360 486 487 497 496; 361 487 488 498 497
266. 362 488 489 499 498; 363 489 490 500 499; 364 490 436 437 500
267. 365 492 491 501 502; 366 491 493 503 501; 367 493 494 504 503
268. 368 494 495 505 504; 369 495 496 506 505; 370 496 497 507 506
269. 371 497 498 508 507; 372 498 499 509 508; 373 499 500 510 509
270. 374 500 437 438 510; 375 502 501 511 512; 376 501 503 513 511
271. 377 503 504 514 513; 378 504 505 515 514; 379 505 506 516 515
272. 380 506 507 517 516; 381 507 508 518 517; 382 508 509 519 518
273. 383 509 510 520 519; 384 510 438 439 520; 385 512 511 521 522
274. 386 511 513 523 521; 387 513 514 524 523; 388 514 515 525 524
275. 389 515 516 526 525; 390 516 517 527 526; 391 517 518 528 527
276. 392 518 519 529 528; 393 519 520 530 529; 394 520 439 440 530
277. 395 522 521 225 105; 396 521 523 228 225; 397 523 524 230 228
278. 398 524 525 232 230; 399 525 526 234 232; 400 526 527 236 234
279. 401 527 528 238 236; 402 528 529 240 238; 403 529 530 242 240
280. 404 530 440 103 242; 407 102 531 536 332; 408 332 536 537 321
281. 409 321 537 538 310; 410 310 538 539 299; 411 299 539 540 288
282. 412 288 540 541 277; 413 277 541 542 266; 414 266 542 543 255
283. 416 244 544 532 103; 417 533 100 127 545; 418 545 127 138 546
284. 419 546 138 149 547; 420 547 149 160 548; 421 548 160 171 549
285. 422 549 171 182 550; 423 550 182 193 551; 424 551 193 204 552
286. 426 553 215 101 534; 428 101 432 571 572; 429 432 433 573 571
287. 430 433 434 574 573; 431 434 435 575 574; 432 435 436 576 575
288. 433 436 437 577 576; 434 437 438 578 577; 435 438 439 579 578
289. 436 439 440 580 579; 437 440 103 581 580; 438 572 571 582 583
290. 439 571 573 584 582; 440 573 574 585 584; 441 574 575 586 585
291. 442 575 576 587 586; 443 576 577 588 587; 444 577 578 589 588
292. 445 578 579 590 589; 446 579 580 591 590; 447 580 581 592 591
293. 448 583 582 593 594; 449 582 584 595 593; 450 584 585 596 595
294. 451 585 586 597 596; 452 586 587 598 597; 453 587 588 599 598
295. 454 588 589 600 599; 455 589 590 601 600; 456 590 591 602 601
296. 457 591 592 603 602; 458 594 593 560 558; 459 593 595 561 560
297. 460 595 596 562 561; 461 596 597 563 562; 462 597 598 564 563
298. 463 598 599 565 564; 464 599 600 566 565; 465 600 601 567 566
299. 466 601 602 568 567; 467 602 603 559 568; 472 103 532 604 581
300. 473 581 604 605 592; 474 592 605 606 603; 475 603 606 569 559
301. 476 534 101 572 607; 477 607 572 583 608; 478 608 583 594 609
302. 479 609 594 558 570; 480 554 610 611 612; 481 610 613 614 611
303. 482 613 615 616 614; 483 615 534 607 616; 484 612 611 617 618
304. 485 611 614 619 617; 486 614 616 620 619; 487 616 607 608 620
305. 488 618 617 621 622; 489 617 619 623 621; 490 619 620 624 623
306. 491 620 608 609 624; 492 622 621 625 556; 493 621 623 626 625
307. 494 623 624 627 626; 495 624 609 570 627; 496 532 628 629 604
308. 497 628 630 631 629; 498 630 632 633 631; 499 632 555 634 633
309. 500 604 629 635 605; 501 629 631 636 635; 502 631 633 637 636
310. 503 633 634 638 637; 504 605 635 639 606; 505 635 636 640 639
311. 506 636 637 641 640; 507 637 638 642 641; 508 606 639 643 569
312. 509 639 640 644 643; 510 640 641 645 644; 511 641 642 557 645
313. 518 195 194 654 655; 519 655 654 205 206; 520 194 196 656 654
314. 521 654 656 207 205; 522 196 197 657 656; 523 656 657 208 207
315. 524 197 198 658 657; 525 657 658 209 208; 526 198 199 659 658
316. 527 658 659 210 209; 528 199 200 660 659; 529 659 660 211 210
317. 530 200 201 661 660; 531 660 661 212 211; 532 201 202 662 661
318. 533 661 662 213 212; 534 202 203 663 662; 535 662 663 214 213

```
319. 536 203 204 664 663; 537 663 664 215 214; 538 227 226 665 666
320. 539 666 665 245 246; 540 226 229 667 665; 541 665 667 247 245
321. 542 229 231 668 667; 543 667 668 248 247; 544 231 233 669 668
322. 545 668 669 249 248; 546 233 235 670 669; 547 669 670 250 249
323. 548 235 237 671 670; 549 670 671 251 250; 550 237 239 672 671
324. 551 671 672 252 251; 552 239 241 673 672; 553 672 673 253 252
325. 554 241 243 674 673; 555 673 674 254 253; 556 243 244 675 674
326. 557 674 675 255 254; 558 204 414 676 664; 559 664 676 423 215
327. 560 414 415 677 676; 561 676 677 424 423; 562 415 416 678 677
328. 563 677 678 425 424; 564 416 417 679 678; 565 678 679 426 425
329. 566 417 418 680 679; 567 679 680 427 426; 568 418 419 681 680
330. 569 680 681 428 427; 570 419 420 682 681; 571 681 682 429 428
331. 572 420 421 683 682; 573 682 683 430 429; 574 421 422 684 683
332. 575 683 684 431 430; 576 422 255 675 684; 577 684 675 244 431
333. 578 255 543 685 675; 579 675 685 544 244; 580 552 204 664 686
334. 581 686 664 215 553
335. START GROUP DEFINITION
336. ELEMENT
337. _FW_BM_700 311 TO 314 321 TO 324 331 TO 334 341 TO 344 351 TO 354 361 TO 364 -
338. 371 TO 374 381 TO 384 391 TO 394 401 TO 404
339. _FW_BM_550 305 TO 310 315 TO 320 325 TO 330 335 TO 340 345 TO 350 355 TO 360 -
340. 365 TO 370 375 TO 380 385 TO 390 395 TO 400
341. _SW_BM_700 11 TO 14 21 TO 24 31 TO 34 41 TO 44 51 TO 54 61 TO 64 71 TO 74 -
342. 81 TO 84 101 TO 104 111 TO 114 131 TO 134 141 TO 144 151 TO 154 161 TO 164 -
343. 171 TO 174 181 TO 184 191 TO 194 201 TO 204 530 TO 537 550 TO 557
344. _SW_BM_550 5 TO 10 15 TO 20 25 TO 30 35 TO 40 45 TO 50 55 TO 60 65 TO 70 75 -
345. 76 TO 80 95 TO 100 105 TO 110 125 TO 130 135 TO 140 145 TO 150 155 TO 160 -
346. 165 TO 170 175 TO 180 185 TO 190 195 TO 200 518 TO 529 538 TO 549
347. _FW_SF_700 321 TO 323 331 TO 333 341 TO 343 351 TO 353 361 TO 363 371 TO 373 -
348. 381 TO 383 391 TO 393
349. _FW_SF_550 315 TO 320 325 TO 330 335 TO 340 345 TO 350 355 TO 360 365 TO 370 -
350. 375 TO 380 385 TO 390
351. _SW_SF_700 10 TO 13 20 TO 23 30 TO 33 40 TO 43 50 TO 53 60 TO 63 70 TO 73 -
352. 80 TO 83 130 TO 133 140 TO 143 150 TO 153 160 TO 163 170 TO 173 180 TO 183 -
353. 190 TO 193 200 TO 203 528 TO 535 548 TO 555
354. _SW_SF_550 5 TO 9 15 TO 19 25 TO 29 35 TO 39 45 TO 49 55 TO 59 65 TO 69 75 -
355. 76 TO 79 125 TO 129 135 TO 139 145 TO 149 155 TO 159 165 TO 169 175 TO 179 -
356. 185 TO 189 195 TO 199 518 TO 527 538 TO 547
357. _RAFT_SHEAR 206 TO 213 216 TO 223 226 TO 233 236 TO 243 246 TO 253 -
358. 256 TO 263 266 TO 273 276 TO 283 438 TO 441 444 TO 451 454 TO 467 -
359. 473 TO 475 477 TO 484 487 488 491 TO 500 503 504 507 TO 511 560 TO 575
360. JOINT
361. MEMBER
362. END GROUP DEFINITION
363. ELEMENT PROPERTY
364. 205 TO 284 295 TO 304 407 TO 414 416 TO 424 426 428 TO 467 472 TO 511 558 -
365. 559 TO 581 THICKNESS 0.75
366. 5 TO 10 15 TO 20 25 TO 30 35 TO 40 45 TO 50 55 TO 60 65 TO 70 75 TO 80 95 -
367. 96 TO 100 105 TO 110 125 TO 130 135 TO 140 145 TO 150 155 TO 160 165 TO 170 -
368. 175 TO 180 185 TO 190 195 TO 200 305 TO 310 315 TO 320 325 TO 330 -
369. 335 TO 340 345 TO 350 355 TO 360 365 TO 370 375 TO 380 385 TO 390 -
370. 395 TO 400 518 TO 529 538 TO 549 THICKNESS 0.55
371. 11 TO 14 21 TO 24 31 TO 34 41 TO 44 51 TO 54 61 TO 64 71 TO 74 81 TO 84 101 -
372. 102 TO 104 111 TO 114 131 TO 134 141 TO 144 151 TO 154 161 TO 164 171 TO 174 -
373. 181 TO 184 191 TO 194 201 TO 204 311 TO 314 321 TO 324 331 TO 334 -
374. 341 TO 344 351 TO 354 361 TO 364 371 TO 374 381 TO 384 391 TO 394 -
```

```
375. 401 TO 404 530 TO 537 550 TO 557 THICKNESS 0.75
376. DEFINE MATERIAL START
377. ISOTROPIC CONCRETE
378. E 2.17185E+07
379. POISSON 0.17
380. DENSITY 23.5616
381. ALPHA 1E-05
382. DAMP 0.05
383. TYPE CONCRETE
384. STRENGTH FCU 27579
385. END DEFINE MATERIAL
386. MEMBER PROPERTY AMERICAN
387. 513 516 PRIS YD 0.5 ZD 0.7
388. MEMBER PROPERTY
389. 512 514 515 517 PRIS YD 0.7 ZD 0.5
390. CONSTANTS
391. MATERIAL CONCRETE ALL
392. SUPPORTS
393. 649 651 653 FIXED
394. 205 TO 284 295 TO 304 407 TO 414 416 TO 424 426 428 TO 467 472 TO 511 558 -
395. 559 TO 581 PLATE MAT DIRECT Y SUBGRADE 12000 PRINT
```

JOINT	INFLUENCE AREA
100	0.368592
101	0.374997
102	0.368592
103	0.374998
127	0.728285
138	0.716420
149	0.704555
160	0.692690
171	0.680825
182	0.668960
193	0.657095
204	0.537731
215	0.422672
244	0.422672
255	0.537731
266	0.657095
277	0.668960
288	0.680825
299	0.692690
310	0.704555
321	0.716420
332	0.728285
342	0.567684
343	1.117570
344	0.652434
345	1.287070
346	0.567684
347	1.117570
348	0.567684
349	1.117570
350	0.567684

351	1.117570
352	0.567684
353	1.117570
354	0.567684
355	1.117570
356	0.652434
357	1.287070
358	0.567684
359	1.117570
360	1.093840
361	1.263340
362	1.093840
363	1.093840
364	1.093840
365	1.093840
366	1.093840
367	1.263340
368	1.093840
369	1.070110
370	1.239610
371	1.070110
372	1.070110
373	1.070110
374	1.070110
375	1.070110
376	1.239610
377	1.070110
378	1.046380
379	1.215880
380	1.046380
381	1.046380
382	1.046380
383	1.046380
384	1.046380
385	1.215880
386	1.046380
387	1.022650
388	1.192150
389	1.022650
390	1.022650
391	1.022650
392	1.022650
393	1.022650
394	1.192150
395	1.022650
396	0.998920
397	1.168420
398	0.998920
399	0.998920
400	0.998920
401	0.998920
402	0.998920
403	1.168420
404	0.998920
405	0.975190
406	1.144690

407	0.975190
408	0.975190
409	0.975190
410	0.975190
411	0.975190
412	1.144690
413	0.975190
414	0.793462
415	0.934462
416	0.793462
417	0.793462
418	0.793462
419	0.793462
420	0.793462
421	0.934462
422	0.793462
423	0.618844
424	0.732094
425	0.618844
426	0.618844
427	0.618844
428	0.618844
429	0.618844
430	0.732094
431	0.618844
432	0.545995
433	0.647995
434	0.545995
435	0.545995
436	0.545995
437	0.545995
438	0.545995
439	0.647995
440	0.545995
531	0.169500
532	0.294000
533	0.169500
534	0.294000
536	0.339000
537	0.339000
538	0.339000
539	0.339000
540	0.339000
541	0.339000
542	0.339000
543	0.282000
544	0.226500
545	0.339000
546	0.339000
547	0.339000
548	0.339000
549	0.339000
550	0.339000
551	0.339000
552	0.282000
553	0.226500

554	0.090000
555	0.090000
556	0.090000
557	0.090000
558	0.165000
559	0.165000
560	0.240000
561	0.285000
562	0.240000
563	0.240000
564	0.240000
565	0.240000
566	0.240000
567	0.284999
568	0.240000
569	0.180000
570	0.180000
571	0.480000
572	0.330000
573	0.570000
574	0.480000
575	0.480000
576	0.480000
577	0.480000
578	0.480000
579	0.570000
580	0.480000
581	0.330000
582	0.480000
583	0.330000
584	0.570000
585	0.480000
586	0.480000
587	0.480000
588	0.480000
589	0.480000
590	0.570000
591	0.480000
592	0.330000
593	0.480000
594	0.330000
595	0.570000
596	0.480000
597	0.480000
598	0.480000
599	0.480000
600	0.480000
601	0.570000
602	0.480000
603	0.330000
604	0.360000
605	0.360000
606	0.360000
607	0.360000
608	0.360000
609	0.360000

610	0.180000
611	0.360000
612	0.180000
613	0.180000
614	0.360000
615	0.180000
616	0.360000
617	0.360000
618	0.180000
619	0.360000
620	0.360000
621	0.360000
622	0.180000
623	0.360000
624	0.360000
625	0.180000
626	0.180000
627	0.180000
628	0.180000
629	0.360000
630	0.180000
631	0.360000
632	0.180000
633	0.360000
634	0.180000
635	0.360000
636	0.360000
637	0.360000
638	0.180000
639	0.360000
640	0.360000
641	0.360000
642	0.180000
643	0.180000
644	0.180000
645	0.180000
664	0.424312
675	0.424312
676	0.623625
677	0.736125
678	0.623625
679	0.623625
680	0.623625
681	0.623625
682	0.623625
683	0.736125
684	0.623625
685	0.225000
686	0.225000
TOTAL	149.502609

396. LOAD 1 DEAD LOAD
**WARNING- JOINT NO. 535 NOT CONNECTED. OK, IF PART OF MASTER/SLAVE.
**WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
MASTER/SLAVE OR IF UNCONNECTED JOINTS.
397. SELFWEIGHT Y -1
398. JOINT LOAD

```
399. 1 FX 0.9 FY -42 FZ -10.1
400. 2 FX 0.3 FY -34.8 FZ 0
401. 3 FX -0.9 FY -56 FZ -7.3
402. 4 FX 1.5 FY -50.3 FZ 0
403. 17 FX 0 FY -67 FZ 8.3
404. 21 FX -1.8 FY -83.9 FZ 9.1
405. LOAD 2 LIVE LOAD
406. JOINT LOAD
407. 1 FX 2.8 FY -14.2 FZ -11
408. 2 FX 1.1 FY -6.7 FZ 0
409. 3 FX -2.9 FY -31.7 FZ -6.1
410. 4 FX 2.2 FY -42.8 FZ 0
411. 17 FX 0.1 FY -41.3 FZ 7.6
412. 21 FX -3.3 FY -56 FZ 9.6
413. ELEMENT LOAD
414. 8 TO 14 18 TO 24 28 TO 34 38 TO 44 48 TO 54 58 TO 64 68 TO 74 78 TO 84 98 -
415. 99 TO 104 108 TO 114 128 TO 134 138 TO 144 148 TO 154 158 TO 164 168 TO 174 -
416. 178 TO 184 188 TO 194 198 TO 204 308 TO 314 318 TO 324 328 TO 334 -
417. 338 TO 344 348 TO 354 358 TO 364 368 TO 374 378 TO 384 388 TO 394 -
418. 398 TO 404 524 TO 537 544 TO 557 PR -16.6
419. 205 TO 284 295 TO 304 558 TO 577 PR GY -39.3
420. JOINT LOAD
421. 442 452 462 472 482 492 502 512 522 MZ -150
422. LOAD 3 SDL1
423. JOINT LOAD
424. 1 FX -1.7 FY -38 FZ 4.5
425. 2 FX -0.2 FY -51.5 FZ 0
426. 3 FX 0.7 FY -33.7 FZ 5.6
427. 4 FX 1.4 FY -51.3 FZ 0
428. 17 FX 0 FY -151.7 FZ 32.4
429. 21 FX -0.2 FY -92.8 FZ 29.6
430. LOAD 4 SDL2
431. JOINT LOAD
432. 1 FX -5.1 FY -255.3 FZ -10.9
433. 2 FX 0.7 FY -201.2 FZ -0.1
434. 3 FX 4.3 FY -234 FZ -6.5
435. 4 FX 2.2 FY -214.8 FZ 0
436. 17 FX 0.1 FY -638.6 FZ 34.3
437. 21 FX -2.2 FY -360.4 FZ 33
438. LOAD 5 WIND LOAD (+X)
439. JOINT LOAD
440. 1 FX 36.403 FY 94.924 FZ -0.206
441. 2 FX 35.971 FY 96.835 FZ 0.028
442. 3 FX 37.926 FY -94.966 FZ 0.62
443. 4 FX 36.128 FY -96.877 FZ -0.055
444. 17 FX 3.206 FY 3.158 FZ -0.188
445. 21 FX 17.056 FY -3.074 FZ -0.199
446. LOAD 6 WIND LOAD (-X)
447. JOINT LOAD
448. 1 FX -36.403 FY -94.924 FZ 0.206
449. 2 FX -35.971 FY -96.835 FZ -0.028
450. 3 FX -37.926 FY 94.966 FZ -0.62
451. 4 FX -36.128 FY 96.877 FZ 0.055
452. 17 FX -3.206 FY -3.158 FZ 0.188
453. 21 FX -17.056 FY 3.074 FZ 0.199
454. LOAD 7 WIND LOAD (+Z)
```

```
455. JOINT LOAD
456. 1 FX -0.145 FY 62.227 FZ 28.649
457. 2 FX 0.214 FY 0.398 FZ 1.257
458. 3 FX -0.191 FY 52.146 FZ 24.676
459. 4 FX 0.275 FY -1.027 FZ 1.382
460. 17 FX 0.004 FY -62.625 FZ 29.244
461. 21 FX -0.156 FY -51.12 FZ 24.303
462. LOAD 8 WIND LOAD (-Z)
463. JOINT LOAD
464. 1 FX 0.145 FY -62.227 FZ -28.649
465. 2 FX -0.214 FY -0.398 FZ -1.257
466. 3 FX 0.191 FY -52.146 FZ -24.676
467. 4 FX -0.275 FY 1.027 FZ -1.382
468. 17 FX -0.004 FY 62.625 FZ -29.244
469. 21 FX 0.156 FY 51.12 FZ -24.303
470. LOAD 9 TEMP LOAD 1
471. JOINT LOAD
472. 1 FX -4.19 FY 0.952 FZ -4.008
473. 2 FX -4.329 FY 0.366 FZ 0.527
474. 3 FX 3.452 FY 0.077 FZ -4.353
475. 4 FX 3.064 FY 0.664 FZ 0.511
476. 17 FX -0.223 FY -1.318 FZ 4.008
477. 21 FX 2.225 FY -0.741 FZ 3.315
478. LOAD 10 TEMP LOAD 2
479. JOINT LOAD
480. 1 FX 4.19 FY -0.952 FZ 4.008
481. 2 FX 4.329 FY -0.366 FZ -0.527
482. 3 FX -3.452 FY -0.077 FZ 4.353
483. 4 FX -3.064 FY -0.664 FZ -0.511
484. 17 FX 0.223 FY 1.318 FZ -4.008
485. 21 FX -2.225 FY 0.741 FZ -3.315
486. LOAD 11 EARTH PRESSURE
487. ELEMENT LOAD
488. 5 6 15 16 25 26 35 36 45 46 55 56 65 66 75 76 95 96 105 106 125 126 135 136 -
489. 145 146 155 156 165 166 175 176 185 186 195 196 305 306 315 316 325 326 335 -
490. 336 345 346 355 356 365 366 375 376 385 386 395 396 518 TO 521 538 TO 540 -
491. 541 PR -7.4
492. 7 8 17 18 27 28 37 38 47 48 57 58 67 68 77 78 97 98 107 108 127 128 137 138 -
493. 147 148 157 158 167 168 177 178 187 188 197 198 307 308 317 318 327 328 337 -
494. 338 347 348 357 358 367 368 377 378 387 388 397 398 522 TO 525 542 TO 544 -
495. 545 PR -22.2
496. 9 10 19 20 29 30 39 40 49 50 59 60 69 70 79 80 99 100 109 110 129 130 139 -
497. 140 149 150 159 160 169 170 179 180 189 190 199 200 309 310 319 320 329 330 -
498. 339 340 349 350 359 360 369 370 379 380 389 390 399 400 526 TO 529 -
499. 546 TO 549 PR -37
500. 11 12 21 22 31 32 41 42 51 52 61 62 71 72 81 82 101 102 111 112 131 132 141 -
501. 142 151 152 161 162 171 172 181 182 191 192 201 202 311 312 321 322 331 332 -
502. 341 342 351 352 361 362 371 372 381 382 391 392 401 402 530 TO 533 -
503. 550 TO 553 PR -51.8
504. 13 14 23 24 33 34 43 44 53 54 63 64 73 74 83 84 103 104 113 114 133 134 143 -
505. 144 153 154 163 164 173 174 183 184 193 194 203 204 313 314 323 324 333 334 -
506. 343 344 353 354 363 364 373 374 383 384 393 394 403 404 534 TO 537 -
507. 554 TO 557 PR -66.6
508. 11 12 21 22 31 32 41 42 51 52 61 62 71 72 81 82 101 102 111 112 131 132 141 -
509. 142 151 152 161 162 171 172 181 182 191 192 201 202 311 312 321 322 331 332 -
510. 341 342 351 352 361 362 371 372 381 382 391 392 401 402 530 TO 533 -
```

```
511. 550 TO 553 PR 7.4
512. 13 14 23 24 33 34 43 44 53 54 63 64 73 74 83 84 103 104 113 114 133 134 143 -
513. 144 153 154 163 164 173 174 183 184 193 194 203 204 313 314 323 324 333 334 -
514. 343 344 353 354 363 364 373 374 383 384 393 394 403 404 534 TO 537 -
515. 554 TO 557 PR 22
516. **
517. ELEMENT LOAD
518. 205 TO 214 407 417 PR GY -127
519. 215 TO 224 408 418 PR GY -129
520. 225 TO 234 409 419 PR GY -132
521. 235 TO 244 410 420 PR GY -134
522. 245 TO 254 411 421 PR GY -136
523. 255 TO 264 412 422 PR GY -138
524. 265 TO 274 413 423 PR GY -140
525. 275 TO 284 414 424 PR GY -143
526. 558 TO 581 PR GY -145
527. 295 TO 304 416 426 PR GY -147
528. 407 TO 414 416 TO 424 426 428 TO 467 472 TO 511 578 TO 581 PR GY -57
529. LOAD 301 DL+SDL1
530. REPEAT LOAD
531. 1 1.0 3 1.0
532. LOAD 302 DL+SDL2
533. REPEAT LOAD
534. 1 1.0 4 1.0
535. LOAD 303 DL + SDL1 + TL1 + LL + EP
536. REPEAT LOAD
537. 1 1.0 3 1.0 9 1.0 2 1.0 11 1.0
538. LOAD 304 DL + SDL1 + TL2 + LL + EP
539. REPEAT LOAD
540. 1 1.0 4 1.0 9 1.0 2 1.0 11 1.0
541. LOAD 305 DL + SDL2 + TL1 + LL + EP
542. REPEAT LOAD
543. 1 1.0 3 1.0 10 1.0 2 1.0 11 1.0
544. LOAD 306 DL + SDL2 + TL2 + LL + EP
545. REPEAT LOAD
546. 1 1.0 4 1.0 10 1.0 2 1.0 11 1.0
547. LOAD 307 DL + SDL1 + EP + 0.75LL + 0.75WL(+X)
548. REPEAT LOAD
549. 1 1.0 3 1.0 11 1.0 2 0.75 5 0.75
550. LOAD 308 DL + SDL1 + EP + 0.75LL + 0.75WL(-X)
551. REPEAT LOAD
552. 1 1.0 3 1.0 11 1.0 2 0.75 6 0.75
553. LOAD 309 DL + SDL1 + EP + 0.75LL + 0.75WL(+Z)
554. REPEAT LOAD
555. 1 1.0 3 1.0 11 1.0 2 0.75 7 0.75
556. LOAD 310 DL + SDL1 + EP + 0.75LL + 0.75WL(-Z)
557. REPEAT LOAD
558. 1 1.0 3 1.0 11 1.0 2 0.75 8 0.75
559. LOAD 311 DL + SDL2 + EP + 0.75LL + 0.75WL(+X)
560. REPEAT LOAD
561. 1 1.0 4 1.0 11 1.0 2 0.75 5 0.75
562. LOAD 312 DL + SDL2 + EP + 0.75LL + 0.75WL(-X)
563. REPEAT LOAD
564. 1 1.0 4 1.0 11 1.0 2 0.75 6 0.75
565. LOAD 313 DL + SDL2 + EP + 0.75LL + 0.75WL(+Z)
566. REPEAT LOAD
```

```
567. 1 1.0 4 1.0 11 1.0 2 0.75 7 0.75
568. LOAD 314 DL + SDL2 + EP + 0.75LL + 0.75WL(-Z)
569. REPEAT LOAD
570. 1 1.0 4 1.0 11 1.0 2 0.75 8 0.75
571. LOAD 315 0.6 (DL + SDL1) + WL(+X) + EP
572. REPEAT LOAD
573. 1 0.6 3 0.6 5 1.0 11 1.0
574. LOAD 316 0.6 (DL + SDL1) + WL(-X) + EP
575. REPEAT LOAD
576. 1 0.6 3 0.6 6 1.0 11 1.0
577. LOAD 317 0.6 (DL + SDL1) + WL(+Z) + EP
578. REPEAT LOAD
579. 1 0.6 3 0.6 7 1.0 11 1.0
580. LOAD 318 0.6 (DL + SDL1) + WL(-Z) + EP
581. REPEAT LOAD
582. 1 0.6 3 0.6 8 1.0 11 1.0
583. LOAD 319 0.6 (DL + SDL2) + WL(+X) + EP
584. REPEAT LOAD
585. 1 0.6 4 0.6 5 1.0 11 1.0
586. LOAD 320 0.6 (DL + SDL2) + WL(-X) + EP
587. REPEAT LOAD
588. 1 0.6 4 0.6 6 1.0 11 1.0
589. LOAD 321 0.6 (DL + SDL2) + WL(+Z) + EP
590. REPEAT LOAD
591. 1 0.6 4 0.6 7 1.0 11 1.0
592. LOAD 322 0.6 (DL + SDL2) + WL(-Z) + EP
593. REPEAT LOAD
594. 1 0.6 4 0.6 8 1.0 11 1.0
595. ****
596. *LOAD COMBINATION FOR CONCRETE DESIGN
597. LOAD 401 1.4 (DL+SDL1)
598. REPEAT LOAD
599. 1 1.4 3 1.4
600. LOAD 402 1.4 (DL+SDL2)
601. REPEAT LOAD
602. 1 1.4 4 1.4
603. LOAD 403 1.4 (DL + SDL1 + TL1) + 1.7 LL + 1.7 EP
604. REPEAT LOAD
605. 1 1.4 3 1.4 9 1.4 2 1.7 11 1.7
606. LOAD 404 1.4 (DL + SDL1 + TL2) + 1.7 LL  + 1.7 EP
607. REPEAT LOAD
608. 1 1.4 3 1.4 10 1.4 2 1.7 11 1.7
609. LOAD 405 1.4 (DL + SDL2 + TL1) + 1.7 LL  + 1.7 EP
610. REPEAT LOAD
611. 1 1.4 4 1.4 9 1.4 2 1.7 11 1.7
612. LOAD 406 1.4 (DL + SDL2 + TL2) + 1.7 LL  + 1.7 EP
613. REPEAT LOAD
614. 1 1.4 4 1.4 10 1.4 2 1.7 11 1.7
615. LOAD 407 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+X)
616. REPEAT LOAD
617. 1 1.2 3 1.2 2 0.5 5 1.6
618. LOAD 408 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-X)
619. REPEAT LOAD
620. 1 1.2 3 1.2 2 0.5 6 1.6
621. LOAD 409 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+Z)
622. REPEAT LOAD
```

```
623. 1 1.2 3 1.2 2 0.5 7 1.6
624. LOAD 410 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-Z)
625. REPEAT LOAD
626. 1 1.2 3 1.2 2 0.5 8 1.6
627. LOAD 411 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+X)
628. REPEAT LOAD
629. 1 1.2 4 1.2 2 0.5 5 1.6
630. LOAD 412 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-X)
631. REPEAT LOAD
632. 1 1.2 4 1.2 2 0.5 6 1.6
633. LOAD 413 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+Z)
634. REPEAT LOAD
635. 1 1.2 4 1.2 2 0.5 7 1.6
636. LOAD 414 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-Z)
637. REPEAT LOAD
638. 1 1.2 4 1.2 2 0.5 8 1.6
639. LOAD 415 0.9 (DL + SDL1) + 1.6 WL(+X) + 1.6 EP
640. REPEAT LOAD
641. 1 0.9 3 0.9 5 1.6 11 1.6
642. LOAD 416 0.9 (DL + SDL1) + 1.6 WL(-X) + 1.6 EP
643. REPEAT LOAD
644. 1 0.9 3 0.9 6 1.6 11 1.6
645. LOAD 417 0.9 (DL + SDL1) + 1.6 WL(+Z) + 1.6 EP
646. REPEAT LOAD
647. 1 0.9 3 0.9 7 1.6 11 1.6
648. LOAD 418 0.9 (DL + SDL1) + 1.6 WL(-Z) + 1.6 EP
649. REPEAT LOAD
650. 1 0.9 3 0.9 8 1.6 11 1.6
651. LOAD 419 0.9 (DL + SDL2) + 1.6 WL(+X) + 1.6 EP
652. REPEAT LOAD
653. 1 0.9 4 0.9 5 1.6 11 1.6
654. LOAD 420 0.9 (DL + SDL2) + 1.6 WL(-X) + 1.6 EP
655. REPEAT LOAD
656. 1 0.9 4 0.9 6 1.6 11 1.6
657. LOAD 421 0.9 (DL + SDL2) + 1.6 WL(+Z) + 1.6 EP
658. REPEAT LOAD
659. 1 0.9 4 0.9 7 1.6 11 1.6
660. LOAD 422 0.9 (DL + SDL2) + 1.6 WL(-Z) + 1.6 EP
661. REPEAT LOAD
662. 1 0.9 4 0.9 8 1.6 11 1.6
663. PERFORM ANALYSIS


        P R O B L E M   S T A T I S T I C S
        -----------------------------------

  NUMBER OF JOINTS        588  NUMBER OF MEMBERS       6
  NUMBER OF PLATES        532  NUMBER OF SOLIDS        0
  NUMBER OF SURFACES        0  NUMBER OF SUPPORTS    251

       Using 64-bit analysis engine.
```

STAAD SPACE -- PAGE NO. 18

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 55, TOTAL DEGREES OF FREEDOM = 2766
TOTAL LOAD COMBINATION CASES = 0 SO FAR.

*WARNING- ZERO STIFFNESS IN DIRECTION 1 AT JOINT 535 EQN.NO. 2272
LOADS APPLIED OR DISTRIBUTED HERE FROM ELEMENTS WILL BE IGNORED.
THIS MAY BE DUE TO ALL MEMBERS AT THIS JOINT BEING RELEASED OR
EFFECTIVELY RELEASED IN THIS DIRECTION.
*WARNING- ZERO STIFFNESS IN DIRECTION 2 AT JOINT 535 EQN.NO. 2273
*WARNING- ZERO STIFFNESS IN DIRECTION 3 AT JOINT 535 EQN.NO. 2274
*WARNING- ZERO STIFFNESS IN DIRECTION 4 AT JOINT 535 EQN.NO. 2275
*WARNING- ZERO STIFFNESS IN DIRECTION 5 AT JOINT 535 EQN.NO. 2276
*WARNING- ZERO STIFFNESS IN DIRECTION 6 AT JOINT 535 EQN.NO. 2277

664. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= OCT 30,2020 TIME= 22:20:46 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://www.bentley.com/en/support/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2017 Bentley Systems, Inc. *
* http://www.bentley.com *
****